Country: People's Republic of China
PROJECT DOCUMENT
United Nations Development Programme
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Project title: Energy Efficiency Improvement in Public Sector Buildings[1] in China (PSBEE)		
Country: People's Republic of China	**Implementing Partner: In China: National Government Offices Administration (NGOA)**	**Management Arrangements:** National Implementation Modality (NIM)
UNDAF/Country Programme Outcome: *Low carbon and other environmentally sustainable strategies and technologies are adapted widely to meet China's commitments and compliance with Multilateral Environmental Agreements*		
UNDP Strategic Plan Output: • Reduction of GHG emissions from the application of building energy efficiency techniques and practices • Mainstreaming environment and energy		
UNDP Social and Environmental Screening Category: *Low*	**UNDP Gender Marker: 1**	
Atlas Project ID/Award ID number: 87742	**Atlas Output ID/Project ID number: 94663**	
UNDP-GEF PIMS ID number: PIMS 5395	**GEF ID number: [2] 6930**	
Planned start date: September 2017	**Planned end date: September 2021**	
LPAC date: 7 July 2017		
Brief project description: The public sector in China include the national and local government-owned and operated institutions and facilities that provide service to the general public. These include buildings of government (central and local) ministries/departments, quasi-government organizations and associations whose operating budget comes from the Chinese government. To facilitate the widespread practice and application of energy conservation and energy efficiency (EC&EE) in the public sector, the National Government Offices Administration (NGOA) established its Energy Conservation Management Division (ECMD) in 2010, making it responsible for advancing, guiding, coordinating and supervising the implementation of EC&EE initiatives in the public sector in China. It has issued a number of administrative actions on EC&EE in the public sector during the 12th Five-Year Plan (12th FYP) period and has so far realized modest achievements. Even though the public sector in China has realized some achievements in EC&EE and in the application of low carbon technologies, energy utilization efficiency and reduction in specific energy consumption in the public sector can be further improved in the coming 13th FYP periods. This existing situation can be greatly improved through improving		

[1] These refer to buildings used, operated and managed by Government (national and local) offices, quasi-government organization and associations whose operations are fully or partly financed by the Chinese government, and includes buildings used as office, hospitals, schools, social/civic/sports & recreation facilities and owned by the Government of China.
[2] Project ID number will be assigned by GEFSEC and to be entered by Agency in subsequent document submissions.

barriers, including: integrating policy and regulatory frameworks, evaluations systems, market-oriented financing mechanism, and improved outreach and professional capacity building.	
FINANCING PLAN	
GEF Trust Fund	USD 8,932,420
UNDP TRAC resources	
Cash co-financing to be administered by UNDP	
(1) Total Budget administered by UNDP	**USD 8,932,420**
PARALLEL CO-FINANCING *(all other co-financing that is not cash co-financing administered by UNDP)*	
UNDP	USD 100,000
National Government Offices Administration (Grant)	*USD 25,810,000*
National Government Offices Administration (In-kind)	*USD 5,000,000*
Provincial Government (Grant)	
- *Jilin Province*	*USD 11,700,000*
- *Jiangxi Province*	*USD 5,700,000*
- *Gansu Province*	*USD 5,790,000*
Private Sector	*USD 16,000,000*
(2) Total co-financing	**USD 70,100,000**
(3) Grand-Total Project Financing (1)+(2)	**USD 79,032,420**

SIGNATURES		
Signature: print name below	**Agreed by Government**	**Date/Month/Year:**
Signature: print name below	**Agreed by Implementing Partner**	**Date/Month/Year:**
Signature: print name below	**Agreed by UNDP**	**Date/Month/Year:**

List of Abbreviations & Acronyms

Abbreviation	Meaning
AAA	Triple A standing in rating system
APR/PIR	Annual Project Report/Project Implementation Review
AWP	Annual Work Plans
CF	Carbon Financing
CO_2	Carbon dioxide
CPAP	Country Program Action Plan
CTA	Chief Technical Advisor
EE	Energy efficiency
EOP	End-of-project
EPC	Engineering, procurement and construction
ERC	Evaluation Resource Center
GDP	Gross domestic product
GE	Green Economy
GEB	Global environmental benefits
GEF	Global Environmental Facility
GHG	Greenhouse gasses
GOC	Government of China
GWh	Gigawatt-hours, or million kWh
kt	Kilotons or 1,000 tons
kWh	Kilowatt-hours
L	Liter
LGOA	Local Government Office Administration
M&E	Monitoring and evaluation
MOF	Ministry of Finance
MOST	Ministry of Science and Technology
MOT	Ministry of Transportation
MTR	Mid-term Review
NDRC	National Development and Reform Commission
NGO	Non-government organization
NGOA	National Government Office Administration
NIM	National Implementation Modality
NPD	National Project Director
NPM	National Project Manager
OFP	Operational Focal Point
PD	Project Design
PIF	Project Identification Form
PMU	Project Management Unit
PPG	Project preparation grant
PPP	Public and Private Partnerships
PPR	Project Progress Reports
PSBEE	Energy Efficiency Improvement in Public Sector Buildings in China
PSC	Project Steering Committee
RE	Renewable Energy
SBAA	Standard Basic Assistance Agreement
TAC	Technical Advisory Committee
TE	Terminal Review
TEU	Twenty-foot equivalent unit (TEU)

toe	Tons of oil equivalent
TOR	Terms of reference
TOR	Terms of Reference
UNDP	United Nations Development Program
WB	World Bank

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II. DEVELOPMENT CHALLENGE

Context and Global Environmental Significance

The public sector in China is comprised of national and local government-owned and operated institutions and facilities that provide service to the general public[3]. These include buildings of government (central and local) ministries/departments, quasi-government organizations and associations whose operating budget comes from the Chinese government. The table below shows the historical energy consumption and corresponding CO_2 emissions in the buildings that comprise the public sector in China. Bulk of the energy consumption is accounted for by coal (50.7%), followed by electricity (40.2), petroleum fuels (7.1%), and natural gas (2.0%). In 2012, the public sector in China consumed about 157 million tce (Mtce) of energy and the corresponding amount of CO_2 emissions is about 462 million tons (Mtons). During the period 2005-2012, the average annual energy consumption of the public sector was about 147.8 Mtce, which translates to an average annual CO_2 emission of about 434.3 Mtons[4].

Year	2007	2008	2009	2010	2011	2012	2013	2014	2015
Energy Usage, Mtce	141	145	149	152	155	157	201	183	183
CO_2 Emissions, Mton	411	427	440	446	456	462	503	458	458

To facilitate the widespread practice and application of energy conservation and energy efficiency (EC&EE) in the public sector, the National Government Offices Administration (NGOA) established its Energy Conservation Management Division (ECMD) in 2010, making it responsible for advancing, guiding, coordinating and supervising the implementation of EC&EE initiatives in the public sector in China. It has issued a number of administrative actions on EC&EE in the public sector during the 12th Five-Year Plan (12th FYP) period and has so far realized modest achievements. Nevertheless, even though the public sector in China has realized some achievements in EC&EE and in the application of low carbon technologies, the demand for energy in the sector is still expected to increase in the coming years. The expected increase is due to: (1) new building floor area; (2) application of energy-using equipment; and, number of building occupants. With the expected increase in the: (1) level of work activities; (2) expansion of the scope and coverage of public services; and, (3) overall socio-economic development, there will also be commensurate increase in the demand for energy in public sector buildings. With the current level of technical capacity, understanding on the application and implementation of EC&EE as well as low carbon technologies, and the general attitude of the public sector towards energy conservation, further improvement in the energy utilization efficiency and reduction in specific energy consumption in the public sector will be very difficult to realize in the coming FYP periods.

Barrier Analysis

The abovementioned situation is further exacerbated by the existence of certain persistent barriers, which are as follows:

[3] According to most recent statistics, there are about 1,904,900 public sector buildings (including 455,900 State buildings, 656,500 education facilities, 285,100 health facilities, 190,500 science facilities, 19,700 cultural facilities, 6,600 recreational facilities, 69,200 other facilities, and 221,400 social facilities) in China as of 2012. In 2012, the total public sector building floor area was about 8.488 billion m2, and consumed about 157 Mtce, accounting for 5.8% of total social energy consumption and 13.684 billion tons of water consumption.

[4] Based on the NGOA internal statistical report on the energy consumption in the public sector in China, the public buildings in the country accounted for about 5.4% of China's total GHG emissions in 2012. Based on energy consumption data in 2010 from the NGOA and the IEA (CO2 Emissions from Fuel Combustion), the GHG emissions from the public buildings accounted for 20 % of the total national commercial & residential sector GHG emissions. Since many of the residential buildings are government-owned, it can be said that the energy consumption in public buildings is 20% of the energy consumption of the buildings sector in China.

(1) ***Inadequate policy and regulatory frameworks that promote and support EC&EE improvement initiatives in the public sector*** - Currently, there are existing general policies and frameworks on EC&EE but there are no specific ones for the public sector, and the implementation and enforcement of these general policies and regulations are inadequate and ineffective. While there are stated (in the 12th FYP) energy targets and the measures that will be taken to realize the target. the implementation plan for the measures to be done to realize the targets is not adequate; EC&EE projects in the public sector are often not sufficiently funded; do not get enough and optimum funding from social funds, and consequently fall short of achieving target energy and energy cost savings. The lax and rather weak implementation of EC&EE policies in the national and local governments further contributes to this major barrier.

(2) ***Absence of an overall evaluation system for reviewing and analyzing the existing energy management procedures and practices in public sector entities and their facilities*** - The administrators and managers of the different public sector buildings in the country are not motivated, and in fact not required/obligated to carry out EC&EE and low carbon technology/technique implementation because their operations costs are uniformly financed from government budgets. Those rather few public sector buildings that have already started doing energy management activities are without the benefit of an effective evaluation system to assist in keeping track of their performance and progress, and provide appropriate recommendations for improving individual EC&EE and energy management efforts. Furthermore, there are also industry associations, the public sector in China does not benefit from an established technology evaluation system, which is currently non-existent.

(3) ***Limited market-oriented[5] financing mechanism for funding EC&EE technologies/products in the public sector*** - The public sector in China is also a big energy consuming segment of the national economy but the application of EC&EE technologies in this sector has been limited. Apart from the low level of knowledge and confidence to apply building-related EC&EE (and low carbon) technologies in the public sector, there is also the problem of lack of funding from other sources such as the financial sector (i.e., private banks and financial institutions) and from energy service companies and entities that provide energy supply outsource services, to supplement limited government and social funds[6]. In regards to engineering, procurement and construction (EPC) in the public sector, China's public buildings represent less than 1% of the country's current EPCs[7].

(4) ***Low level of capacity and awareness of public sector technical and management personnel on the application of EC&EE and low carbon technologies*** - In general, this is true especially in the public sector in provinces and small to medium cities/towns in the country, where generally there are no capacity development programs designed and catered to the public sector on EC&EE and low carbon technologies and techniques/practices applications. Public sector personnel (e.g., government authorities, policy and decision-makers, public sector administrators, buildings and facilities managers, etc.) and the citizenry have low level of

[5] In the context of the project, this refers to financing schemes implemented by commercial banks and applied for EC&EE initiatives in public sector buildings instead of government run and controlled funding schemes.

[6] The administrators and managers of public sector buildings in the country are generally not motivated, and in fact not required/obligated to carry out EC&EE and low carbon technology/technique implementation because their operations costs are uniformly financed from government budgets. Allocating budgets for such endeavors, let alone engaging ESCOs to do, or assist them doing, such efforts, would most likely not be their priority. The generally low level of awareness of public sector personnel (e.g., government authorities, policy and decision-makers, public sector administrators, buildings and facilities managers, etc.) about the various EC&EE and low carbon options and opportunities that can be applicable in public buildings and facilities, and their socio-economic and environmental benefits, prevents them from engaging the services of ESCOs in developing and implementing such initiatives, let alone doing such efforts themselves.

[7] The current barriers to EPCs in the public sector in China are: 1) Lack of public sector involvement as an active EPC customer; 2) Lack of comprehensive M&V protocol and technical and institutional capacity in M&V; 3) Lack of access to external financing and shortage of diversified sources of financing; and (4) Lack of clients' creditworthiness. Refer to LBL/North West Lab Report on promoting EPCs in China. http://www.globalchange.umd.edu/data/epc/EPC_Market_Opportunity_Paper_final0429.pdf

awareness about the various EC&EE and low carbon options and opportunities that can be applicable in public buildings and facilities, let alone the socio-economic and environmental benefits that can be derived from these.

In many developed countries, Energy Service Companies (ESCOs) play a key role in the design, financing, and implementation of EC&EE projects in buildings (commercial and government). In China, local ESCOs are active in such kind of projects in private commercial buildings. ESCOs have not been successful enough in doing business in government buildings[8]. The barriers that prevent ESCOs from engaging in public sector EC&EE projects such as those in national and local government buildings and facilities are mainly linked to the abovementioned 4 main barriers. The non-involvement of the ESCOs (and private sector financing institutions) in public sector EC&EE activities are by and large effects of 3 of the 4 main barriers, and contributes to the barrier of limited market-oriented and/or performance-based financing mechanism for funding EC&EE technologies/products in the public sector. The ESCO industry just like any other profit-oriented industry would definitely be interested in investing resources (expertise, time and money) in profitable EC&EE projects. The lax and rather weak implementation of policies on EC&EE in the national and local government agencies and facilities mainly prevent private sector ESCOs interest and investment in the public sector EC&EE projects. The main barriers for ESCOs entering public building segment are on the demand side: (a) there are no rules and best practices identified to support the use of engineering, procurement and construction (EPC) contracts in public procurement; and, (b) energy budgets are decreased when energy efficiency is achieved (since budgets are determined by the previous years' actual expenses). This reduces the interest of both ESCOs and public building managers in pursuing EPC since the payment allowance reduces public institutions' ability to pay for the services. The project has a potential and ambitions to leverage significant amount of private sector co-financing from Chinese ESCOs by removing the underlying, demand-side barriers, to foster investment in the public buildings sector.

There a number of services that the private sector can provide in the area of EE/RE applications in the public sector buildings. Currently, the services of private engineering firms are being engaged by some public sector buildings that are keen in doing EC&EE. In the past 5 years, the NGOA has contracted private consulting firms to carry out energy auditing and energy management services in some of the major public buildings in the major cities. The ESCO scheme is something that the public sector buildings are keen to carry out but the ESCO-related barriers are hindering the NGOA in doing this. There is also the possibility of engaging the services of the private sector (once the current barriers are removed, and some of the financing schemes will be adopted) to carry out outsourced energy services for the public buildings, including the implementation build-operate-transfer or build-operate-own projects, or variations of these.

III. STRATEGY

Applied Project Strategy

The main strategy that is employed to achieve the project objective is barrier removal. The enabling conditions that will be created through the barrier removal activities of the proposed GEF project

[8] China has a well-established and rapidly growing ESCO sector (Ref: IFC China ESCO Market Study Report): over 2,400 centrally registered companies worth CNY 125 billion. But the industry is still in its early stage of development. Individual company's scale is small and most ESCOs can be categorized as SME or even micro enterprises. Roughly 20% of ESCOs capital come from debt, the rest is equity/self-financing. The business of Chinese ESCOs cover three areas: industry (82%); buildings (15%); and, transportation (3%). China's public buildings represent less than 1% of the country's current EPCs.

will facilitate the achievement of the required outcomes that will contribute in the achievement of the project objective. Specific interventions that will remove barriers to EC&EE and LC initiatives that bring about reduction of CO2 emissions from the operation of energy consuming equipment/appliances and service facilities in public sector buildings will be carried out. Involving the private sector, particularly banks/financial institutions, as well as entities that are willing to finance EC&EE and LC technology applications through ESCO and PPP modalities, forms part of the strategy for the proposed project. The strategy to attract private sector partners and investment is essentially based on 3 core elements:

a) Knowing and understanding of the current energy utilization – This involves laying the foundation for better and more efficient energy management in the public sector through the establishment of a system that generates up to date and accurate information of energy use that can be monitored on-line using modern ICT technology.
b) Creation of the necessary demand for better energy management and increased energy efficiency - This can be done e.g., by introducing mandatory EE targets for the public sector (at the different levels) and mandating public procurement standards incorporating EE, combined with the right incentives (budget process reform, by allowing the public entities to retain all or part of their energy savings).
c) Stimulation of supply of energy management services and EE measures from the market (i.e., the private sector) by promoting EPC contracting between public entities and private ESCOs (assist the public sector with tendering out the EPC contracts) and establish the required appropriate financial mechanism(s) (guarantees of various sorts, or other credit enhancement measures like concessional lending (from public sources); or some sort of combination) allowing the ESCOs to get access to capital on the right terms to finance EE projects in public buildings.

Lastly, the strategy to continuously facilitate the effectiveness of the established enabling conditions through their institutionalization will be applied. Towards the end of the project, as part of the project activities, a suitable follow-up action plan that comprise of the appropriate actions that will ensure sustainability of whatever policies, regulations, and institutional frameworks that will be established and enforced and implemented in the public sector buildings in China will be developed.

Approach to Full Project Design

A logical framework analysis approach was used for the design of the full size project. Within each of the four main barrier categories, stakeholders during a logical framework analysis workshop, identified the key barriers/problems and carried out a problem analysis to establish their cause-and-effect relationships. The inter-relationships of the barriers were then established through a “problem tree” configuration. Problems were then converted into desirable objectives that could realistically be achieved if the respective problems are solved, creating an “Objectives Tree.” The Objective Tree was used in identifying the project goal, objective, outcomes, and outputs. Based on the objective tree, the objective statement on GHG emission reduction was selected at the project goal, considering that such is the goal of GEF-funded climate change mitigation projects. The objective statement that refers to the direct way of achieving the GHG emission reduction goal is selected as the project objective. The objective statements that represent as the direct means of achieving the project objective are the project outcomes. Similarly, the objective statements that represent as the direct means of achieving the project outcomes are the bases for the expected project outputs. Activities were then designed to achieve each of the targeted outputs. Such activities are also based on the various objective statements in the objective tree that represent as means for realizing the project outputs. Details on targeted outcomes and outputs, as well as planned activities are given in Section III of this document.

Alternative Scenario

An alternative scenario that is characterized by more energy conserving operation of existing, and more energy efficient new, buildings in the public sector would be possible if the current and persistent barriers to the widespread practice of EC&EE technologies and techniques/practices are removed. With the removal of barriers, the proper management of energy and resource utilization, the adequate financing and cost-effective application of EC&EE technologies, in public sector buildings will bring about reduced energy consumption and consequently also bring about reduction in CO_2 emissions.

Based on the trend analysis of historical annual energy consumption in public sector buildings, and the forecast potential reduction in energy demand due to the increasing awareness and improving attitude towards EC&EE efforts, a much greener (in terms of energy use) operation of these buildings will bring about a significant reduction in the specific energy consumption. Hence, the envisioned alternative to the current energy utilization practices in these buildings, would be something that would be more rationale, i.e., efficient, effective and wise. There will be more investments in the deployment of EC&EE and LC technologies in the design, construction and operation of public sector buildings. The main motivation and driving force for these will be the enforcement of policies and regulations that are supportive of the widespread practice of EC&EE and LC in these buildings, as well as the tangible benefits that can be derived from them.

Innovativeness and Replication

The PSBEE project is designed to include the following innovative features:

a) Introduction of ICT solutions and "smart" technologies for building energy management to monitor building energy use, spot immediate and most cost-effective opportunities and effectively monitor performance and improvement. Without a good EMIS it is impossible to move on with any market-based instruments because savings should be objectively monitored to be "monetized".
b) Establishment of EE targets/obligations (in particular in the public sector) to boost demand and uptake of EE measures in public sector buildings. This is in line with the fact that if there are no targets/obligations, there is no market solution, because the public sector tends to be very inert (as in most, if not all countries). But, the target system will only work and generate private interest and investment if there is a robust MRV system (i.e., energy savings verification system (ESVS) underpinning it. Hence, the need for a world class EMIS.
c) Promotion of EPC contracting. EPC contracting has proven to be effective to bring the private sector on board. But it is still new in China, and is still very rarely applied in the public sector (even in developed counties) because of the many barriers and inertia typically in place.

The EMIS is in line with what are considered as emerging global "state of the art" in the building sector, including public buildings, is the proliferation of ICT solutions and "smart" technologies for building energy management to monitor building energy use, spot immediate and most cost-effective opportunities and effectively monitor performance/improvement. UNDP has an excellent track record of piloting EMIS, which received nationwide replication in Croatia and is now being rolled out in other countries. The Croatian EMIS has been listed as international best practice for ICT solutions in buildings[9]. These innovative features are also fully consistent with recommendations from Berkeley Lab/Pacific Northwest Lab on promoting EPCs in China (April 2015)[10].

[9] http://www.unece.org/fileadmin/DAM/energy/se/pdfs/geee/pub/ECE_Best_Practices_in_EE_publication.pdf).

[10] (http://www.globalchange.umd.edu/data/epc/EPC_Market_Opportunity_Paper_final0429.pdf)

IV. RESULTS AND PARTNERSHIPS

i. Expected Results:

The PSBEE Project aims to facilitate the energy conserving and energy efficient operation of buildings and building services in the public sector in China, is designed to facilitate the realization of the envisioned alternative scenario in the energy performance of the public sector buildings in China. This project comprise of baseline and incremental activities that will enhance the outputs of the currently implemented and planned programs and activities in the public sector buildings in the country in improving the energy utilization performance of these buildings. These would include, among others, activities on the formulation and enforcement of EC&EE support policies, establishment and implementation of public sector energy performance monitoring and evaluation systems, EC&EE project financing mechanisms, improving the capacity and awareness of the public sector, and the provision of public information services on building EC&EE and low carbon technologies.

Project Goal and Objective

Goal: Well-managed growth rate of energy consumption and associated GHG emissions from the public sector in China.

Objective: Facilitation of the energy conserving and energy efficient operation of buildings and building services in the public sector in China

Project Components, Outcomes, and Outputs

Component 1: Public Sector EC&EE Policy and Regulatory Frameworks

This component comprises activities that collectively address the barriers related to the inadequate policies and regulations that help promote and support EC&EE applications and practices in public sector buildings in China. With the delivery of the expected outputs from these activities, a strict enforcement of approved enhanced policies and rules and regulations on energy efficiency and low carbon operation and maintenance of public sector buildings is the expected outcome.

Outcome 1.1: Strict enforcement of approved enhanced policies and rules and regulations on energy efficiency and low carbon operation and maintenance of public sector buildings.

To achieve Outcome 1.1, a number of outputs have to be delivered through the implementation of the following major activities: (1) Conduct of a comprehensive assessment of applicable foreign and domestic energy conservation and energy efficiency (EC&EE) and low carbon policies and regulations for public sector building ; (2) Formulation, and promotion/advocacy work for the approval, of EC&EE improvement roadmaps; (3) Formulation, and promotion/advocacy work for the approval and enforcement of, policies including the associated guidance and implementing rules and regulations (IRRs) on the: (a) energy monitoring and reporting of public sector buildings, as well as on their energy performance evaluation; (b) energy savings measurement & verification system (ESMVS) for public sector buildings including the associated guidelines and IRRs on the reporting and verification of energy savings realized from registered EC&EE projects; and, (c) improved and revised Regulations on Energy Conservation in Public Sector including the associated new set of guidance and institutional frameworks; (4) Design and implementation of pilot demonstration of the application of EC&EE policies and systems in three to five local government office administration

(LGOA) regions; (5) Development, and promotion/advocacy work for the approval, of the follow-up plan for the replication of the application of the piloted EC&EE improvement policies in public buildings covered by LGOAs in other provinces; and, (6) Development (for approval) of improved and updated public sector building energy standards.

The following paragraphs presents the various outputs that are expected to collectively bring about Outcome 1. For each output, the required activities are enumerated and described:

Output 1.1: Completed comprehensive assessment of applicable foreign and domestic energy conservation and energy efficiency (EC&EE) and low carbon (LC) policies and regulations for public sector buildings. This output is for providing the necessary information for the formulation of the required policies and regulations in promoting and implementing EC&EE and LC measures in the public sector buildings in China. The following are the activities that have to be carried out to deliver this output.

Activity	Description
1.1.1: Gathering of information about EC&EE and LC policies and regulations for public sector buildings from developed and developing countries that have the same circumstances as China	This involves the conduct of a combination of desk researches and study tours in selected countries that enforce specific EC&EE and LC policies in their public sectors. The specific data gathering work during the study tour includes the survey of the effectiveness of the existing enforced policies and regulations in public sector buildings. The study tour includes learning how the relevant policies and regulations are formulated, approved and enforced, as well as learning how to effectively enforce the policies and regulations.
1.1.2: Evaluation of public sector buildings EC&EE and LC policies and regulations from selected countries	This involves the analysis of the information gathered and learned during the study tour to determine the EC&EE and LC policies and regulations that are applicable and appropriate for adoption in China's public sector buildings. Also included in this activity is the identification of all the requirements and feasible implementation arrangements for the cost-effective application of such policies and regulations in China.
1.1.3: Inventory of information on local EC&EE and LC policies and regulations for public sector buildings in China	This involves the conduct of a combination of desk researches and focused stakeholder consultations in LGOAs in selected provinces and SARS in China to determine policies and regulations that are related to EC&EE and LC. The specific data gathering work in the LGOAs includes the survey of the effectiveness of the existing enforced policies and regulations in public sector buildings.
1.1.4: Comparative analysis of local and foreign EC&EE and LC policies and regulations for public sector buildings	This entails the comparative evaluation of the domestic and foreign EC&EE and LC policies and regulations for public sector buildings, based on the data gathered from the LGOAs in selected provinces and SARS, and from other countries. The evaluation is intended to come up with a list of local and foreign EC&EE and LC policies and regulations that are can be appropriately and cost-effectively adopted in the public sector buildings in China.
1.1.5: Presentation and discussions on the results of the comparative analysis and recommended local and foreign EC&EE and LC policies and regulations	This entails the design, organization and conduct of a seminar-workshop that involves the presentation and discussion of the results of the comparative analysis carried out in Activity 1.1.4. At the end of the seminar-workshop, a final list of recommended EC&EE and LC policies and regulations for public sector buildings is adopted for approval by the NGOA/NDRC.

for public sector buildings	
1.1.6: Documentation of the results of the proposed applicable local and foreign EC&EE and LC policies and regulations for adoption in the public sector buildings in China.	This entails the publication (print and web-based) for dissemination of the information on the agreed proposed EC&EE and LC policies and regulations that will be adopted for enforcement in the public sector buildings in China.

GEF support is required for the incremental technical and financial assistance in the conduct of the data gathering and processing work, comparative analysis, presentation and discussion, as well as in the publication and dissemination of the recommended EC&EE and LC policies and regulations.

Output 1.2: Formulated and promoted EC&EE improvement roadmaps. These roadmaps are mainly on the visioning, planning, targeting and implementation of actions towards improving the specific energy performance of public sector buildings in various climate areas China. The following are the activities that have to be conducted to delivering this output:

Activity	Description
1.2.1: Evaluation of the current EC&EE activities in the public sector buildings in China.	This involves a survey of the EC&EE activities in public buildings managed by LGOAs in selected provinces and SARS in China to the current situation in regards the awareness and knowledge of these buildings about EC&EE and the types of EC&EE and energy management activities that are being implemented or being planned for implementation. The survey will also focus on the barriers/issues encountered in achieving the EC&EE objectives (if any), and the effectiveness of the EC&EE initiatives that are implemented in these public sector buildings. An accurate set of information on the current status of costs, performance, technologies applied and limitations of the EC&EE systems deployed in the public sector buildings will be developed.
1.2.2: Defining the desired pathway for the EC&EE applications in the public sector buildings.	This involves modelling and scenario analysis, which are important tools used to define possible future scenarios in regards the energy utilization performance of the public sector buildings. The modelling process will assess important parameters such as the growth if overall floor area of public buildings, projections on building occupancies and activities, and modernization to suggest different energy futures and environmental consequences of those futures. Projections and forecasts will be developed to cover local energy and economic conditions, market realities, and public policy priorities. Based on these projections, appropriate objectives and pathways to reach these will be developed in such a way to local energy supply-and-demand conditions, economic development circumstances, and energy and environmental policies.
1.2.3: Development of the EC&EE improvement roadmaps	This involves the conduct of a series of workshops to solicit ideas from stakeholders through highly structured brainstorming and analysis sessions addressing different aspects of the EC&EE issues/problems. From the workshop results, a draft roadmap document is prepared for each LGOA region (at least 5) presenting a compelling story of possibilities and describes a rational, consistent sequence of activities that can lead to the desired EC&EE objective. A carefully constructed

	review process is implemented to maximize the contributions and support of stakeholders without becoming unmanageable. This will be a series of three expanding review cycles, the last one being an independent review. Each draft is revised after each review cycle with the incorporation of relevant comments and recommendations to produce a final draft roadmap.
1.2.4: Publication of the EC&EE improvement road maps and preparation for their launching and implementation.	This involves the review of the final draft road maps by the NGOA/NDRC or by a select group of roadmap sponsor or select group of experts and NGOA authorities to resolve any final outstanding issues government and approve the publication (print and web-based) of the road maps. This activity also include preparations for the formal launching of the roadmaps. Tentatively, the launching will include a press release, selective electronic distribution of the final roadmap and accompanying materials, and public remarks by roadmap leaders at relevant meetings and conferences. This activity will also include the plans for the implementation of the road map activities, as well as the monitoring of progress and adjustment of the road maps.
1.2.5: Implementation of the EC&EE improvement road maps	This involves carrying out the set of priorities, e.g., research activities, EC&EE technology demonstrations, policy advances, regulatory changes and financial commitments — that are needed over a defined time frame to achieve the roadmap's goals. The defined mechanism of project initiation and management, and stakeholders' engagement to address near-term priorities is a key first step in implementing the roadmap.

GEF support is required for the incremental technical and financial assistance in the conduct of the road mapping process for EC&EE improvements in the public sector buildings.

Output 1.3: Formulated and enforced policies including the associated guidance and implementing rules and regulations (IRRs) on energy monitoring and reporting, energy savings verification, and energy conservation in public buildings. This output is a set of policies, guidance and IRRs on the following: (a) energy monitoring and reporting of public sector buildings, as well as on their energy performance evaluation; (b) energy savings measurement & verification system (ESMVS) for public sector buildings including the associated guidelines and IRRs on the reporting and verification of energy savings realized from registered EC&EE projects; and, (c) improved and revised Regulations on Energy Conservation in Public Sector including the associated new set of guidance and institutional frameworks. The activities that will be carried out to deliver this output are the following:

Activity	Description
1.3.1: Formulation, approval and enforcement of policies and IRRs on energy management information system (EMIS) for public sector buildings and MRV of EC&EE projects in public sector buildings	This involves several sub-activities from the review of existing policies on energy monitoring and reporting as part of the country's national energy conservation law, in general, and the NGOA's energy conservation program to the implementation of the proposed energy management information system (EMIS) for public buildings, evaluation of the results and impacts of the EMIS.
1.3.1.1: Review of	This involves the evaluation of the current policy and the implemented

current policy and implemented system for energy monitoring and reporting of public buildings.	system for energy monitoring and reporting of NGOA-, and LGOA-administered public buildings. It includes the conduct of policy research that will provide the necessary inputs for developing recommendations for potential policy revisions and new policies supporting the development and operationalization of an EMIS for public buildings.
1.3.1.2: Formulation of new policies and regulations on EMIS for public buildings.	This involves the development of new policies and regulations on EMIS for public buildings based on policy research and the NGOA's public buildings energy conservation program. The proposed policies will be for the implementation of appropriate mechanisms (and possible incentives) to enable the effective implementation of the EMIS. The incentives and support mechanisms that will be considered will be based on the findings from the policy researches and may include: (a) technical assistance in the analysis and reporting of energy consumption and energy conservation plans of public buildings; and (b) incentives for the implementation of EC&EE projects for best energy performing public buildings.
1.3.1.3: Presentation and promotion of the recommended policies on EMIS.	This entails the organization and conduct of campaigns to promote the recommended policies on EMIS to NGOA and other relevant government agencies, as well as the key players in the Chinese building industry. Follow-up advocacy/lobbying work with the relevant entities and authorities will be carried out until the proposed policies are approved.
1.3.1.4: Formulation of the supporting implementing rules and regulations (IRRs) for the EMIS.	This entails the formulation of the implementing rules and regulations (IRRs) of the approved policies on the EMIS. It also include the formulation of the implementation guidelines for the provision of the required standard and specific energy data/information that will be reported by the NGOA-, and LGOA-administered public buildings.
1.3.2 Formulation, approval and enforcement of supplementary policies and IRRs to the EMIS for the establishment and operationalization of an Energy Savings Measurement & Verification System (ESMVS).	This involves the conduct of several sub-activities to formulate supporting policies on the energy performance evaluation of public buildings, and for measurement & verification of reported energy savings from EC&EE initiatives implemented by public buildings based on the reports submitted to the EMIS.
1.3.2.1: Formulation of supporting policies and regulations on the evaluation of the energy performance, and verification of reported energy savings (in EMIS), of public buildings.	This involves the formulation and recommendation for approval of supporting policies for a supplementary system to the EMIS that are for: (1) Evaluation of the energy performance of public buildings based on energy reports submitted; and, (2) Measurement and verification of reported energy savings derived from implemented EC&EE and LC initiatives. The verification of reported energy savings from EC&EE initiatives implemented by public buildings based on the reports submitted to the EMIS.
1.3.2.2: Conduct of advocacy and promotional work to secure approval of the ESVS for public buildings.	This involve conduct of advocacy/lobbying work with the relevant government entities and authorities to get the policies approved. Thereafter, work on the design and development of the ESMVS will follow. This supplementary system is important in the provision of feedbacks to the public buildings covered in the EMIS, and particularly in

	cases where incentives are provided for the implementation of EC&EE projects for best energy performing public buildings.
1.3.3: Formulation, approval and enforcement of improved and revised regulations on Energy Conservation in the Public Sector including associated new set of guidance and institutional frameworks.	This involves the implementation of several sub-activities to develop and implement/enforce policies and implementing rules and regulations on Energy Conservation in the Public Sector.
1.3.3.1a: Review of the existing energy conservation policies and regulations as applied to the implementation of EC&EE measures in buildings.	This involves the conduct of a review of current national and local energy conservation policies and regulations as applied to the public sector, in general, and public buildings, in particular. It includes the conduct of policy research that will determine the various items that need revisions to enhance the application of EC&EE and LC initiatives specifically in public buildings, as well as other the necessary inputs for developing recommendations for potential policy revisions and new policies supporting energy conservation in public buildings.
1.3.3.1b: Review of the existing, and potential improvements to, institutional and financial policies regarding the use of savings/revenues generated by public buildings.	This involves a comprehensive review of the existing institutional and financial policies on the use by a public building of savings/revenues generated that it generated, for furtherance of the savings/revenues generation efforts or for funding new initiatives to generate savings/revenues. A special committee that will include, among others MOF and NDRC will be convened by NGOA to discuss this specific issue that demotivates public buildings in implementing EC&EE/LC measures. Policy research will be carried out to determine the pros & cons of the current finance policies as these apply to potential cost savings from EC&EE/LC initiatives that public buildings intend to implement. Policy options will be developed and evaluated to provide inputs to potential special institutional and financial reforms in regards the use of cost savings from EC&EE/LC initiatives in public buildings.
1.3.3.2: Formulation of supporting policies and regulations on energy conservation in the Public Sector.	This involves the development of appropriate policies and cost-effective policy measures on energy conservation in the public sector based on the gap and problem analyses that will be carried out in Activity 1.3.3.1 (and b). This also involves the development of suitable and cost-effective policy reforms regarding retention in the budget (on top of the annual budget) of public buildings the cost savings generated from EC&EE/LC initiatives that they have carried out. Such retained earnings will be for maintaining the implemented initiatives and/or for developing and implementing new EC&EE/LC projects. Allowing the retention of energy cost savings for use in new projects can be considered as financial incentive for public buildings that have performed very well in their EC&EE/LC efforts as reflected in their energy reports to the EMIS, and validated by the ESMVS. Part and parcel of this major activity is the conduct of advocacy work with the relevant government entities and authorities to secure the approval of the policies and have these enforced.
1.3.3.3: Formulation and finalization of the public	This entails the formulation of the implementing rules and regulations (IRRs) of the approved public sector energy conservation policies. It also

sector energy conservation policy measures and associated implementing rules and regulations (IRRs).	include the finalization of the set of sector-wide policy measures based on the approved policies and proposed IRR. The policy measures and the implementation mechanisms will be presented and promoted to NGOA, LGOAs, and relevant government agencies and authorities, as well as the key players in the Chinese building industry. This is to secure support and endorsement of the relevant stakeholders, particularly the NGOA.
1.3.3.4: Evaluation of the results and impacts of public sector energy conservation policies.	The enforcement of the approved new IRRs will be carried out using the new set of guidelines and implementation arrangements that will also be developed after the approval of the new IRRs. The impacts of the public sector energy conservation policies and IRRs will be regularly monitored with the intention of getting comments and suggestions from the management of NGOA-, and LGOA-administered public buildings. Based on the information gathered, improvement or maintenance of the policies, policy measures and IRRs will be recommended.

GEF support is required for the incremental technical and financial assistance in the conduct of all evaluation work that will be carried out in the formulation and promotion of the development of the EMIS and ESVS and the revised regulations on energy conservation in the public sector. GEF assistance is also required for the Incremental work in the promotion for the approval and enforcement of the recommended policies and regulations.

Output 1.4: Completed demonstrations on the application of EC&EE policies and systems in 3 to 5 local government office administration (LGOA) regions. This output comprise of the analyzed results and impacts of the demonstrations on the application of EC&EE policies and systems in 3 LGOA regions. The following major Activity are intended to deliver this output:

Activity	Description
Activity 1.4.1: Solicitation and vetting of ideas for novel policies for discussion among experts and local policy makers.	This involves the design and planning for the piloting of selected recommended EC&EE and LC policies in public sector buildings in China, as well as the design of and reaching of consensus on policy pilots. Based on ideas vetted, the design will first involve liaison with local government and reaching of consensus on cities that will host the policy pilots and on the nature of those pilots. Detailed design of policy pilots will then be carried out. The design shall take into account the investments required as well as the implementation plan for achieving the expected policy impacts. There will be at least 3 local policy pilots.
Activity 1.4.2: Implementation of agreed upon local-level policy pilots.	There will be at least two policies to be piloted. This will include preparation and approval of required government notices or other documentation, outreach to potential public building administrators, and facilitation of the required implementation arrangements for hosting the cities/local governments that will host the policy pilots. Each pilot city will demonstrate the application of a specific public buildings energy conservation policy.
Activity 1.4.3: Design and implementation of plan to monitor results of the two or more local-level policy pilots.	This activity includes the design and use of a monitoring plan for tracking the results of the pilots. It also includes documentation of results and dissemination. The monitoring plan will assess the effectiveness of outreach to potential public building administrators and seek to understand how influential preferential policies were in impacting their EC&EE initiatives. It will further assess the numbers of stakeholders influenced by the relevant pilot policy.

Activity 1.4.4: Design and conduct of survey of cities on the development of local public building EC&EE policies and regulations.	This involves the conduct of a survey that will gather information on how many cities have their own local incentive policy for EC&EE initiatives in public buildings and evaluate whether these are traditional-type incentive policies or policies new to China. The survey will be carried out towards the end-of-project.

Summary of Policy Pilots

EC&EE Policy Piloted	Description of Pilot	Pilot LGOA Regions	Expected Impacts
Energy monitoring and reporting of public sector buildings	Implementation of the policy on requiring public buildings prepare and submit energy consumption reports to LGOA. The energy consumption reports are evaluated by the LGOA, and feedback provided to each reporting public building on their energy performance and EC&EE activities. The pilot will test the feasibility of enforcing and implementing the policy, and provide idea whether to enforce this policy on a mandatory or voluntary basis. The pilot will also be evaluated as to the responsiveness of the public buildings, and the impacts in terms of level of awareness and interest in implementing EC&EE initiatives in their facilities.	Heilongjiang Province and Hunan Province	Increased participation in the energy monitoring and reporting scheme, and indirectly more energy and energy cost savings in the public buildings sector of the LGOA region.
Energy savings measurement & verification system for public sector buildings	Implementation of the policy on measuring and verifying reported energy savings in submitted energy consumption reports to LGOA. The declared energy savings are evaluated by the LGOA, and the reporting public building is informed in the feedback about the validity or non-validity of declared energy savings from EC&EE activities. The pilot will test the feasibility of enforcing and implementing the policy, for use in the provision of financial incentives for best performing buildings and evaluate the responsiveness of the public buildings on the incentive scheme.	Hainan Province and Guizhou Province	Increased level of awareness and interest of public buildings in implementing EC&EE initiatives, and increased participation in the energy monitoring scheme. Indirectly more energy and energy cost savings in the public buildings sector of the LGOA region.
Building energy auditing in the public building sector	Implementation of the policy on the conduct of energy audits in public buildings to assist them in identifying energy saving opportunities that they can implement to save energy and energy costs. The procedures in the energy audit system that will be developed will be applied in the pilot. The pilot will test the feasibility of enforcing and implementing the policy	Shanxi Province	Increased level of awareness and interest of public buildings in implementing EC&EE initiatives, and increased participation in the energy auditing

	smoothly, and evaluate the reactions and response of the public buildings, and the impacts in terms of level of interest in carrying out energy audits to assist them in planning and implementing EC&EE initiatives in their facilities. This will provide also the idea whether to enforce this policy on a mandatory or voluntary basis.		scheme. Indirectly more energy and energy cost savings in the public buildings sector of the LGOA region.

GEF support is required for the technical assistance in the design and implementation of the policy pilots, and the monitoring and evaluation plan, as well as in the documentation, publication and dissemination of the results and recommendations from the policy pilots. It is also required for the technical services required in the design and conduct of the survey.

Output 1.5: Developed and approved follow-up plan for the replication of piloted EC&EE improvement policies in public buildings in other provinces. This output is an approved follow-up plan for the replication of the piloted EC&EE improvement policies in public buildings that other LGOAs can consider in the localities that they control. The following activities are meant to deliver this output:

1.

Activity	Description
1.5.1: Design of the follow-up plan to promote and implement the replication of the successful applications of the piloted EC&EE and LC policies to other LGOA regions in the country.	This involves the development and adoption of a replication plan for the application of the EC&EE and LC concept to other LGOA regions where the best practices and lessons learned from the pilots can be replicated and promoted. This will require support from central and local government agencies, and potentially private sectors, and other academic institutions for the replication of feasible design and application EC&EE and LC systems design and operation. This includes the formulation of recommendations and feasible strategies and incentives for the providing motivation and promotion of EC&EE and LC policies and regulations for public sector buildings, enforcement of policies and policy support activities, and implementation of financing schemes for related projects that promote EC&EE and LC initiatives in other Chinese cities.
1.5.2: Promotion of the successful applications of the piloted EC&EE and LC policies to other major Chinese provinces and cities.	This involves the selection of the cities/towns where the replications will be carried out. It includes the conduct of promotional campaigns in the selected cities of the application of the piloted EC&EE and LC policies and their beneficial impacts in terms of energy savings (and the corresponding GHG emissions reduction from the reduced electricity demand). A monitoring and evaluation system for the follow-up and replication program will also be designed and adopted by the NGOA before the end of the project.

GEF support is required for the incremental technical and financial assistance in the facilitation and completion of the inputs to the comprehensive study and preparation of the EC&EE and LC policies replication/adoption follow-up plan and proposals to relevant policy making bodies.

Output 1.6: Developed improved and updated public sector building energy standards. This output is an updated code of practice for the application EE features, as well as EE and RE (where applicable and feasible) systems in public sector buildings. The following activities are intended to deliver this output:

Activity	Description
Activity 1.6.1: Assessment of best practices on EE public sector building design and operation.	This involves the development of a methodology for the assessment of the different public sector building designs in China and in other countries (e.g., those with the same climate zones as in China) including systems and equipment that are used will be prepared. This includes procedures for drawing out samples of buildings and their energy systems, the parameters in assessing system performance, the procedures for the conduct of performance evaluation and testing as well as the identification of various testing equipment for the assessment. Upon the approval of the said methodology, the procurement of testing equipment as well as the conduct of actual performance assessment shall follow. Based on the results of the assessment, best practices on EE building and building energy system design and installation shall be identified and compiled according to criteria set for cost, system performance and/or quality. A major component of this activity is the evaluation of existing building energy standards that are applicable in the public buildings in the different climate regions of China. Existing standards from other countries will be used as reference for the development of the building energy performance standards (and labeling) in the public sector. Where possible, international standards will be used as model for the development of the standards that can be adapted as long as these are in accord with the current EE situation and circumstances of the public sector in China. The most appropriate scheme (or combination of schemes) that can be adapted to the current EE situation and circumstances of the public sector in China will be used[11]. The most appropriate scheme (or combination of schemes) that can be adapted to the current EE situation and circumstances of the public sector in China will be used.
Activity 1.6.2: Assessment and setting of improved public sector building EE codes.	This involves the identification and assessment of standards and best practices on the performance, design, construction and installation of energy systems within China and in other countries[12]. It also includes the conduct of a comparative analysis between existing domestic building design and their energy systems and existing building energy performance standards available. The analysis will also include cost factors. Based on the analyses, a set of recommended designs and standards for public sector buildings will be prepared including the corresponding testing and evaluation scheme for the measurement of building energy performance. This will also involve the setting up of the specifications for an energy efficient public building.

[11] Existing building energy standards program such as the US EPA's Energy Star will be used as reference for the design of the building energy performance standards (and labeling) program, or where possible be used as model for a similar appropriate program that is adapted to the current EE situation and circumstances of the public sector in China. The NGOA can refer to other schemes that can be used aside from the USEPA Energy Star to be used as reference model for the planned public sector building energy standards. Other schemes that China can consider include: LEED (US), Green Globes (US), Living Building Challenge (US), Beam (Hong Kong), BREEAM (UK, EU), CASBEE (Japan), Green Mark Scheme (Singapore), Green Star SA (South Africa), Pearl Rating (UAE), etc.

[12] The STAP has recommended consulting the Global Building Performance Network that has developed a Buildings Energy Code portal. http://www.gbpn.org/beet-3/?utm_source=newsletter&utm_medium=email&utm_campaign=Greater%20Energy%20Savings%20from%20Buildings:%20New%20Web%20Portal%20to%20Support%20International%20Collaboration%20on%20Building%20Energy%20Codes

	This major activity will also include the mechanisms for determining and designating the public buildings that are energy efficient (i.e., EE Public Buildings) based on the set EE building specifications. This will be done in coordination with the implementation of Activity 2.4.4.
Activity 1.6.3: Enforcement of public sector building energy codes and standards	This involves the conduct of stakeholder consultations with building practitioners, building service providers, building materials suppliers, and public building owners/administrators on the setting up of the energy performance standards, best practices and the recommended testing procedures. This also includes work towards the registration of said design standards and practices with the relevant government institutions. Once approved, a promotional program for the adoption and widespread use of the prescribed standards and best practices will be developed and implemented. The dissemination of the standards and best practices will be carried out under the promotional programs of the PSBEE Project. The enforcement of the EE Public Buildings standards will also be widely promoted since this is viewed as one very essential incentive will encourage and motivate public buildings to seriously embrace sustainable energy principles and invest on EC&EE and LC initiatives.

GEF support is required for the incremental technical assistance in the assessment and setting up of the recommended public sector building EE codes and also to cover for the costs for the consultations and promotion of the codes.

Component 2: Energy Performance Monitoring and Evaluation System for Public Sector Buildings

This component of the project will enable the comprehensive tracking and control of the energy performance of the buildings in the public sector in China. This consists of activities that will deliver outputs that will contribute to better control and enhanced management of the energy performance of public sector buildings. The following major activities will deliver the expected outputs that will help bring about such expected outcome: (1) Review and verification, including the provision of supplemental baseline energy information in the various major segments of the public sector; (2) Design, establishment and operationalization of a public sector building energy audit system; (3) Conduct of energy audits of public sector buildings in at least four buildings per building type, in two public sector sub-sectors; (4) Design, establishment and operationalization of a public sector buildings energy (supply and consumption) management information system (EMIS); (5) Design, establishment and operationalization of a supplementary system of the EMIS for the measurement & verification of reported energy savings in EMIS energy report submissions; and, (6) Design and operationalization of a Public Buildings EMIS database. The EMIS will cover all levels of government, and that there should also be a hierarchy of targets from national to local with a clear system for their enforcement.

The following are the deliverables that will be delivered through the implementation of "technical assistance" activities:

Output 2.1.1: Reviewed and verified supplemental baseline energy information in the various major types of buildings within the public sector. This output is a comprehensive mapping of the EC&EE initiatives carried out by NGOA-, and LGOA-administered public buildings, inclusive of the historical sub-sectoral energy consumption and energy performance. The following activities are intended to deliver this output:

Activity	Description
2.1.1.1: Conduct of research on active energy management information system (EMIS) for buildings in other countries	This involves the conduct of comprehensive research about the features of existing EMIS for the buildings sector in other countries (e.g., UNDP-GEF pilot BEMS in Croatia). It also includes study visits to countries where such system is implemented (e.g., UNDP-GEF pilot BEMS in Croatia), particularly those that cover all levels of government, with hierarchy of targets from national to local governments, and with well-delineated institutional arrangements for their enforcement.
2.1.1.2: Conduct of a comprehensive review of the current and planned EC&EE activities in various types of public buildings.	This involves a survey of all registered NGOA-, and LGOA-administered public buildings in the country focusing in particular on their installed energy systems and annual energy consumption and budgets. The existing public building registry will be the basis for identifying the various buildings that will be covered in the survey. Information on EC&EE activities of public buildings that have been energy audited will also be used. The procedures and mechanics for the data gathering work will be designed and established. Among the data that will be gathered for each building and included in a database are the specifications of the installed energy systems, annual energy production and/or consumption, EC&EE activities implemented, annual EC&EE investments, and annual energy savings realized. These data will be used as part of the initial inputs to the EMIS (Output 1.3; Activity 1.3.1.4)
2.1.1.3: Organization and conduct of a workshop on EC&EE in public buildings.	This involves the conduct of a consultative workshop with the public building managers/administrators to discuss and understand the current status of the energy utilization performance in the public buildings sector in China, and the various actions that need to be done collectively to address the barriers/issues to improve the average specific energy consumption (SEC) of each building type (in different climate zones of the country), and to inform and get their support in the survey, which will serve as the pre-cursor for the EMIS (Activity 1.3.1.7).

GEF support is required for the incremental financial assistance in the review of ongoing and planned EC&EE initiatives in the public buildings sector to come up with the comprehensive mapping of the EC&EE initiatives in the country.

Output 2.1.2: Established public sector building energy audit system. This is the system by which regular energy audits will be carried out in public buildings in conjunction with the findings and recommendations from the established EMIS and ESVS. The following activities are intended to deliver this output:

Activity	Description
2.1.2.1: Review of building energy audit practices and needs	This involves the evaluation of the current practices, methodologies, levels of knowledge and capacity needs in the field of energy auditing in buildings (public/institutional and commercial). It is aimed at finding ways of improving the current awareness of the public buildings sector, and its general attitude towards, energy auditing. A survey of public buildings that have undergone energy auditing, as well of energy auditing firms, will be conducted to establish the sector's regard to energy auditing and the quality of energy audits done in and for public buildings. Also included is the conduct of a seminar-workshops on energy efficiency and energy

	audit techniques, as part of the efforts to encourage the practice of energy conservation and energy auditing in public buildings. The seminar-workshop will also serve as a forum for the discussion of current issues regarding EC&EE practices and energy auditing issues in public buildings (e.g., techniques, barriers/constraints, and costs), with the aim of coming up with relevant policy recommendations to the NGOA concerning the practice of EC&EE and energy audits in public buildings. Based on the recommendations of the capacity needs assessment, technical personnel from NGOA will be trained in all aspects of energy management and energy auditing, to become qualified building energy auditors.
2.1.2.2: Development of a public sector Building Energy Audit System (BEAS)	This involves the development of a standard energy audit procedures and tools that will have to be followed by energy consulting firms and ESCOs in providing energy audit services to public buildings. This includes (1) specifications for each type of energy audit that can be provided by energy consultants (energy surveys, preliminary audits, detailed audits) particularly as to the objective, scope, duration, frequency, audit team composition, preparations required, audit kit, audit report contents, types of recommendations, and application of recommendations; and, (2) standardized auditing tools (e.g., checklists, calculation sheets and audit forms) for energy auditing purposes. In this regard, this activity will also include the development of an accreditation scheme for energy auditors who have to comply with the established energy auditing standards and guidelines. Training will also be provided to technical personnel of NGOA and the LGOAs on advanced energy auditing techniques, and the provision of energy auditing courses tailor-made for technical people in specialty type public buildings. The BEAS will also supplement the EMIS and the ESVS that will be established under Component 1 of this project.
2.1.2.3: Building Energy Audit System (BEAS) Evaluation	This involves the conduct of an evaluation of the performance and impact of the BEAS through the conduct of a post energy audit survey among the public buildings that were audited. The evaluation will determine, among others, the actual results/impacts to each company of the application of the audit recommendations, the reasons behind the implementation, and non-implementation of audit recommendations, and the building management's comments about, and suggestions to improve, the system. Based on the findings and recommendations of the evaluation, improvements will be employed. It is expected that the energy audit program will be an effective tool for the public buildings sector to improve its energy performance. In that regard, this activity will also include the development of a self-sustainable scheme that will facilitate the continuance of the energy audit program even after the GEF support. This includes further enhancement of the energy audit business of local ESCOs and local engineering consulting firms. Affiliated ESCOs working in the industry sector are also expected to carry out the design and engineering of energy saving schemes/projects that are recommended for implementation in the energy audits.

GEF support is required for the incremental technical assistance in the design and setting up of the appropriate energy audit system for the public buildings sector.

Output 2.1.3: Established public sector buildings energy management information system (EMIS). This is the system by which the data/information on energy supply and demand and energy utilization performance of the various building types, i.e., public buildings sub-sectors, are collated, processed, analyzed, reported and stored. The following activities are intended to deliver this output:

<table>
<tr><th>Activity</th><th>Description</th></tr>
<tr><td>2.1.3.1: Design of the EMIS for public buildings.</td><td>This involves the design and development of the EMIS, including the standard and specific energy data/information that will be reported by the NGOA-, and LGOA-administered public buildings, and the templates/forms that will be used. This will make use of the information that will be available from Output 2.1. The system will be web-based to enable online reporting of the required data/information. The reported energy data/information as well as the results of the analyses of these data/information will be encoded into the public sector energy database that will be designed, installed, operated and maintained by NGOA.</td></tr>
<tr><td>2.1.3.2: Finalization of the EMIS framework and mechanisms based on the approved policies and IRRs.</td><td>After gathering the comments and recommendations from the pertinent stakeholders and the EMIS implementing entity (NGOA), the EMIS framework and mechanisms based on the approved policies and IRRs will be finalized. This will be presented and promoted to NGOA, LGOAs, and relevant government agencies and authorities, as well as the key players in the Chinese building industry through advocacy/lobbying work for its approval and enforcement.</td></tr>
<tr><td>2.1.3.3: Implementation of the EMIS, including provision of technical assistance to public buildings requesting assistance in the reporting process.</td><td>This involves the setting-up, launching and operationalization of the EMIS on a pilot basis covering selected NGOA-, and LGOA-administered public buildings in Beijing and other major cities. The initial number of buildings covered in the pilot starting Year 2 is as follows:
<table>
<tr><th rowspan="2">Location</th><th rowspan="2">Building Type</th><th colspan="4">Number of Buildings Covered</th></tr>
<tr><th>Yr1</th><th>Yr2</th><th>Yr3</th><th>Yr4</th></tr>
<tr><td rowspan="4">National (Beijing)</td><td>Office</td><td>0</td><td>10</td><td>50</td><td>100</td></tr>
<tr><td>Hospital</td><td>0</td><td>5</td><td>15</td><td>25</td></tr>
<tr><td>School</td><td>0</td><td>5</td><td>15</td><td>25</td></tr>
<tr><td>Others</td><td>0</td><td>0</td><td>15</td><td>50</td></tr>
<tr><td rowspan="4">City/Municipality</td><td>Office</td><td>0</td><td>100</td><td>500</td><td>1000</td></tr>
<tr><td>Hospital</td><td>0</td><td>50</td><td>100</td><td>150</td></tr>
<tr><td>School</td><td>0</td><td>50</td><td>100</td><td>150</td></tr>
<tr><td>Others</td><td>0</td><td>0</td><td>50</td><td>100</td></tr>
<tr><td rowspan="4">County</td><td>Office</td><td>0</td><td>50</td><td>100</td><td>300</td></tr>
<tr><td>Hospital</td><td>0</td><td>25</td><td>50</td><td>100</td></tr>
<tr><td>School</td><td>0</td><td>25</td><td>50</td><td>100</td></tr>
<tr><td>Others</td><td>0</td><td>0</td><td>25</td><td>100</td></tr>
</table>
Part of this major activity is assisting buildings that request assistance in complying with the reporting process. The reports will be analyzed by a team in NGOA and feedbacks will be provided to the reporting public buildings on their energy performance and recommendations in improving or sustaining their current performance. The implementation and results of the EMIS will be regularly monitored and comments/recommendations from the management of each reporting public building will be gathered and used for improving the system.</td></tr>
</table>

	Among the major impact that will be investigated is the share of energy efficient public buildings in the Chinese buildings sector. See also Activity 2.1.3.
2.1.3.4: Preparation of annual reports on the status and trends in the energy supply, demand and consumption in the public buildings sector in China.	This involves the preparation of an annual report on the energy performance of the public buildings sector. From the submitted reports, feedbacks on the reports, etc., a summary report will be prepared to present the overall and sub-sectoral energy performance in the public buildings sector. Among the major information in the annual reports are the following: (a) Annual energy consumption (by building type); (b) Annual average SEC value (by building type); (c) Annual % share of "EE Public Buildings"; (d) Annual EC&EE investments in the public buildings sector (million RMB); and, (e) Annual energy savings in the public buildings sector. Apart from the management of each reporting public buildings, the report will be available to interested entities in the Chinese buildings sector and building industry. This major activity will also include the designation of the public buildings that are energy efficient, i.e., EE Public Buildings based on the specifications that are set in Activity 1.6.2.
2.1.3.5: Evaluation of the results and impacts of the EMIS	This involves the conduct of an evaluation of the performance and impacts of the implemented EMIS. It also include the organization and conduct of a workshop to discuss the findings of the evaluation, and to discuss and agree on recommendations for improving and updating the EMIS, as well as the policies and IRRs that support its implementation.

GEF support is required for the incremental technical and financial assistance in the design, development, establishment and initial operation of the EMIS, including in the purchase and set up the computer hardware/software for the operation of the system, and in the formulation of support policies and regulations.

Output 2.1.4: Established energy savings measurement & verification system (ESMVS) in public buildings sector. This is system for the evaluation of the energy performance of public buildings based on energy reports submitted; and, verification of reported energy savings derived from implemented EC&EE/LC initiatives. The following activities are intended to deliver this output:

Activity	Description
2.1.4.1: Design of the ESMVS for public buildings.	This involves the design of the supplementary system for the EMIS that will be for the evaluation of the energy performance of public buildings based on energy reports submitted; and, verification of reported energy savings derived from implemented EC&EE/LC initiatives. The design includes the specific procedures and criteria to: (1) analyze and report the energy performance of; and, (2) measure & verify reported energy savings by, a public building covered by the EMIS. The ESMVS will be applied in the implementation of the financial incentive scheme that will be developed and implemented in Component 3. The ESMVS results and analyses will be encoded into the EMIS database that will be operated and maintained by NGOA.
2.1.4.2: Formulation and finalization of the ESMVS mechanisms and	This entails the formulation of the implementing rules and regulations (IRRs) of the approved policies on the ESVS. It also include the finalization of the ESVS mechanisms based on the approved policies and IRR. This will

supporting implementing rules and regulations (IRRs) based on the approved policies.	be presented and promoted to NGOA, LGOAs, and relevant government agencies and authorities, as well as the key players in the Chinese building industry through advocacy/lobbying work for its approval and enforcement.
2.1.4.3: Implementation of the ESMVS, including evaluation of its results and impacts.	This will involve the operationalization of the ESMVS as a supplementary system to the EMIS. This will be used mainly to assess and verify the energy savings information provided in the energy reports. Along with the EMIS, the implementation and results of the ESMVS will be regularly monitored and comments and suggestions from the management of each reporting public building will be gathered and used for improving the system. Also included is the conduct of the performance evaluation of the system along with that of the EMIS, with the intention of improving it, as well as the policies and IRRs that support its implementation.
2.1.4.4: Publication and dissemination of results of the ESMVS	This involves the preparation of annual reports on the overall effectiveness of the ESMVS for the public buildings sector based on the results of the analyses of energy consumption information from the reports submitted to the EMIS, as well as in the implementation of the financial incentive scheme that is developed and implemented in Component 3. Also included is the formulation of recommendations (if any) for the improvement of the system and its implementation.

GEF support is required for the incremental technical assistance in the design and development of the ESMVS, as well as in the formulation of the relevant support policies.

The following are the deliverables that will be delivered through the implementation of "investment" activities:

Output 2.2.1: Completed energy audits of public sector buildings. These are the energy audits that will be conducted to determine specific EC&EE/LC measures that can be implemented in representative public building types, as well as determine the realistically achievable energy savings from implemented EC&EE/LC measures. The following activities are intended to deliver this output:

Activity	Description
2.2.1.1: Development and implementation of Public Building Energy Audit Program	This involves the development of the energy audit program based on the BEAS. The program will involve annual scheduling of energy audits of public buildings, the prioritization could be based on their energy utilization performances and extent of energy saving potentials (as per the EMIS). The same criteria will be used in deciding the type of energy audit that will be carried out in each public building. The programming and planning for individual energy audits shall be in consultation with the management and technical personnel of each public building. All aspects of the energy audit program from the conduct of the audits, to the evaluation of audit findings, drawing of conclusions and formulating of recommendations, up to the preparation of energy audit reports, will be carried out. Since sufficient data are already available in the EMIS (energy supply and consumption profiles of the buildings), energy surveys will no longer be necessary and only preliminary and detailed (investment-grade) energy audits will be conducted.
2.2.1.2: Conduct of	This involves the implementation of the scheduled energy audits in

<table>
<tr><td>scheduled energy audits</td><td>selected public buildings, according to the annual energy audit program. Depending on the agreed energy audit program, the coverage of the energy audits will be: government (national, municipal, county) office buildings, hospital buildings and school buildings. The number and types of energy audits that will be conducted per years are as follows:

<table>
<tr><td colspan="3">Preliminary Energy Audits</td></tr>
<tr><td>Location</td><td>Building Type</td><td>Min. Number (per year)</td></tr>
<tr><td rowspan="3">National (Beijing)</td><td>Office</td><td>6</td></tr>
<tr><td>Hospital</td><td>2</td></tr>
<tr><td>School</td><td>2</td></tr>
<tr><td rowspan="3">City/Municipality</td><td>Office</td><td>10</td></tr>
<tr><td>Hospital</td><td>4</td></tr>
<tr><td>School</td><td>4</td></tr>
<tr><td rowspan="3">County</td><td>Office</td><td>4</td></tr>
<tr><td>Hospital</td><td>1</td></tr>
<tr><td>School</td><td>1</td></tr>
</table>

Each PEA will be completed within 1 to 2 months. PEA reports will be prepared and submitted to NGOA.
For the detailed energy audits (DEAs), the public building type that will be selected will be based on the results of the PEA. The main criteria for the building selection for the conduct of DEA is the magnitude of the feasible energy saving potentials as determined in the PEA. The number of DEAs to be conducted each year is as follows:

<table>
<tr><td colspan="3">Detailed Energy Audits</td></tr>
<tr><td>Location</td><td>Building Type</td><td>Min. Number (per year)</td></tr>
<tr><td>National (Beijing)</td><td>TBD</td><td>2</td></tr>
<tr><td>City/Municipality</td><td>TBD</td><td>3</td></tr>
<tr><td>County</td><td>TBD</td><td>1</td></tr>
</table>

Each DEA is expected to be completed within 3 months. DEA reports will be prepared and submitted to NGOA.

Data/information on the results and recommendations of the energy audits will be included in the EMIS database.</td></tr>
<tr><td>2.2.1.3: Publication and dissemination of results of the energy audit program</td><td>This involves the preparation of annual reports on the results of the various energy audits that are implemented each year. Each energy audit report will be summarized to highlights the identified and recommended energy saving opportunities (ESOs), and the implemented energy saving measures (e.g., investments made, EC&EE and RE technologies applied, actual energy savings, and economics). These reports will be published and disseminated to NGOA-, and LGOA-administered public buildings.</td></tr>
<tr><td>2.2.1.4: Enhancement of the Public Building Energy Audit Program</td><td>This involves the conduct of evaluation of the energy audit program in terms of the magnitude of energy and energy cost saving potentials (and associated GHG emission reductions) and the actual energy and energy cost savings realized from implemented energy saving opportunities recommended (in PEAs) and engineered (in DEAs). This is supplemented by information from post energy audit surveys among the public buildings that were audited. Also included in this activity is the development and</td></tr>
</table>

	implementation of the energy audit (PEA and DEA) program for the other public building types (e.g., civic centers, recreational centers, transport terminal buildings, and government-owned residential and commercial buildings) similar to the process done for the 3 priority public building types. These energy audits will be carried out either towards the end of Year 2 or start of Year 3. Similar to the earlier public building types, the aim is to facilitate the practice of energy auditing in these types of buildings.

GEF support is required for the incremental technical and financial assistance in the planning and conduct of programmed energy audits in selected public buildings (various types in different climate zones of the country) including in the purchase of sets of energy audit instruments that will be used and as part of the capacity development interventions in the field of energy auditing.

Output 2.2.2: Established Public Buildings EMIS database. The PBEMIS database shall serve as the repository of all data/information about the energy supply and demand, consumption and energy utilization performance of the public buildings sector in China. The following activities are intended to deliver this output:

Activity	Description
2.2.2.1: Conduct of research/study on the requirements and procedures for processing, verification, and encoding, as well as data updating.	This involves the identification and design of the most feasible, reliable and cost-effective means of data processing, verification, and encoding, and maintenance. This will also take into consideration similar procedures implemented in the EMISs from other countries (Output 2.1). The procedures for data sourcing, collection and categorization are defined in the EMIS design.
2.2.2.2: Design and development of the public buildings EMIS database.	This is the public buildings sector energy management information system database, which include the profiles of all NGOA-, and LGOA-administered public buildings. The database shall be housed in the NGOA, which will be responsible for its operation and upkeep. The database modules will be based on the parameters that are defined in the EMIS design. Apart from the information from the energy consumption reports of public buildings, results of analyses of energy data, energy audit results, and the results of energy saving measurements and verifications of selected public buildings, also included in the database are information of up to date building EC&EE and LC technologies development and applications in China and in other countries.
2.2.2.3: Capacity development in the use of the PBEMIS Database.	This involves the design, preparation and conduct of the following events for the NGOA and LGOAs: (a) Workshop to review the operation, performance of the PBEMIS database; (b) Conference on the operationalization and maintenance of the PBEMIS database, and how public buildings can make use of publicly available information in the database; and, (c) Training on the utilization and maintenance of the

	PBEMIS database (including reporting procedures to EMIS).

GEF support is required for the incremental technical and financial assistance in the design, development, establishment and initial operation of the PBEMIS Database, including in the purchase and set up the computer hardware/software for the operation of the system.

Component 3: EC&EE Improvement Promotion and Demo Programs for Public Sector Buildings

This major component of the project is meant to address the low level of knowledge and experience in the application and benefits of EC&EE (as well as low carbon) technologies, as well as the financing of these initiatives in public sector buildings. There are two sets of activities within this component. The first set addresses the inadequate resources for implementing EC&EE interventions, and the other set will focus on facilitating the implementation of these technologies. The first set of activities are intended to deliver outputs that collectively contribute to the increased availability of resources (technical capacity, information and financing) for EC&EE initiatives in public sector buildings and facilities. The proposed activities include: (1) Establishment and operationalization of a system for providing information about the features, technical specifications and costs of new EC&EE low carbon technologies (including products) for the public sector; (2) Publication of a directory of recommended applicable and cost-effective new EC&EE and low carbon (LC) technologies (systems and products) for public sector building administrators/managers; (3) Conduct of a comprehensive assessment of market-based financing scheme options and design of appropriate market-based financing of new EC&EE/LC technology (system and product) applications; and, (4) Identification and selection of 5 EC&EE/LC projects that will be financed through market-based financing scheme in public buildings in selected public sub-sectors in three to five cities or regions; (5) Conduct of feasibility analyses and design of EC&EE/LC technology application demonstrations[13].

The second set of activities are meant to deliver outputs that will collectively contribute to the realization of increased application of EC&EE technologies in public sector buildings and facilities. These activities include: (1) Establishment and operationalization of a scheme for market-based financing of new EC&EE/LC technology (system and product) applications[14]; (2) Implementation of five EC&EE/LC projects financed through market-based financing scheme in public buildings in selected public sub-sectors in two to five cities or regions; (3) Implementation of selected EC&EE (and RE) technology application demonstrations in public sector buildings (for two sub-sectors); (4) Publication and dissemination of reports on the impacts of the EC&EE/LC project financing and demonstration program; and, (5) Development (for approval and enforcement) of relevant sustainable follow-up plans for the replication of the demonstrated applicable and feasible EC&EE technologies in the public sector buildings in other provinces.

[13] For delivering Outputs 3.1.3 to 3.1.5, the engagement of the services of local ESCOs will be considered. This will facilitate the involvement of the private sector in the energy efficiency improvement efforts in public sector buildings, and effectively mobilize market-based finance for supporting public sector EC&EE initiatives through, among others, the local ESCO industry. As in previous GEF-funded energy (EE and RE) and climate change mitigation projects in China, local engineering and consulting firms and ESCOs, will be utilized in the implementation of the planned activities of the project, which include interventions that will enable to design and development of the various proposed EE and energy management systems (which are also those suggested by the reviewer), as well as in the implementation of such systems.

[14] The NGOA recognizes the importance of the involvement of the private sector, particularly banks/financial institutions, as well as entities that are willing to finance EC&EE (and low carbon) technology applications through ESCO and PPP modalities. Hence, private sector involvement is part of the strategy for this project. Realizing the fact that finances are limited in the public sector, the project proponents have already become aware of the importance of private sector investment as well as other private sector-led market-based financing for supporting public sector EC&EE initiatives. Preliminary discussions with commercial banks and ESCOs indicate that these 2 stakeholder groups are interested in working on EC&EE projects in public buildings if there will be adequate enabling conditions that will motivate and make them confident in venturing into investing in such projects. The creation of such enabling conditions by addressing barriers and making good use of the opportunities for private sector involvement is covered in Components 1 and 4 of this project.

Component 3 activities are intended are to: (a) increase the availability of resources (technical capacity, information and financing) for EC&EE initiatives in public sector buildings and facilities; and, (b) Increased application of EC&EE technologies in public sector buildings and facilities.

Component 3.1: Facilitation of EC&EE/LC Technologies Application Demonstrations – The expected outcome from this sub-component is the increased availability of resources (technical capacity, information and financing) for EC&EE initiatives in public sector buildings and facilities.

Output 3.1.1: Established scheme for providing information about the features, technical specifications and costs of new EC&EE low carbon technologies (including products) for the public sector. This is the system by which the sourcing of EC&EE/LC technology-related information for the PBEMIS Database will be done. The following activities are intended to deliver this output:

Activity	Description
3.1.1.1: Conduct of analyses of data/information in the PBEMIS Database.	This involves the evaluation of the current stock of supplementary data/information in the database on various important issues, trends and developments (international and domestic) on EC&EE and LC applications in buildings. The study will also evaluate how the database can accommodate other important information that the public sector needs for their EC&EE and LC initiatives, such as forecast demand and application trends for EC&EE and LC technologies; technology costs, socio-economic and environmental impacts of technology applications; future potential technological and application advances, etc.
3.1.1.2: Conduct of a workshop on the sharing of EC&EE and LC technologies as applied to public buildings.	This involves the organization and conduct of a workshop on information sharing on public buildings EC&EE and LC technology development and application. This is to get the cooperation and support in the sourcing of new information (e.g., EC&EE technology database subscriptions and links) on the creation of a scheme for the uptake of new information to the PBEMIS Database for use by the public buildings sector.
3.1.1.3: Design and establishment of a scheme for the sourcing of EC&EE/LC technology-related information for the PBEMIS Database	This involves the setting up of arrangements regarding the data/information sourcing, including requirements and terms and conditions, as discussed and agreed during the workshop. If necessary further discussion meetings (online) will be carried out to reach agreement.

GEF support is required for the incremental technical and financial assistance in the design and setting-up of the scheme, and initial operation of the scheme for the information sourcing for the PBEMIS Database, including the subscription to EC&EE/LC technology databases.

Output 3.1.2: Published directory of recommended applicable and cost-effective new EC&EE/LC technologies (systems and products) for public sector building administrators/managers. This is a convenient information resource on the technology suppliers, including technical specifications and prices of the required hardware and software involved. The following activities are intended to deliver this output:

Activity	Description
3.1.2.1: Conduct of	This involves the conduct of research and techno-economic feasibility

analyses of EC&EE and LC technologies that are feasible and applicable in public buildings in China.	analyses of various EC&EE and LC technologies applied in buildings. These include locally available as well as those developed and are commercially available in other countries. The technologies that are found to be applicable, feasible and replicable in public buildings in the different climate zones in China will be identified. The evaluation also involves the determination of the suppliers and installer of the systems that comprise each technology.
3.1.2.2: Development of a compendium/directory of EC&EE and LC technology suppliers	Based on the results from Activity 3.1.2.1, a directory of EC&EE and LC technology suppliers (local and abroad) will be developed. The directory will include the profiles of the suppliers including their contact details, as well as the technical specifications and prices of the required hardware and software for each technology application. A vetting system will be carried out to ensure that the identified suppliers are reliable. The NGOA will be responsible for sharing the information in the directory to the public.
3.1.2.3: Conduct of a commercial exhibition and conference on EC&EE and LC technologies included in the directory.	This involves the organization and conduct of an international conference and exhibition of EC&EE and LC technologies as applied in buildings. It will be participated in by the public buildings administrators/managers, building practitioners and the technology developers and suppliers. The technology developers and suppliers will showcase their technology hardware/software, as well as conduct presentations about their technologies as applied in buildings for saving energy, energy cost and reducing GHG emissions.
3.1.2.4: Review and updating of the EC&EE and LC technologies directory.	This involves the conduct of a survey among public building managers/administrators and building practitioners about the performance (e.g., level of expertise, service provision, after sales services, pricing, etc.) of the technology suppliers in the directory. The survey will also enquire about the operational performance of the technology (including components) supplied. A separate research and survey (for technology developers/suppliers) will also be conducted to determine new applicable EC&EE and LC technologies that can be included in the directory. The results of these surveys and research will be the bases for the updating of the directory.

GEF support is required for the incremental technical and financial assistance in the review and vetting of technology suppliers, and in the publication of the directory.

Output 3.1.3: Completed assessment of market-based financing scheme options. These will be the potential market-based financing mechanisms that would be recommended for the financing of EC&EE/LC technology application projects in the public buildings sector. The following activities are intended to deliver this output:

Activity	Description
3.1.3.1: Review of existing operational financing schemes for EC&EE/LC projects in China.	This involves a comprehensive review of the financing schemes that GOC, local governments and the private sector (e.g., commercial banks, ESCOs) on the implementation of EC&EE/LC projects, purchase and installation of EC&EE appliances or equipment. The review will assess the feasibility of applying the same strategies and mechanics in coming up with financial assistance schemes for supporting the EC&EE/LC technology application initiatives in the public buildings sector. It will

	also identify appropriate policies that will support the financing of such initiatives in public buildings in China.
3.1.3.2: Review of potential market-based financing schemes for EC&EE/LC projects in the public buildings sector	This involves research and assessment of potential financing schemes that can be considered for funding EC&EE/LC projects in public buildings. These could be the same as the existing ones in the country, or new innovative ones based on current market trends and financing strategies. The potential schemes that can be considered include: (1) Energy savings trading scheme[15], which will be in partnership with one of the existing carbon exchanges to set up the scheme for public buildings (and potentially private commercial buildings; (2) ESCO support scheme, wherein funds to be managed by a financial institution will be available for ESCOs to borrow for their use in financing EE/RE projects in public buildings[16]; (3) Usual commercial banking schemes but specifically designed for public buildings (involving subsidies from the NGOA); and, (4) Outsourcing of services to private sector, which is a form of public-private partnership, which can include private sector operating power generation facilities that are currently owned and operated by some public sector buildings. Also for consideration is the awarding of financial incentives to public buildings that have performed very well in their EC&EE/LC efforts as reflected in their energy reports to the EMIS, and validated by the ESMVS. The retention of energy cost savings in the coffers of public buildings (for use in developing new EC&EE/LC projects) can be considered as financial incentive that have performed very well.
3.1.3.3: Promotion of potential schemes for EC&EE/LC project financing in public buildings.	This involves the organization and conduct of a seminar-workshop for NGOA, LGOAs, public building managers and administrators, as to design of a scheme that will award financial incentives on the financing of EC&EE/LC efforts projects in public buildings. The recommended financial and incentive schemes will be presented and comments and suggestions from the workshop participants will be solicited. Based on the findings and recommendations from the seminar-workshop, a report on the potential options for financing of EC&EE/LC projects (pros/cons, requirements, etc.) will be prepared, published and disseminated.

GEF support is required for the incremental technical and financial assistance in the evaluation of the financing options and in the promotion of these options to the public buildings sector and financial sector in China.

Output 3.1.4: Designed market-based financing of new EC&EE/LC technology (system and product) applications. This is the design of the selected most appropriate and feasible market-based financing scheme for the financing of EC&EE/LC projects for the public buildings sector in China. The following activities are intended to deliver this output:

Activity	Description
3.1.4.1: Selection of market-based financing	This involves decision making amongst the NGOA (and LGOAs) and partner financial institutions (government and private sector) on the

[15] This is one of the potential application of the ESMVS.

[16] The typical ESCO arrangements with the client building will apply (performance based), but the lending scheme with the financial institution can be through commercial lending or also performance-based.

scheme options	market-based financing scheme(s) that will be considered, promoted and implemented. The final decision will be endorsed by the stakeholders and the Project Steering Committee (PSC).
3.1.4.2: Design of selected market-based financing scheme(s) for EC&EE/LC projects in the public buildings sector	This involves the design of all aspects of the selected market-based financing scheme(s) that will be supported by each partner financial institution. These includes all aspects of the financing scheme implementation (e.g., capitalization by financing institution, financing terms and conditions, financing eligibility criteria, etc.). The design will also include the preparation of business plan for the scheme implementation.
3.1.4.3: Design of an EC&EE/LC incentive scheme for public buildings.	This involves the design of a scheme that will award financial incentives to public buildings that have performed very well in their EC&EE/LC efforts as reflected in their energy reports to the EMIS, and validated by the ESMVS. This can be implemented either by the government (national/local) or interested banks/financial institutions. This includes the preparation of the business plan for the scheme implementation.

GEF support is required for the incremental technical assistance in the design of the most appropriate market-based financing scheme and the EC&EE/LC incentive scheme for public buildings.

Output 3.1.5: Selected EC&EE/LC projects (total of 20) in public buildings in selected public sub-sectors. This is the agreed line-up of demo public building EC&EE/LC technology application projects that will be showcased in the PSBEE Project. The following activities are intended to deliver this output:

Activity	Description
3.1.5.1: Review of the potential demo/pilot projects.	This involves the evaluation of the information from the PBEMIS database about potential EC&EE/LC projects stated by public buildings in their energy reports to EMIS. A buildings energy technology information exchange (BETIX) system (see Output 4.3) will also be utilized to find out other potential EC&EE/LC technology application projects that can be considered.
3.1.5.2 Pre-feasibility assessments of the potential projects.	This involves the carrying out of preliminary evaluation of the financial and technical viability of each project, and the magnitude of potential energy savings and GHG emission reductions that can be realized. The assessments will also determine potential public buildings that will be interested in implementing such projects, and also determine the levels of commitment of the potential owners/implementers to be the demo/pilot host. A set of criteria will be developed for use in the final selection of the demos/pilots and the demo/pilot host entities. The pre-feasibility analyses that were conducted during the project preparation phase will be reviewed to determine and verify project implementation requirements. Further feasibility assessments will be carried out by the demonstration hosts.
3.1.5.3: Conduct of preliminary discussions for financing of demo projects.	This entails preliminary discussions and agreements between the public buildings that will host the demo projects with the relevant entity or entities in-charge of the implementation of the financing schemes and the EC&EE/LC incentive schemes on the financing of the demo projects.
3.1.5.4: Finalization of the line-up of confirmed	This involves the application of the set criteria in Activity 3.1.5.2 in the identification of the final line-up of demo projects and hosts. Based on

EC&EE demos/pilots.	the stakeholder discussions during the PPG phase, it was agreed that overall, there will be 20 demonstrations. For the EC&EE/LC technology application demonstrations in public buildings, 10 demos will be implemented - Provincial Government buildings: 3 office buildings, 3 university buildings, 1 hospital building; and County Government buildings: 1 university building and 2 hospital buildings. For the EC&EE/LC demonstration projects financed through market-based financing scheme in public buildings in selected public buildings, there will be 10 demos – National Government building: 1 office building; Provincial Government buildings: 2 office buildings, 2 university buildings, and 1 hospital building; Municipal Government buildings: 1 office building and 1 hospital building; Country Government buildings: 1 office building, and 1 hospital building.

GEF support is required for the incremental technical assistance in the pre-feasibility analyses of the potential demonstrations leading up to the final line up of showcase projects.

Output 3.1.6: Completed feasibility analyses and design of 20 EC&EE and LC technology application demonstrations. This is the set of completed comprehensive feasibility analyses of the selected EC&EE/LC technology application demonstrations. The following activities are intended to deliver this output:

Activity	Description
3.1.6.1: Review of pre-feasibility analyses of identified demo/pilot projects.	This involves the review of the pre-feasibility analyses that were conducted to deliver Output 3.1.5. The review will also involve the evaluation of data/information that are lacking or are in need of updating, as well as verify the project implementation requirements.
3.1.6.2: Conduct of comprehensive feasibility analyses.	This involves the conduct of more detailed analyses of the feasibility of each demo project. The detailed feasibility assessments will be carried out by the demonstration hosts but technical assistance can be provided to them in the feasibility analyses if requested. More data/information will be gathered, also as part of the due diligence that financing institutions and the government entity/entities managing the financing and incentive schemes will require.
3.1.6.3: Design of the EC&EE/LC demonstrations.	This will involve carrying forward the existing feasibility studies for the identified demo/pilot projects to detailed technical design and engineering, budget cost calculation, design of ownership and management models, cost-benefit analysis, design of operation and maintenance concept, and assessment of financing aspects. It will also include the establishment of baseline data for each demo project such as the baseline lighting energy consumption of the relevant application. The operating performance targets for each demo project will also be established. This activity can also be carried out in conjunction with the review/conduct of the feasibility analyses. The line-up of EC&EE/LC demonstrations is presented in Annex K.
3.1.6.4: Promotion of the demonstrations.	This involves the preparation and conduct of a seminar-workshop to promote the demonstration. This activity will be for purposes of informing the stakeholders in the public buildings sector, the financial sector, and building industry in China what these are all about, the

	objectives, and the expected results and impacts.
3.1.6.5: Finalization of the design of demo/pilot projects.	The demonstration hosts will be assisted (if requested) in the comprehensive engineering design of the demo projects. Assistance will also be provided in the processing of financing applications of the demo hosts for the financing of the operation and maintenance of their respective demo project. This is for the host companies that are availing of the financing from banks/financial institutions for their EC&EE/LC projects that will serve as demos for this UNDP-GEF project.

GEF support is required for the incremental technical and financial assistance in the comprehensive feasibility analyses of the demonstrations, as well as in the engineering design of these showcase projects, and in the promotion of these demonstrations.

Component 3.2: Implementation of EC&EE/LC Technologies Application Demonstrations – The expected outcome from this sub-component is increased application of EC&EE technologies in public sector buildings and facilities.

Output 3.2.1: Established scheme for market-based financing of new EC&EE/LC technology (system and product) applications. This is scheme that will be promoted to the public buildings sector for the financing of their EC&EE/LC technology application projects. The following activities are intended to deliver this output:

Activity	Description
3.2.1.1: Establishment and operationalization of the selected financing scheme(s).	This involves the setting-up, capitalization, launching and operation by the government (national and local) and partner banks/financial institutions. This also includes the EC&EE/LC incentive scheme for public buildings. All requirements for the setting up of financing mechanisms in China will be complied with to facilitate smooth implementation. The operation of these schemes will be regularly monitored and comments/recommendations from the beneficiaries will be gathered and used for improving (when necessary) them.
3.2.1.2: Conduct of promotional campaigns for the financing/incentive schemes.	This entails the confirmed partnership between the NGOA (and LGOAs) with relevant government entities and partner banks/financial institutions). This activity will be for the promotion to the public building management of the financing scheme(s), and EC&EE/LC incentive scheme, and will be facilitated by financial institutions that will be implementing the schemes.
3.2.1.3: Capacity development for public buildings.	This involves the design, planning and conduct of capacity development activities for public buildings that lack technical capacity in the development and implementation of building EC&EE initiatives. The capacity development will be on: (1) Designing, planning and implementing EC&EE/LC application projects that are supported by financing schemes; and, (2) Financial feasibility analysis of EC&EE/LC projects.

GEF support is required for the incremental technical and financial assistance in the establishment of the market-based financing scheme, including the implementation of the promotional campaigns and capacity development for the uptake of this scheme.

Output 3.2.2: Implemented 10 EC&EE/LC projects financed through market-based financing scheme in public buildings in selected public sub-sectors. These are the 10 EC&EE/LC technology application demonstrations that will be partly/fully funded by through the established market-based financing scheme. These demo projects are presented in Annex K. The following activities are intended to deliver this output:

Activity	Description
3.2.2.1: Implementation of the demo EC&EE/LC projects that are financed through market-based financial schemes.	This involves the actual engineering, installation, operation and maintenance of the systems involved in the demo EC&EE/LC projects that are financed through market-based financing mechanisms, or where possible, by the project owners/proponents themselves. Generally, the implementation of these demo projects will be carried out by the project owner/proponent with the support of the PSBEE project team to ensure that all requirements for the successful implementation of the demo are in place. Each demo project will be regularly monitored by the owner/proponent and the PSBEE project personnel using a common M&E system that will be designed and employed for this purpose.
3.2.2.2: Preparation of the demo project profiles (as case studies).	This involves the documentation of the results of the implemented demo EC&EE/LC technology application projects that are financed through market-based financing schemes. An inventory of the demo project results (e.g., energy use, operational performance, economic performance) will be prepared, which will be compared to the results of similar EC&EE/LC technology application projects that were implemented in other countries. Each project report will be summarized into project profiles (or case studies) following an agreed presentation format. These project profiles will be inputted into the PBEMIS database and BETIX. The project profiles will be stored in a specific module of the PBEMIS Database.
3.2.2.3: Conduct of an overall performance evaluation of the demo projects.	This involve a comprehensive analysis of the results from each demo project. The analysis will be on the energy performance, as well as the economic feasibility performance of each demo project, which is funded through market-based financing schemes. The analysis will also focus on the process of financing such projects, particularly when financed through market-based financing mechanisms. The effectiveness of such type of financing will be evaluated. The results of the analysis and pertinent conclusions and recommendations will be publicized, including the demo project results and recommendations. Technical guidance materials (web-based and printed media) on specific EC&EE/LC technology applications in public buildings (apart from those that will be showcased) will also be prepared and disseminated. The results of the demonstrations will all be inputted in the PBEMIS Database and BETIX.

GEF support is required for the incremental technical and financial assistance in the monitoring and evaluation of the operational performance of the implemented commercial finance-supported demo projects, and in the preparation of the demo project profiles.

Output 3.2.3: Implemented 10 EC&EE/LC technology application demonstrations in public buildings. These are the other EC&EE/LC technology application demo projects that will be partly

financially supported by the PSBEE and the NGOA/LGOA. These demo projects are presented in Annex K. The following activities are intended to deliver this output:

Activity	Description
3.2.3.1: Implementation of the EC&EE/LC demonstrations.	This involves the actual engineering, installation, operation and maintenance of the systems involved in the EC&EE/LC demonstration. The implementation of the demo project will be carried out by the demo host with the support of the PSBEE project team to ensure that all requirements for the successful implementation of the demo are in place. Each demo project will be regularly monitored by the host and the PSBEE project personnel using a common M&E system that will be designed and employed for this purpose.
3.2.3.2: Preparation of the demo project profiles (as case studies).	This involves the documentation of the results of the implemented EC&EE/LC technology application demonstrations. An inventory of the demo project results (e.g., energy use, operational performance, economic performance) will be prepared. The results will be compared to the results of similar EC&EE/LC technology application projects that were implemented in other countries. Each project report will be summarized into project profiles (or case studies) following an agreed presentation format. These project profiles will be inputted into the PBEMIS database and BETIX. The project profiles will be stored in a specific module of the PBEMIS Database.
3.2.3.3: Conduct of an overall performance evaluation of the demo/pilots projects.	This involve a comprehensive analysis of the results from each demonstration. The analysis will be on the energy performance, as well as the economic feasibility performance of each demo. The results of the analysis and pertinent conclusions and recommendations will be publicized, including the demo project results and recommendations. Technical guidance materials (web-based and printed media) on specific EC&EE/LC technology applications in public buildings (apart from those that will be showcased) will also be prepared and disseminated. The results of the demonstrations will all be inputted in the PBEMIS Database and BETIX.

GEF support is required for the incremental technical and financial assistance in the monitoring and evaluation of the operational performance of the demo projects, and in the preparation of the demo project profiles.

Output 3.2.4: Published reports on the impacts of the EC&EE/LC project financing and demonstration program. These provide the useful feedback information to the NGOA and the UNDP-GEF about the impacts of the promoted/supported financing schemes (1 report) and the demonstrations (1 report). The following activities are intended to deliver this output:

Activity	Description
3.2.4.1: Organization and conduct of a seminar-workshop on the results of the demonstrations.	This involves the demo hosts presenting their respective project highlighting the specific aspects of the EC&EE/LC technology applied in their buildings, the scheme, the investment involved, results, energy savings achieved, actual project economics, estimated GHG emission reductions, and their respective recommendations for the EC&EE/LC technology demonstration under the PSBEE Project. The workshop will

	assess the overall performance of the demonstration program and will come up with recommendations to the PSC concerning the relevance of such kind of programs to support the EC&EE/LC efforts of the public buildings sector in China.
3.2.4.2: Documentation, publication and dissemination of the seminar-workshop proceedings.	This involves the dissemination of information about the results of the EC&EE/LC technology demonstrations to the public. It is for the purpose of influencing the NGOA and LGOAs, and the management of the public buildings that they administer, and buildings industry in China about the potentials and benefits of the application of EC&EE/LC technologies in public buildings. This will also provide important opportunities for networking, for high level discussions, exposure to regional and possibly overseas expertise etc. The publication will also include information about the status of the PSBEE project. It will also include the detailed information on the results and evaluation of the impacts of the implemented demos. Among the impacts that will be tracked is the increased uptake of EC&EE/LC technologies in the public buildings sector in China.

GEF support is required for the incremental technical and financial assistance in the monitoring and evaluation of the operational performance of the 5 demo projects, and in the preparation of the demo project profiles.

Output 3.2.5: Developed sustainable follow-up plans for the replication of the demonstrated applicable and feasible EC&EE/LC technologies in the public sector buildings in 5 other provinces.
These outlines how the positive impacts of the financing and demonstration interventions will be scaled-up or replicated, and how the program will be adjusted accordingly to address any negative result and impact. The following activities are intended to deliver this output:

Activity	Description
3.2.5.1: Evaluation of additional capacity development needs on EC&EE/LC technologies.	This involves the assessment of other capacity development requirements of the public buildings sector on the application of EC&EE/LC technologies taking into account the results and recommendations of the demos that were conducted.
3.2.5.2.: Design of sustainable follow-up plans.	This involves the action planning for the continuous application of proven EC&EE/LC technologies in public buildings in 5 other LGOA regions/provinces in China. Based on the results of Activity 3.2.5.1, a sustainable follow-up program will be designed and developed for financially supporting the application of EC&EE/LC technologies in public buildings in the other Chinese provinces.
3.2.5.3: Promotion of the sustainable follow-up program.	This involves the organization and conduct of a seminar-workshop to: (a) Present and solicit support from the government (NGOA and LGOAs), financial institutions and the buildings industry in the approval and implementation of the developed sustainable follow-up program; and, (b) Introduce new innovative ways of planning, financing and implementing EC&EE/LC technology applications in public buildings.

GEF support is required for the incremental technical and financial assistance in the evaluation of the capacity development needs and the development of the sustainable follow-up plans for demo project replications or scale up.

Component 4: Public Sector EC&EE Capacity and Awareness Enhancement Program

The achievement of enhanced awareness and knowledge of public sector authorities and personnel and the citizenry on the cost-effective application of EC&EE technologies is the expected outcome from the outputs that will be delivered under this project component. This project component is intended to remove the low level of technical capacity and awareness within the public sector in the application and practice of EC&EE and low carbon technologies/techniques. The Intended activities for delivering the expected outputs are the following: (1) Organization and conduct of project promotional campaigns and workshops in target segments of the public sector; (2) Design and conduct of trainings for NGOA and LGOA authorities and technical staff on the implementation of the various EC&EE programs: *(a) energy audit; (b) EMIS; (c) ESMS; and, (d) EC&EE/LC demonstration; (e) energy management and technology*; (3) Establishment and operationalization of an information network for the promotion and dissemination of knowledge on public sector EC&EE/LC technology applications; (4) Design, establishment and operationalization of public sector EC&EE/LC management and education training centers in the different climate regions; and, (5) Design, conduct and post-evaluation of EC&EE/LC capacity development programs for the public sector.

The following are the deliverables that will be delivered through the implementation of "technical assistance" activities:

Output 4.1.1: Completed project promotional campaigns and workshops in target segments of the public sector. These are the completed advocacy campaigns to various stakeholders in the public buildings sector in embracing and practicing EC&EE/LC initiatives. The following activities are intended to deliver this output:

Activity	Description
4.1.1.1: Conduct of capacity needs assessment.	This involves the evaluation of the current levels of understanding and knowledge about EC&EE in the public buildings sector of China. The evaluation will cover both NGOA-, and LGOA-administered public buildings. Five consultation workshops will be conducted. Among these, is one for the local buildings industry, which will also be consulted to ascertain the level of knowledge of the public buildings sector about the EC&EE technologies that are available, a for the public buildings sector and also the buildings industry's level of knowledge and expertise (or level of service provision) about such technologies. The financial sector will also be consulted to determine their level of understanding and interest in the benefits of EC&EE initiatives for the public buildings sector and the buildings industry.
4.1.1.2: Design and conduct of promotional campaigns.	This involves the organization and conduct of specific promotional campaigns to specific segments of the public buildings sectors on the implementation of EC&EE projects. Five specific campaigns will be designed depending on the capacity needs of each sub-sector of the public buildings sector. These specific campaigns will include one for the buildings industry to enhance their support to the EC&EE efforts of the public buildings sector. The financial sector (including ESCOs) will also have a specific campaign program designed for them to encourage their support to finance EC&EE initiatives in the public buildings sector. Each specific promotional campaigns will be conducted at determined suitable start dates and will run throughout the planned campaign periods.

4.1.1.3: Evaluation of the implemented promotional campaigns.	This involves the review of the results of the implemented promotional campaigns to establish and evaluate the results. The review will be followed up by an evaluation of the impacts of the campaigns in terms of, among others, the uptake of EC&EE initiatives in public buildings, planned EC&EE/LC technology application projects, and planned financing of such projects by financial institutions.

GEF support is required for the incremental technical and financial assistance in the design and conduct of the EC&EE/LC promotional campaigns, as well as in their evaluation.

Output 4.1.2: Completed trainings for NGOA and LGOA authorities and technical staff on the implementation of the various EC&EE/LC programs. These are the completed capacity development activities for the NGOA and LGOA personnel to enhance their capabilities in the promotion and implementation of EC&EE/LC initiatives in their administered public buildings. The following activities are intended to deliver this output:

Activity	Description
4.1.2.1: Design of capacity development program for NGOA and LGOA.	This involves the design of a capacity building program that will comprise of training courses for NGOA and LGOA officials and technical personnel in the various aspects of EC&EE program design, development, planning, financing, implementation, monitoring and evaluation. The number and types of training courses that will be designed will depend on the results and recommendations of the capacity needs assessment in Activity 4.1.1. Estimate is 5 training courses (7 days/course). Relevant training materials will also be prepared. The capacity development program will also include 2 specialized study tours in other countries where there are ongoing successful EC&EE/LC programs for public buildings. These will be for selected technical personnel from NGOA and 5 major LGOAs, and NGOA-and LGOA-administered public buildings.
4.1.2.2: Conduct of the planned training courses.	This involves the implementation of the EC&EE/LC training program (in-country and abroad) as per the planned schedule. The in-country training program will consist consists of classroom instructions, on-the-job training, study tour, etc. For the in-country training, the trainees will be given assignments after each session, which they will complete after returning home and bring to the subsequent session. In the end, there will be assessments to determine whether they have completed the training courses satisfactorily. There are 2 EC&EE/LC Study Tours, one for NGOA & LGOA technical personnel; and one for technical personnel of NGOA-, and LGOA-administered public buildings.
4.1.2.3: Post-Evaluation of the capacity development program.	This involves the design of a specially designed post-training survey to be conducted a year after the completion of the capacity development program. This is also for determining the impacts of the training courses and also for designing follow-up training programs that will be carried out regularly by NGOA and the LGOAs after the PSBEE Project.

GEF support is required for the incremental technical and financial assistance in the design and conduct of the capacity development programs, as well as their evaluation.

Output 4.1.3: Completed and post-evaluated EC&EE/LC capacity development programs for the public buildings sector. These are the completed capacity development programs that will be implemented in the public buildings sector. The following activities are intended to deliver this output:

Activity	Description
4.1.3.1: Design of EC&EE/LC capacity development program for the public buildings sector.	This involves the design of capacity building programs for the public buildings sector in the different climate regions of the country. The programs will be based on the capacity development assessment that is carried out in Activity 4.1.1, and will comprise of training courses for the technical personnel of LGOA-administered buildings (including building managers) in the various aspects of EC&EE/LC technology application project design, planning, implementation, operation and monitoring and evaluation. Relevant training materials will also be prepared. The training courses are expected to be administered by the EC&EE/LC management cum education training centers that will be established in selected LGOA regions (Output 4.4). The capacity development program will also include specialized study tours in other countries where there are ongoing successful EC&EE/LC programs for public buildings. These will be for selected technical personnel from NGOA-, and LGOA-administered public buildings.
4.1.3.2: Conduct of the programmed EC&EE/LC training courses.	This involves the implementation of the EC&EE/LC training program (in-country and abroad) as per the planned schedule. The training program could consists of classroom instructions, on-the-job training, study tour, etc. The trainees will be given assignments after each session, which they will complete after returning home and bring to the subsequent session. In the end, there will be assessments to determine whether they have completed the training courses satisfactorily.
4.1.3.3: Post-Evaluation of the EC&EE/LC capacity development program.	This involves the design of a specially designed post-training survey to be conducted a year after the completion of the capacity development program. This is also for the purpose of determining the impacts of the implemented training courses and also for designing follow-up training programs that will be carried out regularly by the EC&EE/LC management cum education training centers after the completion of the PSBEE Project.

GEF support is required for the incremental technical and financial assistance in the design, development, and conduct of the capacity development programs, including their post-evaluation.

The following are the deliverables that will be delivered through the implementation of "investment" activities:

Output 4.2.1: Established information network for the promotion and dissemination of knowledge on public sector EC&EE and LC technology applications. This is the information exchange network for the sharing of information on building EC&EE/LC technology applications within and outside the country. The following activities are intended to deliver this output:

Activity	Description
4.2.1.1: Conduct of	This involves the evaluation of the current stock of information (i.e.,

capacity and information needs assessments of the public buildings sector.	type, quality and quantity) about building EC&EE/LC technologies available in the public buildings sector, including in the LGOAs. A comprehensive report about the results and recommendations of the assessments will be prepared for purposes of identifying the sort of technical and information support that each LGOA region should be provided.
4.2.1.2: Development, establishment and operationalization of a buildings energy technology information exchange (BETIX) service.	This involves the design and development of the agreed system of EC&EE and LC technology information sharing in Activity 3.1.1.2, including the management and operational arrangements for the system. This system will enable sharing of latest EC&EE and LC technology and market development information among public building managers/administrators, building practitioners in China and building materials and building energy technology developers and suppliers in other countries.
4.2.1.3: Sustaining and strengthening the BETIX service.	This involves the organization and conduct of workshops to strengthen technical and information exchanges among the public building managers and administrators, building practitioners in the buildings sector in China and in other countries. These will be on the: (a) Review of the operation, performance and impacts of the BETIX service to identify potential improvements; and, (b) Sustenance and improvement of the utilization and maintenance of the BETIX service.

GEF support is required for the incremental technical and financial assistance in the design, development, establishment and initial operation of the BETIX, including in the purchase and set up the computer hardware/software for the operation of the system.

Output 4.2.2: Established public sector EC&EE/LC management cum education training centers in the different climate regions. These are strategically sited centers for the promotion, technical assistance and capacity development in the field of EC&EE/LC technology applications in public buildings. The following activities are intended to deliver this output:

Activity	Description
4.2.2.1: Assessment of current capabilities and resources for regional capacity development.	This involves the evaluation of the current capabilities and resources available in the LGOA regions for the provision of capacity development services. The specific targets are LGOAs in the different climate regions of China. With the results and recommendations of other capacity and information needs assessments (Activities 4.1.1 and 4.3.1), a comprehensive report about the potential cities where the public buildings sector education training centers can be located.
4.2.2.2: Selection of LGOAs where EC&EE/LC management cum education training centers will be established.	Based on the results of the assessments in Activity 4.4.1, a set of criteria will be formulated for the selection of LGOAs that will host such centers. With the set of criteria, the 5 LGOAs that will host these management cum education training centers will be selected. Commitments of the selected LGOAs for hosting the centers will be secured.
4.2.2.3: Establishment and operationalization of EC&EE/LC management cum education training centers.	This involves the design, establishment and operationalization of the center in each LGOA. The host LGOA is expected to support the continuous operation of the center. This activity will also involve discussions and technical assistance in the development and implementation of the business plan of each center.

4.2.2.4: Performance evaluation of the EC&EE/LC management cum education training centers.	This involves the conduct of a survey of the performance of each center in terms of the delivery of services to assist public buildings in the LGOA region in their EC&EE/LC efforts, as well as in their obligations towards the EMIS. The survey will be conducted among the managers and administrators of the public buildings administered by the LGOA. The survey evaluation will provide recommendations as to the: (a) potential improvements in the operation, performance and impacts of the center's services; and, (b) Sustenance and improvement of the utilization and maintenance of the center's service.

GEF support is required for the incremental technical and financial assistance in the design, development, establishment and initial operation of the EC&EE/LC management cum education training centers, including in the purchase and set up of the required technical advisory and education paraphernalia.

ii. Partnerships:

Key partners for the proposed project and how the proposed PSBEE project will work in coordination and in partnership with them are described below.

- National Government Office Administration (NGOA): The NGOA represents the GOC in this project as UNDP implementing partner (development and implementation). It will coordinate and liaise with all other pertinent GOC agencies that will be involved in all aspects of the barrier removal activities of the project such as in policy formulation and enforcement. NGOA will also be responsible in facilitating and coordinating the design and implementation of the demo EE technology applications in selected public buildings under its management. It will also be responsible in the regular monitoring of the progress of the project activities (in terms of delivery of outputs and utilization of budget funds) to manage the whole project implementation.

- Local Government Office Administration (LGOA) Units: The LGOAs in each province is responsible for coordinating and co-funding the design and implementation of the demo EE technology applications in selected public buildings under its management. It will also be responsible in the regular monitoring of the progress of the project activities under its purview. It will coordinate and liaise with all other relevant provincial government agencies that will be involved in specific aspects of the barrier removal activities of the project such as in policy formulation and enforcement. It will also be responsible for facilitating the co-funding of specific activities of the project that will be carried out within the provincial public buildings and in the establishment and operation of the EC&EE/Low Carbon (EC&EE/LC) management centers.

- City Governments: Similar with that at the provincial level particularly when the LGOA is located in a city. It will also be responsible for facilitating the co-funding of specific activities of the project that will be carried out within the city public buildings, and in the establishment and operation of the EC&EE/LC management centers.

- County Governments: The main responsibility of government entities at this level would be on energy consumption data provision and in the promotion and implementation of EC&EE techniques and practices in community buildings. These can also benefit from the EC&EE/LC management centers operated by the LGOAs.

- Partnership and Coordination with ongoing EC&EE Projects in China - The design of the PSBEE Project can make use of whatever lessons learned from recently completed and ongoing GEF-funded EC&EE projects in China, as well as any new EE/RE projects that may be online by the time this project will be up and running by 2017. Among the ongoing projects are the following:

    - GEFID 4621 China ADB - Hebei Energy Efficiency Improvement and Emission Reduction Project: During the project implementation, the project team will consult with the implementers of this project on the planning, implementation and monitoring of public building EC&EE activities in Hebei Province and possibly in other provinces similar to Hebei by making use of the principles, techniques and strategies that will be applied in improving EE in large industries that are applicable in public buildings.
    - GEFID 4869 China World Bank - Urban-Scale Building Energy Efficiency and Renewable Energy: The project team will consult with the implementers of this project in regards the planning, implementation and monitoring of demonstration building EC&EE activities financed through market-based financing mechanisms or involve the promotion of commercial bank/financial institution financing of EC&EE in public buildings, particularly on EC&EE techniques and strategies applied in this project in improving policy making and implementation of low-carbon and EE technologies in public and commercial buildings.
    - GEFID 5669 China UNDP - Enabling Solid State Lighting Market Transformation Promotion of Light Emitting Diode Lighting: Since the improvement of lighting systems in public buildings is among the EC&EE measures that will be promoted, the project team will coordinate with the implementers of this project planning, implementation and monitoring of demonstration building EC&EE projects that will showcase the application of energy efficient LED lamps and LED lighting systems in public buildings.
    - GEFID 5360 China UNDP - Promoting Energy Efficient Electric Motors in Chinese Industries: Among the demonstrations that will be implemented in the project is the application of EE electric motors in public buildings. In this regard, the project team will coordinate with the implementers of this project the planning, implementation and monitoring of demonstration building EC&EE projects that will showcase the application of energy efficient motors and motor systems in public buildings.

The establishment of the coordination and reporting arrangements with the relevant agencies and ongoing projects/programs will be carried out during the project to help in identifying the relevant activities that will build on their respective achievements. The UNDP country office in China will be fully involved in the project development through its participation in the various stakeholder and co-financing consultation meetings and technical workshops during project development, and in the multipartite review meetings.

iii. Stakeholder engagement:

The main stakeholders of this project are the National Government Offices Administration (NGOA) and the local government office administration (LGOA) units in the provinces. The other stakeholders are those involved in the education and health sectors, building construction industry, and those in the private sector mainly banks/financial institutions, ESCOs and service companies.

Stakeholder	Roles and Responsibilities in Project Implementation
National Government Offices Administration	Responsible for communication and coordination with MOF and UNDP on the project implementation, liaison with local governments, in-charge of project management.
Ministry of Education of the	Provision of assistance in the co-funding and implementation and

People's Republic of China	management arrangements of demonstrations for the promotion of EC&EE and low-carbon technology applications in school buildings and educational facilities
Ministry of Health of the People's Republic of China	Provision of assistance in the co-funding and implementation and management of demonstrations for the promotion of EC&EE and low-carbon technology applications in hospital buildings and medical facilities
National Energy Conservation Center	Provision of assistance in the implementation of demonstrations for the promotion of EC&EE and low-carbon technology applications in national and local government office buildings and building facilities
Standardization Administration of China	Provision of assistance in the formulation of EC&EE and low-carbon design standards for public sector buildings and provision of technical advice in the various EC&EE/LC standards setting activities for public sector buildings
Dept. of Energy Conservation in Local Governments Offices Administration	Provision of assistance in the design and implementation of the technical assistance, capacity development and demonstration activities of the project
Management/Administration of designated pilot public sector buildings	Provision of advice and co-funding in design and implementation of the technical assistance and capacity development activities of the project
Certification and verification entities	Provision of information in the design and implementation of the development of standards for energy-efficient and low-carbon management and evaluation technique.
Energy Service Companies (ESCOs)[17]	Provision of technical and funding assistance in the design of the activities on EC&EE project financing, feasibility analyses of demonstrations, as well as in the design of EC&EE training programs.

The following summarizes the roles and responsibilities at the different government levels in the development and implementation of the proposed project:

iv. Mainstreaming gender:

The proposed GEF project presents opportunities for the involvement of professional women working in both management and technical departments of public sector agencies/institutions who can play important roles in the operation and management of installed and operated energy management, monitoring, analysis and evaluation, and information dissemination systems in the NGOA and LGOAs. The project will not only provide an opportunity to further assess the role of women in deployment of low carbon technologies and mitigation options, and come up with gender-sensitive policies in the energy consuming operations and processes within the public sector in China, but will definitely be cognizant of the potential contributions of women in the management and implementation of climate change mitigation measures. It should be noted that adequate

[17] ESCOs that will be engaged in the project design are expected to provide the relevant information in the design of the activities on EC&EE project financing, feasibility analyses of demonstrations, as well as in the design of EC&EE training programs. Obviously, since ESCOs are business entities, this would mean that specific ESCOs can be hired to carry out specific project design activities. In this project, interventions will be carried out to encourage and facilitate ESCOs to be actively involved in the design, financing and implementation of EC&EE projects in public sector buildings and facilities. ESCOs could also be involved in the project implementation to carry out project design, finance, implementation, monitoring, verification of energy savings, post-evaluation, and certification. In addition to ESCOs (i.e., ESCOs that design, finance and implement EC&EE projects), the services of local (and foreign) energy engineering firms and energy consulting firms can also be tapped to do the same jobs.

consideration was accorded to women and indigenous people in the design and preparation of this project when there were opportunities to involve them.

v. South-South and Triangular Cooperation (SSTrC):

China and UNDP-China are exponents of South-South and Triangular Cooperation (SSTrC). The PSBEE Project provides another appropriate opportunity for both partners to pursue and promote this noteworthy endeavor of sharing knowledge among developing countries. The SSTrC features of the project shall include exchange and sharing of information between China and other interested developing countries on policies and strategies, regulations, institutional arrangements, financing, and sourcing for the application of EC&EE/LC technologies and techniques in the public buildings sector. During the project implementation, there will be 2 potential approaches that will be considered in regards SSTrC. Firstly, the project will consider opportunities for China to leverage the experience of other developing nations as it promotes EC&EE/LC in the PSBEE Project. China has done a lot in regards energy conservation in buildings (though not specifically on public buildings). Moreover, China also produce and export building materials and appliances/equipment used in buildings. Such presence in the buildings industry not only in the country but also in developing countries is a significant potential avenue by which to leverage South-South cooperation in the area of EC&EE/LC applications in public buildings.

Secondly, the project will consider China's sharing of its experience and knowledge base that will be further developed through the PSBEE Project with other Asian countries. Presently, China has been doing this with select number of Asian countries, and at the same also provide funding in the form of equipment/appliances that partner countries can utilize for their application of EC&EE/LC measures. The support frameworks and tools that will be developed and implemented under the PSBEE will be shared with other countries through the building energy information exchange (BETIX) system that will also be developed under the project and through various events held in the region, With that, China would be able to share its experiences and lessons learned with other developing countries.

V. FEASIBILITY

Cost Efficiency & Effectiveness

A number of relevant activities of the baseline projects have been subsumed in the proposed GEF project. Along with specific incremental activities, these serve as showcase pieces of cost-effective implementation of EC&EE and LC initiatives in public sector buildings. Some of them have been modified to incorporate additional EC&EE/LC features, and these improved/modified activities are now part and parcel of the proposed GEF project. The project has been designed to incorporate features associated with the concepts and techniques of EC&EE and LC operations of public sector buildings in China. The facilitation and enabling of the application of EC&EE and LC technologies and techniques are to be carried out through the removal of barriers associated with the lack of support policies, regulations and institutional mechanisms, limited capacity and knowledge about the application, design, financing and operation of energy efficient building services. Most of the barrier removal activities make up the incremental activities that the project will carry out since the NGOA and LGOAs will not be (or at best in a rather limited manner) addressing these barriers in the baseline projects. Hence, incremental support activities are necessary to facilitate the demonstrations of the application of EC&EE and LC technologies in the public sector buildings. Without the incremental barrier removal and enabling activities the achievement of the anticipated alternative scenario that will bring about an average annual %energy use reduction of about 11.4% will not be realized. Also, the incremental activities for establishing and enforcing policy and regulatory frameworks that are supportive (through effective institutional arrangements, financial/fiscal incentives, information sharing, etc.) of EC&EE efforts in public sector buildings are necessary to sustain the realization of energy efficient public sector buildings in other provinces in China. With the incremental cost for removing the barriers, the expected potential additional global environmental benefits (in terms of avoided CO_2 emissions linked from the electricity and fossil fuels that will be saved) would be realized. These CO_2 emissions reductions are from the energy consumption reduction in public sector buildings in China. The emission reductions would come from: (1) Direct emission reductions from completed EC&EE technology application demonstrations and replications in selected public sector buildings managed by NGOA and some LGOAs; and, (2) Indirect emission reductions from follow-up EC&EE and low carbon technology application projects in public sector buildings under the LGOAs in other provinces.

Comparing the forecast energy consumption trend in the baseline scenario (i.e., with implementation of NGOA EC&EE activities) from the expected energy consumption trend that will be facilitated by the proposed GEF project, an average % annual energy savings of about 12.9% during the period 2015-2025 can be realized with the GEF project. The GEF project is expected to be implemented during the 4 year period from 2018 to 2021. Throughout the project's 4 year implementation period, the expected average annual incremental energy savings would be about 27.7 Mtce corresponding to an average annual incremental CO_2 emission reduction of about 29.7 Mtons CO_2[18]. Assuming that a conservative estimate of about 10% of the potential cumulative CO_2 emission reduction throughout the 4 year period is directly attributable to the interventions that will be carried under the project, this translates to a modest unit abatement cost (UAC) of about US$ 0.75/ton CO_2 (i.e., GEF$ per ton CO_2). This measure of the project's cost effectiveness (i.e., UAC) will be tracked using a monitoring and evaluation system that the proposed project will be developed during the project. This UAC figure will be regularly re-evaluated and updated during the project implementation particularly in quantifying the potential energy savings from projected replications, and in coming up with the CO_2 emission reduction estimates.

[18] This is based on the comparison of the forecast annual CO_2 emission reductions (Baseline Scenario – Own Efforts vs. Alternative Scenario – w/ GEF Support) during the project implementation period.

Risk Management:

During the project implementation, the risks that might prevent the project objectives from being achieved are listed as follows:

Project Risks					
Description	**Type**	**Impact & Probability**	**Mitigation Measures**	**Owner**	**Status**
Municipality, city & county governments in LGOA regions may not support the project implementation promptly and sufficiently	Political	Lack of political leadership and commitment will make it hard for the project to get direction and support for policy, legal, and institutional framework and for budgetary allocation for investment in EC&EE/LC applications in public buildings. Probability = 2 Impact = 2 Risk = 2	NGOA and LGOAs will be supported with government-wide stakeholder engagement and outreach program to create awareness and advocate at the highest political and decision-making levels within and among the local government institutions	UNDP-China, NGOA	Improving
Inadequate communication and cooperation between national and local government institutions to effectively plan and implement EC&EE/LC measures in the public buildings sector.	Organizational	Lack of proper coordination will create confusion over the responsibilities of each institutions that are key players of the project, thus negatively affecting efforts and investments in EC&EE/LC technologies in the public sector Probability = 3 Impact = 3 Risk = 3	The project will support NGOA in conducting stakeholder mapping through its leading of the Low Carbon Growth Technical Working Group, which is comprised of development partners and public sector, private sector, and CSO stakeholders. It will further support NGOA in strengthening the existing coordination mechanism within the LGOA regions.	UNDP-China, NGOA	Improving
The committed level of co-financing for specific activities of the project may not become fully available in time.	Financial	Lack of ongoing budget allocations would result in ad hoc budgetary allocation, which would limit sustainable financing of renewable energy and energy efficiency initiatives. Probability = 3 Impact = 3 Risk = 3	The project team shall secure central government funding prior to project launching. During project implementation, the project team will closely monitor and ensure the timely availability of co-financing from project partners and co-financers. In case of co-financing shortfall for a particular activity, the NGOA will rely on the support of either the GOC or the local governments.	UNDP-China, NGOA	Improving

Technical capacity on EC&EE/LC in the public buildings sector is inadequate	Organizational	Limited pool of national experts will affect technical oversight for and operation and maintenance of EC&EE/LC investments. Probability = 2 Impact = 2 Risk = 2	The project will undertake capacity assessment and target capacity development in NGOA and LGOA regions on EC&EE/LC implementation and O&M. This is expected to generate interest and mobilize support and partnership to promote cost-effective investment in technologies and human resources.	UNDP-China, NGOA	Improving
Low level of participation of the management and personnel of public sector buildings	Operational	The main impact will be delays in the scheduled implementation of project activities and could lead also to complications in the schedules of the co-financed activities. Probability = 3 Impact = 3 Risk = 3	As the national energy management administrator of the public sector, NGOA has enough management and coordination capabilities for ensuring the participation and cooperation of the administrators/managers of public sector buildings to participate in this project. Fast track recruitment arrangements will be employed to hire the relevant project staff.	UNDP-China, NGOA	Improving
Relevant national/local government agencies' failure to approve and enforce formulated policies and regulations	Regulatory	The project seeks to remove policy barriers to EC&EE/LC applications in public buildings. But without the enforcement of the proposed policies, not only will the barriers be really removed but the resulting institutional competition is likely to affect leadership and ownership of project results and negatively impact long-term sustainability Probability = 3 Impact = 3 Risk = 3	Policy change through planned intensive communication and advocacy activities will be pursued. The formulated and approved policies and regulations that will be delivered by the project will be enforced by the NGOA. Policy makers will be adequately briefed on the cost advantages of EC&EE/LC and showing them proof of concept with the project demos to ensure they will support EC&EE/LC applications in the public buildings sector.	UNDP-China, NGOA	Improving
Sustaining the outcomes and benefits of GEF investment on the activities implemented will not be fully sustained.	Financial	All the enabling frameworks and conditions that the project will establish will all be for naught if these will be not institutionalized and discontinued after the end of project	Sustainable follow-up plan will be developed as part of the project activities (to be supported by the local governments and LGOAs) for the replication of the demonstrated applicable and feasible EC&EE technologies in the public	UNDP-China, NGOA	Improving

		Probability = 2 Impact = 2 Risk =2	sector buildings in other provinces.		

Social and environmental safeguards:

The PSBEE Project's social and environmental risk rating is "low." Annex F shows the Social and Environmental Screening carried out for the project. The screening results show that there are no clear direct impacts and risks. There may be some negative impacts but most likely limited in scale and perhaps with a reasonable degree of certainty can be avoided through compliance with the appropriate regulations; or managed through application of standard best practice. There may be possible negative environmental and social impacts, and/or risks associated with the project (or specific project component), but these are predominantly indirect. For example, EE retrofits that will be done in the demos may result in negative impacts if the disposal or the management of the handling of replaced non-EE items are done improperly. Of all of the project activities, the demos and associated activities may present some potential risk of negative social and environmental impacts if factors that would prevent from such risks happening are not considered during their implementation. Thus, the project management unit have to be vigilant enough in facilitating close monitoring of these demos both during the planning and implementation stages. For demonstrations that will involve outdoor installations (but within the boundaries of the host demo public building) any required environmental and safety requirements will be complied with. To ensure the lowest level of environmental and social risk, additional social and environmental assessments (if warranted and necessary) will be carried out as part of feasibility studies that will address technical and design aspects of the demos. Such assessments will include consultations with surrounding communities particularly in LGOA regions to ensure that there are no negative environmental or social impacts of the demos. Such assessment for the initial demos can be used as model for subsequent replications of the demos.

An important aspect of project design for the project activities (including the demos) to reduce social risk is the promotion of the EC&EE/LC technology applications to all segments of the country's buildings sector and building industry, as well as the financial sector. In such sectors in China, the potential of increased women's involvement is significant. For example, the building materials supply industry is where potential businesses owned and operated by women can be further developed. Providing equal opportunities for both men and women in the technical, entrepreneurial and management aspects of EC&EE/LC projects is covered in the project's capacity development activities.

The EC&EE/LC technology application demos as applied in public building retrofits will be indoor, i.e., installed within the demo buildings based on the demo design and on the request of the demo hosts. The project's demo program design reduces the social risk that benefits might only accrue to the public buildings in major cities such as national government buildings in Beijing. The demos are mostly in LGOA-administered office, school and hospital buildings in provincial, municipal and country governments. In all of the demos, a means for citizens reporting environmental grievances will be set up. Environmental and social grievances will be reported to the GEF in the annual PIR reporting process.

Sustainability and Scaling Up:

Sustainability: The enabling conditions that will be created through the barrier removal activities of the proposed GEF project (e.g., adoption of supportive policies/regulations and institutional mechanisms to facilitate the widespread application of EC&EE technologies in the public sector

buildings will, among others, help ensure their sustainability. The strategy to continuously facilitate the effectiveness of these enabling conditions involves their development, establishment and institutionalization. Towards the end of the project, as part of the project activities, a suitable follow-up action plan will be developed. Activities will be carried out aimed towards securing the approval and enforcement of the plan after project completion using the institutional arrangement that will be developed for such purpose. This is to ensure avoidance of the recurrence of the barriers and the continuance of the enabling environments that will be created and/or facilitated by the project. The appropriate actions that will ensure sustainability of whatever policies, regulations, and institutional frameworks that will be established and enforced and implemented in the public buildings sector in China will be determined during the project preparation phase (i.e., during the PPG exercise)[19].

Potential for Scaling-up: The public sector in China is quite big. The proposed GEF project will only involve representative segments of this sector for testing and showcasing the impacts of effective models and schemes for supporting EC&EE initiatives in public sector buildings. The pilot schemes and demonstrations that form part of the interventions to remove barriers can be replicated in other provinces where the LGOAs are keen in improving the energy utilization performance of the public sector buildings they manage. The policies and implementing rules and regulations that will be developed and adopted in selected public sector buildings can be replicated or further enhanced to further enhance the efforts to conserve energy and resources used in these buildings. The successful demonstrations of market-financed EC&EE projects can also be scaled up by interested private sector entities[20].

[19] The following actions are recommended to ensure sustainability of the : (a) Defining the specific requirements for policy implementation; (b) Defining the specific requirements such as implementing rules and regulations in the enforcement of newly established and enforced policies and systems; (c) Defining specific actions to assist key players/stakeholders in the public buildings sector to comply with the abovementioned requirements; (d) Establishment and implementation of some sort of a committee that will oversee proper policy implementation and compliance to policies and implemented systems; and, (e) Evaluation of policy and regulatory framework performance and impacts, and coming up with recommended revisions for improved policy and regulatory implementations.

[20] The extent of potentials for scaling up and replication can be clearly ascertained and defined with a clear understanding of: (a) the barriers preventing the private sector involvement in EC&EE project development and implementation in public sector buildings; (b) the requirements of the private sector in investing in EC&EE projects in public sector buildings; (c) the actual level of interest of the private sector in venturing in building EC&EE projects in the public sector; and, (d) the opportunities are available for both the public sector building administrators/managers and the private sector entities (e.g., commercial banks and ESCOs) in jointly developing and implementing building EC&EE and low carbon initiatives. Considering the commitment of the GOC (NGOA) to improve the energy performance of the public sector, the enforcement of the enabling conditions that the project will facilitate (e.g., financial incentives, policies, etc.), the sheer number of public sector buildings (1,904,900 buildings as of 2012), and increasing demand for energy in these buildings since 2005 (8.2% in 2012), and estimated at 20% in 2020, the potential for scaling up what will be demonstrated and promoted under the project is huge.

VI. PROJECT RESULTS FRAMEWORK

PSBEE Project Planning Matrix

This project will contribute to achieving the following Country Programme Outcome as defined in CPAP or CPD: *More people enjoy a cleaner, healthier environment as a result of improved environmental protection and sustainable green growth*
Country Programme Outcome Indicators: *Indicator 2.1 National carbon dioxide emissions per unit GDP (kg/Yuan)*
Primary applicable Key Environment and Sustainable Development Key Result Area (same as that on the cover page, circle one): *Inclusive and sustainable solutions adopted to achieve increased energy efficiency and universal modern energy access.*
Applicable GEF Strategic Objective and Program: *Promote market transformation for energy efficiency in industry and the building sector*
Applicable GEF Expected Outcomes: *Reduction of GHG emissions from the widespread application of public building energy efficiency*
Applicable GEF Outcome Indicators: *Cumulative CO2 emissions reduction by end-of-project (EOP)*

Project Strategy	**Objectively Verifiable Indicators**				**Means of Gauging Success**	**Critical Assumptions**
	Indicator	**Baseline**	**Mid-term Target**	**End-of-Project Target**		
GOAL: Reduction of greenhouse gas emissions through the widespread application of public building energy efficiency	• Cumulative CO2 emissions reduction, ktons • Annual growth rate of GHG emissions, %	• 0 • 1.8	• 18.6 • 1.2	• 55.7 • 0.6	• Project M&E reports • Project Activity reports • Annual Energy Supply and Consumption Reports from LGOAs and individual public buildings	• Continued commitment, support and active participation of Government of China through NGOA, LGOA, enterprises and the public
OBJECTIVE: Facilitation of the energy conserving and energy efficient operation of buildings and building services in the public sector in	• Cumulative fossil fuel savings due to project intervention, ktce • No. of new jobs created with the application of EC&EE technologies and techniques in the public buildings sector in China	• 0 • 0	• 2,174 • At least 700	• 6,523 • 3,600	• Reports from NGOA on energy used and saved • Consolidated report from annual reports of NGOA, LGOA	• High level of commitment of stakeholders, realization of committed co-financing from NGOA and LGOAs in the implementation of project activities and

China						monitoring systems
Component 1: Public Sector EC&EE Policy and Regulatory Frameworks						
Outcome 1: Strict enforcement of approved enhanced policies and rules and regulations on energy efficiency and low carbon operation and maintenance of public sector buildings	• No. of approved and enforced policies, and associated guidance and implementing rules and regulations (IRRs)[21]. • No. of improved and updated public sector building energy standards.	• 0 • 0	• 2 • 1	• 8 • 2	• Project M&E Reports • Project Activity Reports • Annual reports from LGOAs and the NGOA • Documents on public sector building EC&EE policies and energy standards	• High level of commitment of stakeholders (including the necessary co-financing from government agencies NGOA, LGOA in the implementation of project activities and monitoring systems
Component 2. Energy Performance Monitoring and Evaluation System for Public Buildings						
Outcome 2: Better control and enhanced management of the energy performance of public sector buildings	• No. of official building energy audits completed each year under the energy audit system starting Year 1. • No. of public buildings that regularly submit energy supply and consumption reports annually to the EMIS. • No. of public buildings that are classified as energy efficient.	• 0 • 0 • 0	• 8 • At least 1,000 • At least 1,000	• 12 • 2,200 • 2,200	• Energy audit reports • Reports from the Energy Monitoring and Information System (EMIS) • Evaluation reports on energy efficiency performance of public buildings from NGOA • Project M&E reports • Project activity reports	• High level of commitment of stakeholders (including the necessary co-financing from government agencies NGOA, LGOA in the implementation of project activities and monitoring systems
Component 3: EC&EE Improvement Promotion and Demo Programs for Public Sector Buildings						

[21] These are policies on: (a) energy monitoring and reporting of public sector buildings, as well as on their energy performance evaluation; (b) energy savings verification system (ESVS) for public sector buildings including the associated guidelines and IRRs on the reporting and verification of energy savings realized from registered EC&EE projects; and, (c) improved and revised Regulations on Energy Conservation in Public Sector including the associated new set of guidance and institutional frameworks. Also include developed and approved follow-up plan for the replication of piloted EC&EE improvement policies in public buildings in other provinces.

Outcome 3.1: Increased availability of resources (technical capacity, information and financing) for EC&EE initiatives in public sector buildings and facilities	• No. of public buildings utilizing the established system for providing information about the features, technical specifications and costs of new EC&EE low carbon technologies each year starting Year 2 • No. of banks that are providing financing for EC&EE/LC projects through the market-based financing scheme for public buildings	• 0 • 2	• At least 35 • 3	• 100 • 5	• Reports from information system • Reports from partner banks implementing the market-based financing scheme • Reports on the operations of the EC&EE/LC management centers • Project M&E reports • Project activity reports	• Steady economic growth in China • Government and LGOAs provide support in gathering and providing necessary data on measuring and monitoring energy and environmental impacts • Public and commercial banks have public and socio-economic development financing portfolios
Outcome 3.2: Increased application of EC&EE technologies in public sector buildings and facilities	• No. of EC&EE/LC projects financed through the market-based financing scheme for public buildings. • No. of successfully implemented EC&EE/LC technology application demonstrations in public buildings. • No. of EC&EE/LC projects implemented in public buildings in other provinces that replicate the demonstrations.	• 0 • 0 • 0	• At least 10 • At least 10 • At least 5	• 20 • 20 • 10	• Project documents of EC&EE/LC projects financed through the public buildings financing scheme • Project documents of replication EC&EE/LC projects • Performance and evaluation reports of the various EC&EE/LC demo projects • Project M&E reports • Project activity reports	• Steady economic growth in China • Demo projects are implemented and completed as per schedule • Local Government and private industry sector fully support and commit to the replication of successful results of the project
Component 4: Public Sector EC&EE Capacity and Awareness Enhancement Program						
Outcome 4: Enhanced	• No. of trained public	• 0	• At least	• 3,600	• Certifications of LGOA	• NGOA commitment

awareness and knowledge of public sector authorities and personnel and the citizenry on the cost-effective application of EC&EE technologies	building personnel that are qualified and skilled in the design and cost-effective implementation and evaluation of EC&EE projects. • No. of public buildings with established energy management programs and are implementing EC&EE/LC projects.	• 0	1,500 • At least 1,000	• 2,200	and public building personnel • Reports on energy management programs and the planned and implemented EC&EE/LC projects of public buildings • Reports on the operations of the public sector education training centers • Project M&E reports • Project activity reports.	and support for EC&EE continues.

VII. MONITORING AND EVALUATION (M&E) PLAN

The project results as outlined in the project results framework will be monitored annually and evaluated periodically during project implementation to ensure the project effectively achieves these results. The project monitoring and evaluation plan will also facilitate learning and ensure knowledge is shared and widely disseminated to support the scaling up and replication of project results.

Project-level monitoring and evaluation will be undertaken in compliance with UNDP requirements as outlined in the UNDP POPP and UNDP Evaluation Policy. While these UNDP requirements are not outlined in this project document, the UNDP Country Office will work with the relevant project stakeholders to ensure UNDP M&E requirements are met in a timely fashion and to high quality standards. Additional mandatory GEF-specific M&E requirements (as outlined below) will be undertaken in accordance with the GEF M&E policy and other relevant GEF policies[22].

In addition to these mandatory UNDP and GEF M&E requirements, other M&E activities deemed necessary to support project-level adaptive management will be agreed during the Project Inception Workshop and will be detailed in the Inception Report. This will include the exact role of project target groups and other stakeholders in project M&E activities including the GEF Operational Focal Point and national/regional institutes assigned to undertake project monitoring. The GEF Operational Focal Point will strive to ensure consistency in the approach taken to the GEF-specific M&E requirements (notably the GEF Tracking Tools) across all GEF-financed projects in the country. This could be achieved for example by using one national institute to complete the GEF Tracking Tools for all GEF-financed projects in the country, including projects supported by other GEF Agencies.[23]

M&E Oversight and monitoring responsibilities:

Project Manager: The Project Manager is responsible for day-to-day project management and regular monitoring of project results and risks, including social and environmental risks. The Project Manager will ensure that all project staff maintain a high level of transparency, responsibility and accountability in M&E and reporting of project results. The Project Manager will inform the Project Board, the UNDP Country Office and the UNDP-GEF RTA of any delays or difficulties as they arise during implementation so that appropriate support and corrective measures can be adopted.

The Project Manager will develop annual work plans based on the multi-year work plan included in Annex A, including annual output targets to support the efficient implementation of the project. The Project Manager will ensure that the standard UNDP and GEF M&E requirements are fulfilled to the highest quality. This includes, but is not limited to, ensuring the results framework indicators are monitored annually in time for evidence-based reporting in the GEF PIR, and that the monitoring of risks and the various plans/strategies developed to support project implementation (e.g. gender strategy, KM strategy etc..) occur on a regular basis.

Project Steering Committee (PSC): The PSC will take corrective action as needed to ensure the project achieves the desired results. The PSC will hold project reviews to assess the performance of the project and appraise the Annual Work Plan for the following year. In the project's final year, the PSC will hold an end-of-project review to capture lessons learned and discuss opportunities for scaling up and to highlight project results and lessons learned with relevant audiences. This meeting

[22] See https://www.thegef.org/gef/policies_guidelines
[23] See https://www.thegef.org/gef/gef_agencies

will also discuss the findings outlined in the project terminal evaluation report and the management response.

Project Implementing Partner: The Implementing Partner is responsible for providing any and all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary and appropriate. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes, and is aligned with national systems so that the data used by and generated by the project supports national systems.

UNDP Country Office: The UNDP Country Office will support the Project Manager as needed, including through annual supervision missions. The annual supervision missions will take place according to the schedule outlined in the annual work plan. Supervision mission reports will be circulated to the project team and PSC within one month of the mission. The UNDP Country Office will initiate and organize key GEF M&E activities including the annual GEF PIR, the *independent mid-term review* and the independent terminal evaluation. The UNDP Country Office will also ensure that the standard UNDP and GEF M&E requirements are fulfilled to the highest quality.

The UNDP Country Office is responsible for complying with all UNDP project-level M&E requirements as outlined in the UNDP POPP. This includes ensuring the UNDP Quality Assurance Assessment during implementation is undertaken annually; that annual targets at the output level are developed, and monitored and reported using UNDP corporate systems; the regular updating of the ATLAS risk log; and, the updating of the UNDP gender marker on an annual basis based on gender mainstreaming progress reported in the GEF PIR and the UNDP ROAR. Any quality concerns flagged during these M&E activities (e.g. annual GEF PIR quality assessment ratings) must be addressed by the UNDP Country Office and the Project Manager.

The UNDP Country Office will retain all M&E records for this project for up to seven years after project financial closure in order to support ex-post evaluations undertaken by the UNDP Independent Evaluation Office (IEO) and/or the GEF Independent Evaluation Office (IEO).

UNDP-GEF Unit: Additional M&E and implementation quality assurance and troubleshooting support will be provided by the UNDP-GEF Regional Technical Advisor and the UNDP-GEF Directorate as needed.

Audit: The project will be audited according to UNDP Financial Regulations and Rules and applicable audit policies on NIM implemented projects.[24]

Additional GEF monitoring and reporting requirements:

Inception Workshop and Report: A project inception workshop will be held within two months after the project document has been signed by all relevant parties to, amongst others:

a) Re-orient project stakeholders to the project strategy and discuss any changes in the overall context that influence project strategy and implementation;
b) Discuss the roles and responsibilities of the project team, including reporting and communication lines and conflict resolution mechanisms;
c) Review the results framework and finalize the indicators, means of verification and monitoring plan;

[24] See guidance here: https://info.undp.org/global/popp/frm/pages/financial-management-and-execution-modalities.aspx

d) Discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E budget; identify national/regional institutes to be involved in project-level M&E; discuss the role of the GEF OFP in M&E;
e) Update and review responsibilities for monitoring the various project plans and strategies, including the risk log; Environmental and Social Management Plan and other safeguard requirements; the gender strategy; the knowledge management strategy, and other relevant strategies;
f) Review financial reporting procedures and mandatory requirements, and agree on the arrangements for the annual audit; and
g) Plan and schedule PSC meetings and finalize the first year annual work plan.

The Project Manager will prepare the inception report no later than one month after the inception workshop. The inception report will be cleared by the UNDP Country Office and the UNDP-GEF Regional Technical Adviser, and will be approved by the PSC.

GEF Project Implementation Report (PIR): The Project Manager, the UNDP Country Office, and the UNDP-GEF Regional Technical Advisor will provide objective input to the annual GEF PIR covering the reporting period July (previous year) to June (current year) for each year of project implementation. The Project Manager will ensure that the indicators included in the project results framework are monitored annually in advance of the PIR submission deadline so that progress can be reported in the PIR. Any environmental and social risks and related management plans will be monitored regularly, and progress will be reported in the PIR.

The PIR submitted to the GEF will be shared with the PSC. The UNDP Country Office will coordinate the input of the GEF Operational Focal Point and other stakeholders to the PIR as appropriate. The quality rating of the previous year's PIR will be used to inform the preparation of the subsequent PIR.

Lessons learned and knowledge generation: Results from the project will be disseminated within and beyond the project intervention area through existing information sharing networks and forums. The project will identify and participate, as relevant and appropriate, in scientific, policy-based and/or any other networks, which may be of benefit to the project. The project will identify, analyze and share lessons learned that might be beneficial to the design and implementation of similar projects and disseminate these lessons widely. There will be continuous information exchange between this project and other projects of similar focus in the same country, region and globally.

GEF Focal Area Tracking Tools: The following GEF Tracking Tool(s) will be used to monitor global environmental benefit results: *list the required GEF Tracking Tool(s), as agreed with the UNDP-GEF Regional Technical Advisor*. The baseline/CEO Endorsement GEF Focal Area Tracking Tool(s) – submitted as Annex D to this project document – will be updated by the Project Manager/Team (not the evaluation consultants hired to undertake the *MTR* or the TE) *(indicate other project partner, if agreed)* and shared with *the mid-term review consultants* and terminal evaluation consultants before the required *review/*evaluation missions take place. The updated GEF Tracking Tool(s) will be submitted to the GEF along with the completed *Mid-term Review report* and Terminal Evaluation report.

Independent Mid-term Review (MTR): An independent mid-term review process will begin after the second PIR has been submitted to the GEF, and the MTR report will be submitted to the GEF in the same year as the 3rd PIR. The MTR findings and responses outlined in the management response will be incorporated as recommendations for enhanced implementation during the final half of the project's duration. The terms of reference, the review process and the MTR report will follow the

standard templates and guidance prepared by the UNDP IEO for GEF-financed projects available on the UNDP Evaluation Resource Center (ERC). As noted in this guidance, the evaluation will be 'independent, impartial and rigorous'. The consultants that will be hired to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. The GEF Operational Focal Point and other stakeholders will be involved and consulted during the terminal evaluation process. Additional quality assurance support is available from the UNDP-GEF Directorate. The final MTR report will be available in English and will be cleared by the UNDP Country Office and the UNDP-GEF Regional Technical Adviser, and approved by the PSC.

Terminal Evaluation (TE): An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The terminal evaluation process will begin three months before operational closure of the project allowing the evaluation mission to proceed while the project team is still in place, yet ensuring the project is close enough to completion for the evaluation team to reach conclusions on key aspects such as project sustainability. The Project Manager will remain on contract until the TE report and management response have been finalized. The terms of reference, the evaluation process and the final TE report will follow the standard templates and guidance prepared by the UNDP IEO for GEF-financed projects available on the UNDP Evaluation Resource Center. As noted in this guidance, the evaluation will be 'independent, impartial and rigorous'. The consultants that will be hired to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. The GEF Operational Focal Point and other stakeholders will be involved and consulted during the terminal evaluation process. Additional quality assurance support is available from the UNDP-GEF Directorate. The final TE report will be cleared by the UNDP Country Office and the UNDP-GEF Regional Technical Adviser, and will be approved by the PSC. The TE report will be publically available in English on the UNDP ERC.

The UNDP Country Office will include the planned project terminal evaluation in the UNDP Country Office evaluation plan, and will upload the final terminal evaluation report in English and the corresponding management response to the UNDP Evaluation Resource Centre (ERC). Once uploaded to the ERC, the UNDP IEO will undertake a quality assessment and validate the findings and ratings in the TE report, and rate the quality of the TE report. The UNDP IEO assessment report will be sent to the GEF IEO along with the project terminal evaluation report.

Final Report: The project's terminal PIR along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the PSC during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

Mandatory GEF M&E Requirements and M&E Budget:

GEF M&E requirements	Primary responsibility	Indicative charged to the Project Budget[25] (US$)		Time frame
		GEF grant	Co-financing	
Inception Meeting/Workshop	UNDP Country Office	20,000	30,000	Within 2 months of ProDoc signing
Inception Report	Project Manager	10,000	None	Within 2 weeks of inception meeting
Standard UNDP monitoring and reporting as outlined in the UNDP POPP	UNDP CO	None	None	Quarterly, annually

[25] Excluding project team staff time and UNDP staff time and travel expenses.

GEF M&E requirements	Primary responsibility	Indicative charged to the Project Budget (US$)		Time frame
		GEF grant	Co-financing	
Monitoring of indicators in project results framework	Project Manager	20,000 (US$ 5,000/year)	40,000	Annually
GEF Project Implementation Report (PIR)	Project Manager and UNDP CO and UNDP-GEF team	None	None	Annually
NIM Audit as per UNDP audit policies	UNDP CO	16,000 (US$ 4,000/year)	20,000	Annually or as per UNDP Audit policies
Lessons learned and knowledge generation	Project Manager; relevant contractors and consultants	Covered by various capacity development and information dissemination activities	Covered by capacity development & information dissemination activities	Annually
Monitoring of environmental and social risks, and corresponding management plans	Project Manager UNDP CO	None	20,000	On-going
Addressing environmental and social grievances	Project Manager UNDP CO, UNDP BPPS as needed	None for time of project manager and UNDP CO	As needed	As needed
Project Steering Committee (PSC) meetings	PSC UNDP CO Project Manager	40,000 (US$ 10,000 per year)	25,000	At minimum annually
Supervision missions	UNDP CO	None[26]	25,000	Annually
Oversight missions	UNDP-GEF team	None	As needed	Troubleshooting as needed
Knowledge management	Project Manager	Covered by capacity development & information dissemination activities	Covered by capacity development & information dissemination activities	On-going
GEF Secretariat learning missions/site visits	UNDP CO, Project Manager and UNDP-GEF team	None	10,000	To be determined.
Mid-term GEF Tracking Tool	Project Manager, MTR consultants	10,000	10,000	Before mid-term review mission takes place.
Independent Mid-term Review (MTR) and management response	UNDP CO, PMO, and UNDP-GEF team	55,000	35,000	Between 2nd and 3rd PIR.
Terminal GEF Tracking Tool	Project Manager, TE consultants	10,000	10,000	Before terminal evaluation mission takes place
Independent Terminal Evaluation (TE) included in UNDP evaluation plan, and management response	UNDP CO, PMO, and UNDP-GEF team	55,000	35,000	At least three months before operational closure
Translation of MTR and TE reports into English	UNDP Country Office	NA	NA	----

[26] The costs of UNDP Country Office and UNDP-GEF Unit's participation and time are charged to the GEF Agency Fee.

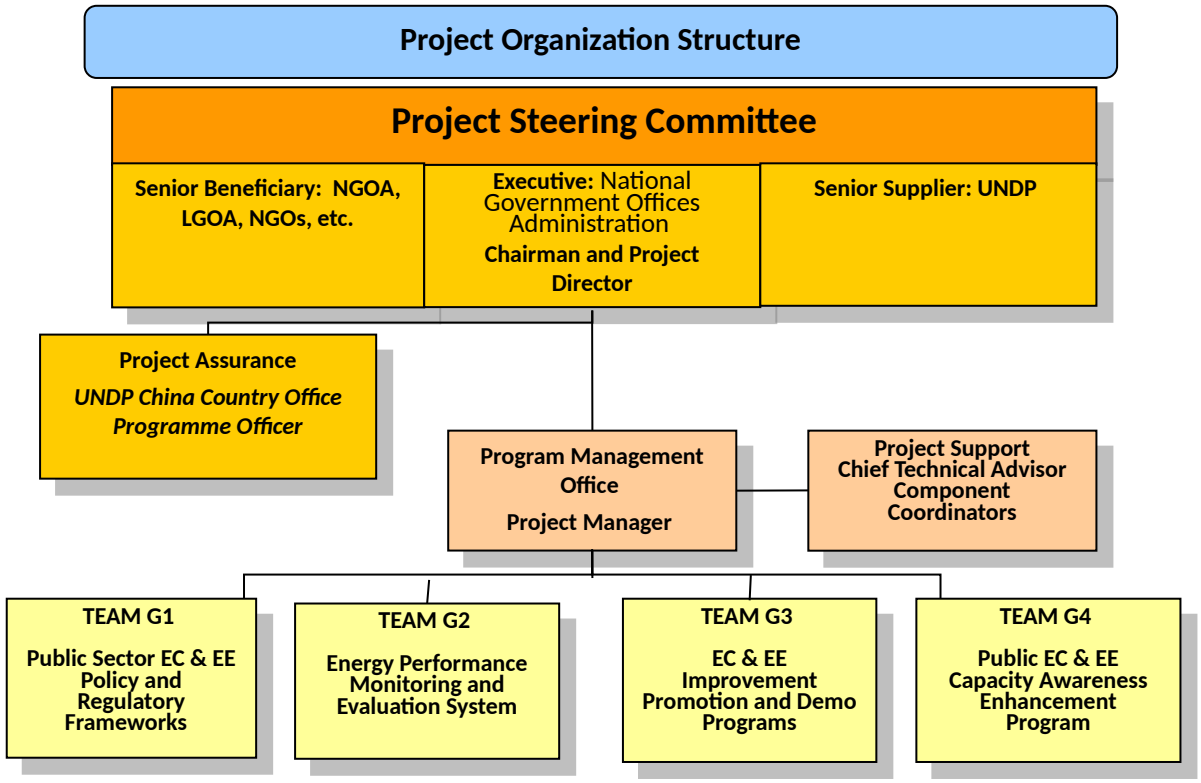
GEF M&E requirements	Primary responsibility	Indicative charged to the Project Budget (US$)		Time frame
		GEF grant	Co-financing	
TOTAL indicative COST Excluding project team staff time, and UNDP staff and travel expenses		**236,000**	**260,000**	----

VIII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

Roles and responsibilities of the project's governance mechanism: As practiced in all UNDP/GEF-supported projects, UNDP always endeavors to seek adaptive management approach in the implementation of projects. Based on the partnerships defined and firmed up during the project development, the management arrangements have always been anchored on co-operation and mutual sharing of benefits where accountability and responsibility for implementing the project and achieving the project outputs.

The project will be implemented following UNDP's national implementation modality, according to the Standard Basis Assistance Agreement between UNDP and the Government of China, and the Country Programme.

- The **Implementing Partner** for this project is the China National Government Offices Administration. The Implementing Partner is responsible and accountable for managing this project, including the monitoring and evaluation of project interventions, achieving project outcomes, and for the effective use of UNDP resources.
- The project organization structure is as follows:



A Project Steering Committee (PSC) will be established and will be composed of the representatives of the UNDP-China, and the representatives of the participating government agencies: National Government Offices Administration (NGOA), representatives from the Local Government Office Administration (LGOA), and the Director of the Project Management Unit (PMU). The National Project Director (NPD) and the National Project Manager (NPM) will come from the Executing Agency which is the NGOA.

The PSC will play the role of policy and decision making body for the project implementation. The PSC members will also be invited to participate in the annual project review meetings. The Chairperson of the PSC is the designated NPD. The NPM provides the management oversight to the whole project and assists the NPD in the operational and industry sectoral aspects of the project implementation. The PMU will be established to be responsible for day-to-day coordinating and implementing the activities of the project and carrying out the prescribed monitoring system. The PMU Director will serve as the Secretary of the PSC.

PSBEE will be implemented through a National Implementation Modality (NIM) on behalf of the Government of China through NGOA as the UNDP's Implementing Partner. The Implementing Partner will assume the overall responsibility of ensuring that all activities are executed accordingly and as per the approved Project Document. The NPD will be responsible for the achievement of the project objectives through institutional coordination with the key stakeholder members of the PSC and overall alignment of the PSBEE Project with the relevant national EE programs of China. He/she takes overall responsibility for all projects' timely reporting, including the submission of Annual Work Plans (AWP), Annual Project Report/Project Implementation Review (APR/PIR) and financial reports to be prepared with the support of the PMU. He/She will ensure the delivery of the project outputs and the judicious use of the project resources. This will ensure that expected outputs are delivered using the most efficient and cost□effective implementation strategies and procedures. The NPM shall also see to it that the Executing Agency (NGOA) supports the project in the planning, coordination, secretariat, administration and financial management of the project in coordination with UNDP China Country Office.

The PSC will be responsible for the following: (a). Reviewing of annual progress reports for necessary guidance; (b) Reviewing and approving the annual work plans and budgets; (c) Providing guidance on the effectiveness of PSBEE project implementation, and its linkages to corporate UNDP policy decisions, and other UNDP initiatives; and, (d) Monitoring and evaluating the implementation of PSBEE towards the intended outputs, after two (2) years of project execution. As a minimum, the PSC will meet at least once a year, allowing for the stakeholders to review the progress with the project implementation and to agree on a coordinated annual project implementation strategy and plan. The first PSC meeting should be held within the first 12 months following the inception workshop.

UNDP-China, in close coordination with the UNDP-GEF Regional Technical Advisor for Climate Change in the Asia-Pacific Region will carry out the GEF oversight and ensure that expected project performance standards are met. It will be also responsible for monitoring and evaluation (M&E), including organizing project reviews, approving annual implementation work plans and budget revisions, monitoring progress, identifying problems, suggesting actions to improve project performance, facilitating timely delivery of project inputs, and provide linkages to other sub-regional, Asia-Pacific regional and global initiatives. All M&E functions will be carried out in line with standard UNDP and UNDP-GEF procedures. UNDP China will also provide country office support for all the activities of the project as coordinated with the Executing Agency.

A Project Management Unit (PMU) will be established and is responsible for the day-to-day management of all the project activities including those on capacity building, demonstration sub-projects and dissemination activities at the national level. The PMU will be managed by a Project Manager (NPM), who will be supported by the Chief Technical Advisor (CTA), administrative staff and at least three Coordinators for the three (3) Components. The CTA provides strategic technical support and advice to ensure that the project is implemented according to the agreements in the Project Document and the standards of UNDP and GEF in project implementation.

A Technical Advisory Committee (TAC) will be established with the main responsibility of providing the necessary expert advice in the implementation of technical aspects of the implementation of the various project components.

The PSBEE Project will be implemented for four (4) years. It is anticipated that the project will start within the first half of 2017 and will be completed by the second half of 2020.

The PSBEE Management Team will prepare its overall country 4-year work plan at the inception stage of the project based on the project activities that are described in this Project Document., specifying the level of activities that will be carried out for the year, the targets to be achieved, and the corresponding inputs (in terms of manpower and budget). During the inception stage, the Executing Agency with assistance of the PMU will prepare its first year work plan and submit this to the UNDP China to be approved for the allocation of funds for the implementation initial year activities. Succeeding annual work plans, based on the results of the previous year and the planned activities for the current year, will be prepared and submitted for approval and budget allocations at the start of each year.

Also during the inception stage, the PSBEE Management Team will prepare its overall 4–year M&E plan based on the overall country work plan, and also based on the PSBEE Project Planning Matrix, PSBEE Annual Targets and PSBEE Project Monitoring Plan. The M&E plan will consist of success indicators (output and impact) with realistic targets and time lines, and backed up with clear means of verification, and assumptions. Each activity/task will be carried out and monitored in terms of the appropriate output indicators (for the activity deliverables) and the impact indicators (for the impacts). The targets will be reviewed each year and any necessary revision or adjustment of these, as well as the assumptions will be done on a continuous basis during the life of the project as part of adaptive management.

To accord proper acknowledgement to GEF for providing funding, a GEF logo will appear on all relevant publications and documents produced by the project, including among others, project hardware purchased with GEF funds. Any citation from any of the PSBEE Project publications and documents will also accord proper acknowledgment to GEF. The UNDP logo should be more prominent and separated from the GEF logo if possible, as UN visibility is important for security purposes.

The **project assurance** role will be provided by UNDP country office. Additional quality assurance will be provided by UNDP Regional Technical Advisor as needed. The project assurance role supports the project steering committee by carrying out objective and independent project oversight and monitoring functions. The role ensures appropriate project outputs and milestones are managed and completed and proper use of UNDP/GEF Funds. Project assurance has to be independent of the project manager; therefore the project steering committee cannot delegate any of its assurance responsibilities to the project manager.

UNDP Direct Project Services (DPS) as requested by Government: The UNDP, as GEF Agency for this project, will provide project management cycle services for the project as defined by the GEF Council. In addition the Government of China may request UNDP direct services for specific projects, according to its policies and convenience. The UNDP and Government of China acknowledge and agree that those services are not mandatory, and will be provided only upon Government request. If requested, the services would follow the UNDP policies on the recovery of direct costs. These services (and their costs) are specified in the Letter of Agreement (Annex _xxx_). As is determined by the GEF Council requirements, these service costs will be assigned as Project Management Cost, duly identified in the project budget as Direct Project Costs. Eligible Direct Project Costs should not be charged as a flat percentage. They should be calculated on the basis of estimated actual or transaction based costs and should be charged to the direct project costs account codes: "64397-Services to projects – CO staff" and "74596 – Services to projects – GOE for CO".

Audit Clause - Audit will be conducted according to UNDP Financial Regulations and Rules and applicable Audit policies. The Government of China will provide the UNDP Resident Representative with certified periodic financial statements, and with an annual audit of the financial statements relating to the status of UNDP (including GEF) funds expended on the project according to the established procedures set out in the appropriate UNDP programming and finance manuals. The audit will be conducted by the legally recognized auditor of the Government of China, or by a commercial auditor engaged by the Government.

MONITORING FRAMEWORK AND EVALUATION

The project will be monitored through the following M& E activities. The M& E budget is provided in the table below.

Project start:

A Project Inception Workshop will be held within the first 2 months of project start with those with assigned roles in the project organization structure, UNDP country office and where appropriate/feasible regional technical policy and program advisors as well as other stakeholders. The Inception Workshop is crucial to building ownership for the project results and to plan the first year annual work plan.

The Inception Workshop should address a number of key issues including:

1. Assist all partners to fully understand and take ownership of the project. Detail the roles, support services and complementary responsibilities of UNDP CO and RCU staff vis-à-vis the project team. Discuss the roles, functions, and responsibilities within the project's decision-making structures, including reporting and communication lines, and conflict resolution mechanisms. The Terms of Reference for project staff will be discussed again as needed.
2. Based on the project results framework and the relevant GEF Tracking Tool if appropriate, finalize the first annual work plan. Review and agree on the indicators, targets and their means of verification, and recheck assumptions and risks.
3. Provide a detailed overview of reporting, monitoring and evaluation (M&E) requirements. The Monitoring and Evaluation work plan and budget should be agreed and scheduled.
4. Discuss financial reporting procedures and obligations, and arrangements for annual audit.
5. Plan and schedule PSC meetings. Roles and responsibilities of all project organization structures should be clarified and meetings planned. The first PSC meeting should be held within the first 12 months following the inception workshop.

An Inception Workshop report is a key reference document and must be prepared and shared with participants to formalize various agreements and plans decided during the meeting.

Quarterly:

The following are monitoring and reporting activities that have to be carried out every quarter:

- Progress made shall be monitored in the UNDP Enhanced Results Based Management Platform.
- Based on the initial risk analysis submitted, the risk log shall be regularly updated in ATLAS. Risks become critical when the impact and probability are high. Note that for UNDP GEF projects, all financial risks associated with financial instruments such as revolving funds, microfinance schemes, or capitalization of ESCOs are automatically classified as critical on the basis of their innovative nature (high impact and uncertainty due to no previous experience justifies classification as critical).
- Based on the information recorded in Atlas, a Project Progress Reports (PPR) can be generated in the Executive Snapshot.
- Other ATLAS logs can be used to monitor issues, lessons learned etc… The use of these functions is a key indicator in the UNDP Executive Balanced Scorecard.

Annually:

Annual Project Review/Project Implementation Reports (APR/PIR): This key report is prepared to monitor progress made since project start and in particular for the previous reporting period (30 June to 1 July). The APR/PIR combines both UNDP and GEF reporting requirements.

The APR/PIR includes, but is not limited to, reporting on the following:

- Progress made toward project objective and project outcomes - each with indicators, baseline data and end-of-project targets (cumulative)
- Project outputs delivered per project outcome (annual).
- Lesson learned/good practice.
- AWP and other expenditure reports
- Risk and adaptive management
- ATLAS QPR
- Portfolio level indicators (i.e. GEF focal area tracking tools) are used by most focal areas on an annual basis as well.

Periodic Monitoring through site visits:

UNDP CO and the UNDP RCU will conduct visits to project sites based on the agreed schedule in the project's Inception Report/Annual Work Plan to assess first hand project progress. Other members of the PSC may also join these visits. A Field Visit Report/BTOR will be prepared by the CO and UNDP RCU and will be circulated no less than one month after the visit to the project team and PSC members.

Mid-term of project cycle:

The project will undergo an independent Mid-Term Evaluation at the mid-point of project implementation (insert date). The Mid-Term Evaluation will determine progress being made toward the achievement of outcomes and will identify course correction if needed. It will focus on the effectiveness, efficiency and timeliness of project implementation; will highlight issues requiring

decisions and actions; and will present initial lessons learned about project design, implementation and management. Findings of this review will be incorporated as recommendations for enhanced implementation during the final half of the project's term. The organization, terms of reference and timing of the mid-term evaluation will be decided after consultation between the parties to the project document. The Terms of Reference for this Mid-term evaluation will be prepared by the UNDP CO based on guidance from the Regional Coordinating Unit and UNDP-GEF. The management response and the evaluation will be uploaded to UNDP corporate systems, in particular the UNDP Evaluation Office Evaluation Resource Center (ERC). The relevant GEF Focal Area Tracking Tools will also be completed during the mid-term evaluation cycle.

End of Project:

An independent Final Evaluation will take place three months prior to the final PSC meeting and will be undertaken in accordance with UNDP and GEF guidance. The final evaluation will focus on the delivery of the project's results as initially planned (and as corrected after the mid-term evaluation, if any such correction took place). The final evaluation will look at impact and sustainability of results, including the contribution to capacity development and the achievement of global environmental benefits/goals. The Terms of Reference for this evaluation will be prepared by the UNDP CO based on guidance from the Regional Coordinating Unit and UNDP-GEF.

The Terminal Evaluation should also provide recommendations for follow-up activities and requires a management response which should be uploaded to PIMS and to the UNDP Evaluation Office Evaluation Resource Center (ERC). The relevant GEF Focal Area Tracking Tools will also be completed during the final evaluation.

During the last three months, the project team will prepare the Project Terminal Report. This comprehensive report will summarize the results achieved (objectives, outcomes, outputs), lessons learned, problems met and areas where results may not have been achieved. It will also lay out recommendations for any further steps that may need to be taken to ensure sustainability and replicability of the project's results.

Learning and knowledge sharing:

Results from the project will be disseminated within and beyond the project intervention zone through existing information sharing networks and forums.

The project will identify and participate, as relevant and appropriate, in scientific, policy-based and/or any other networks, which may be of benefit to project implementation though lessons learned. The project will identify, analyze, and share lessons learned that might be beneficial in the design and implementation of similar future projects.

Finally, there will be a two-way flow of information between this project and other projects of a similar focus.

Communications and visibility requirements:

Full compliance is required with UNDP's Branding Guidelines. These can be accessed at http://intra.undp.org/coa/branding.shtml, and specific guidelines on UNDP logo use can be accessed at: http://intra.undp.org/branding/useOfLogo.html. Amongst other things, these guidelines describe when and how the UNDP logo needs to be used, as well as how the logos of donors to UNDP projects needs to be used. For the avoidance of any doubt, when logo use is required, the UNDP logo

needs to be used alongside the GEF logo. The GEF logo can be accessed at: http://www.thegef.org/gef/GEF_logo. The UNDP logo can be accessed at http://intra.undp.org/coa/branding.shtml.

Full compliance is also required with the GEF's Communication and Visibility Guidelines (the "GEF Guidelines"). The GEF Guidelines can be accessed at: http://www.thegef.org/gef/sites/thegef.org/files/documents/C.40.08_Branding_the_GEF%20final0.pdf. Amongst other things, the GEF Guidelines describe when and how the GEF logo needs to be used in project publications, vehicles, supplies and other project equipment. The GEF Guidelines also describe other GEF promotional requirements regarding press releases, press conferences, press visits, visits by Government officials, productions and other promotional items.

Where other agencies and project partners have provided support through co-financing, their branding policies and requirements should be similarly applied.

IX. FINANCIAL PLANNING AND MANAGEMENT

The total cost of the project is USD 79,032,420. This is financed through a GEF grant of USD 8,932,420, and USD 70,100,000 in co-financing. UNDP, as the GEF Implementing Agency, is responsible for the execution of the GEF resources and the cash co-financing transferred to UNDP bank account only.

Parallel co-financing: The actual realization of project co-financing will be monitored during the *mid-term review* and terminal evaluation process and will be reported to the GEF. The planned parallel co-financing will be used as follows:

Co-financing source	Co-financing type	Co-financing amount	Planned Activities/Outputs	Risks	Risk Mitigation Measures
NGOA	Cash and in-kind	US$ 30.81 million	National level policy, regulatory and institutional framework enabling activities; EC&EE/LC technology application demonstrations; establishment, operation and maintenance of systems such as EMIS, PBEMIS, BEAS, ESMVS, and BETIX.	NGOA management changes and new management is not supportive of PSBEE activities. Committed co-financing for specific project activities may not become fully available in time.	Advocacy to support justification to NGOA's bottom line for continuously supporting the project. The project team shall secure GOC funding prior to project launching.
Local Governments	Cash	US$ 23.19 million	Local level policy, regulatory and institutional framework enabling activities; EC&EE/LC technology application demonstrations and replications; participation in the operation of systems such as EMIS, PBEMIS Database, BEAS, ESMVS, and the EC&EE/LC management cum training centers.	PG administration decides to use funds for other purposes. Committed co-financing for specific project activities may not become fully available in time.	High level liaison by LGOA with PG administration; involvement of PG representative PSC. The project team shall secure GOC funding prior to project launching.
Private Companies	Cash	US$ 16.00 million	Financing of EC&EE/LC demo and replication projects; capacity building	Committed financing may actually turn out to be leveraged financing. Will only commit if the project or a specific activity of the project is/are on.	The project team shall secure commitments from private entities during project inception phase.
UNDP	Cash	US$ 100,000	National level policy and planning work; project management	NA	NA
TOTAL		US$ 70,100,000			

Budget Revision and Tolerance: As per UNDP requirements outlined in the UNDP POPP, the PSC will agree on a budget tolerance level for each plan under the overall annual work plan allowing the project manager to expend up to the tolerance level beyond the approved project budget amount for the year without requiring a revision from the PSC. Should the following deviations occur, the Project Manager and UNDP Country Office will seek the approval of the UNDP-GEF team as these are considered major amendments by the GEF: a) Budget re-allocations among components in the project with amounts involving 10% of the total project grant or more; b) Introduction of new budget items/or components that exceed 5% of original GEF allocation.

Any over expenditure incurred beyond the available GEF grant amount will be absorbed by non-GEF resources (e.g. UNDP TRAC or cash co-financing).

Refund to Donor: Should a refund of unspent funds to the GEF be necessary, this will be managed directly by the UNDP-GEF Unit in New York.

Project Closure: Project closure will be conducted as per UNDP requirements outlined in the UNDP POPP.[27] On an exceptional basis only, a no-cost extension beyond the initial duration of the project will be sought from in-country UNDP colleagues and then the UNDP-GEF Executive Coordinator.

Operational completion: The project will be operationally completed when the last UNDP-financed inputs have been provided and the related activities have been completed. This includes the final clearance of the Terminal Evaluation Report (that will be available in English) and the corresponding management response, and the end-of-project review Project steering committee meeting. The Implementing Partner through a Project steering committee decision will notify the UNDP Country Office when operational closure has been completed. At this time, the relevant parties will have already agreed and confirmed in writing on the arrangements for the disposal of any equipment that is still the property of UNDP.

Financial completion: The project will be financially closed when the following conditions have been met: a) The project is operationally completed or has been cancelled; b) The Implementing Partner has reported all financial transactions to UNDP; c) UNDP has closed the accounts for the project; d) UNDP and the Implementing Partner have certified a final Combined Delivery Report (which serves as final budget revision).

The project will be financially completed within 12 months of operational closure or after the date of cancellation. Between operational and financial closure, the implementing partner will identify and settle all financial obligations and prepare a final expenditure report. The UNDP Country Office will send the final signed closure documents including confirmation of final cumulative expenditure and unspent balance to the UNDP-GEF Unit for confirmation before the project will be financially closed in Atlas by the UNDP Country Office.

[27] see https://info.undp.org/global/popp/ppm/Pages/Closing-a-Project.aspx

X. TOTAL BUDGET AND WORK PLAN

***CEO endorsement template**: The TBWP table of the ProDoc needs to be align with Tables B and D*

Total Budget and Work Plan			
Atlas[28] Proposal or Award ID:	00087742	Atlas Primary Output Project ID:	00094663
Atlas Proposal or Award Title:	Energy Efficiency Improvement in Public Sector Buildings [29] in China		
Atlas Business Unit	CHN10		
Atlas Primary Output Project Title	Energy Efficiency Improvement in Public Sector Buildings [30] in China		
UNDP-GEF PIMS No.	PIMS 5395		
Implementing Partner	*In China: National Government Offices Administration*		

GEF Component / Atlas Activity	Responsible Agency	Source of Fund	Budget Code	Description	Annual Expenses				Total	Budget Notes
					Year 1	Year 2	Year 3	Year 4		
Component 1: Public Sector EC&EE Policy and Regulatory Frameworks										
Outcome 1: Strict enforcement of approved enhanced policies and rules and regulations on energy efficiency and low carbon operation and maintenance of public sector buildings.	NGOA	GEF	71200	International Consultants	30,000	30,750	22,500	-	83,250	1
	NGOA	GEF	71300	National Consultants	6,250	24,750	16,000	13,000	60,000	2
	NGOA	GEF	71600	Travel	13,000	27,300	22,500	10,100	72,900	3
	NGOA	GEF	72200	Equipment & Furniture	-	40,000	10,000	-	50,000	4
	NGOA	GEF	74200	Audio Visual & Print Prod Costs	4,000	16,000	20,500	9,500	50,000	5
	NGOA	GEF	74500	Miscellaneous Cost	2,150	3,150	3,450	3,450	12,200	6
	NGOA	GEF	72100	Sub-contract Services (Comp.)	100,000	238,668	126,000	37,332	502,000	7
	NGOA	GEF	75700	Training/Workshop/Meetings	7,950	34,150	23,650	7,500	73,250	8
	NGOA	GEF	72500	Office Supplies Cost	2,000	3,000	2,750	2,250	10,000	9
	NGOA	GEF	72400	Communications Cost	2,250	3,250	3,050	2,450	11,000	10

[28] See separate guidance on how to enter the TBWP into Atlas

[29] These refer to buildings used, operated and managed by Government (national and local) offices, quasi-government organization and associations whose operations are fully or partly financed by the Chinese government, and includes buildings used as office, hospitals, schools, social/civic/sports & recreation facilities and owned by the Government of China.

[30] These refer to buildings used, operated and managed by Government (national and local) offices, quasi-government organization and associations whose operations are fully or partly financed by the Chinese government, and includes buildings used as office, hospitals, schools, social/civic/sports & recreation facilities and owned by the Government of China.

Component 1 Total					**167,600**	**421,018**	**250,400**	**85,582**	**924,600**	
Component 2: Energy Performance Monitoring and Evaluation System for Public Sector Buildings										
Outcome 2: Better control and enhanced management of the energy performance of public sector buildings.	NGOA	GEF	71200	International Consultants	55,000	83,750	40,000	8,750	187,500	11
	NGOA	GEF	71300	National Consultants	55,000	72,500	45,000	4,500	177,000	12
	NGOA	GEF	71600	Travel	90,000	109,500	59,800	20,300	279,600	13
	NGOA	GEF	72200	Equipment & Furniture	75,000	375,000	25,000	-	475,000	14
	NGOA	GEF	74200	Audio Visual & Print Prod Costs	9,200	21,000	12,500	7,500	50,200	15
	NGOA	GEF	74500	Miscellaneous Cost	3,000	4,250	4,250	3,000	14,500	16
	NGOA	GEF	72100	Sub-contract Services (Comp.)	40,000	350,000	118,200	104,100	612,300	17
	NGOA	GEF	75700	Training/Workshop/Meetings	12,500	40,000	20,000	10,000	82,500	18
	NGOA	GEF	72500	Office Supplies Cost	1,500	4,750	4,750	3,500	14,500	19
	NGOA	GEF	72400	Communications Cost	2,000	4,750	3,750	1,500	12,000	20
Component 2 Total					**343,200**	**1,065,500**	**333,250**	**163,150**	**1,905,100**	
Component 3: EC&EE Improvement Promotion and Demo Programs for Public Sector Buildings										
Outcome 3.1: Increased availability of resources (technical capacity, information and financing) for EC&EE initiatives in public sector buildings and facilities. Outcome 3.2: Increased application of EC&EE technologies in public sector buildings and facilities.	NGOA	GEF	71200	International Consultants	7,500	64,500	61,250	43,750	177,000	21
	NGOA	GEF	71300	National Consultants	2,000	30,150	78,750	58,000	168,900	22
	NGOA	GEF	71600	Travel	5,000	67,500	62,200	34,700	169,400	23
	NGOA	GEF	72200	Equipment & Furniture	-	1,500,000	500,000	-	2,000,000	24
	NGOA	GEF	74200	Audio Visual & Print Prod Costs	4,000	15,600	33,000	18,500	71,100	25
	NGOA	GEF	74500	Miscellaneous Cost	500	3,500	3,750	3,750	11,500	26
	NGOA	GEF	72100	Sub-contract Services (Comp.)	-	160,000	175,800	55,800	391,600	27
	NGOA	GEF	75700	Training/Workshop/Meetings	2,500	17,500	30,000	32,500	82,500	28
	NGOA	GEF	72500	Office Supplies Cost	500	3,250	3,750	4,500	12,000	29
	NGOA	GEF	72400	Communications Cost	500	3,250	3,750	4,000	11,500	30
Component 3 Total					**22,500**	**1,865,250**	**952,250**	**255,500**	**3,095,500**	
Component 4: Public Sector EC&EE Capacity and Awareness Enhancement Program										
Outcome 4: Enhanced awareness and knowledge of public sector authorities and personnel and the citizenry on the cost-effective application of EC&EE technologies.	NGOA	GEF	71200	International Consultants	15,000	47,500	52,500	20,000	135,000	31
	NGOA	GEF	71300	National Consultants	8,000	33,500	49,500	12,500	103,500	32
	NGOA	GEF	71600	Travel	8,000	32,000	53,000	15,000	108,000	33
	NGOA	GEF	72200	Equipment & Furniture	-	1,025,000	325,000	-	1,350,000	34
	NGOA	GEF	74200	Audio Visual & Print Prod Costs	1,000	27,500	53,000	11,500	93,000	35
	NGOA	GEF	74500	Miscellaneous Cost	1,500	13,500	14,000	12,000	41,000	36

	NGOA	GEF	72100	Sub-contract Services (Comp.)	-	125,000	198,698	33,699	357,397	37
	NGOA	GEF	75700	Training/Workshop/Meetings	-	139,425	195,575	17,500	352,500	38
	NGOA	GEF	72500	Office Supplies Cost	1,200	5,000	10,300	7,500	24,000	39
	NGOA	GEF	72400	Communications Cost	600	4,200	7,903	4,800	17,503	40
Component 4 Total					**35,300**	**1,452,625**	**959,476**	**134,499**	**2,581,900**	
ALL Components Total					**568,600**	**4,804,393**	**2,495,376**	**638,731**	**8,507,100**	
Project Management										
	NGOA	GEF	71200	International Consultants	-	26,250	-	26,250	52,500	41
	NGOA	GEF	71300	National Consultants	-	6,750	-	6,750	13,500	42
	NGOA	GEF	71400	Contractual Services (Individual)	47,520	47,520	47,520	47,520	190,080	43
	NGOA	GEF	71400	Contractual Services (Individual)	21,120	21,120	21,120	21,120	84,480	44
	NGOA	GEF	71600	Travel	1,500	6,050	1,500	6,050	15,100	45
	NGOA	GEF	72200	Equipment & Furniture	5,000	-	2,500	-	7,500	46
	NGOA	GEF	74200	Audio Visual & Print Prod Costs	2,000	3,000	2,000	3,000	10,000	47
	NGOA	GEF	74500	Miscellaneous Cost	1,000	1,000	1,000	894	3,894	48
	NGOA	GEF	74100	Professional Services (Audit)	4,000	4,000	4,000	4,000	16,000	49
	NGOA	GEF	75700	Training/Workshop/Meetings	4,000	5,000	3,000	3,000	15,000	50
	NGOA	GEF	72500	Office Supplies Cost	2,500	2,000	2,000	2,000	8,500	51
	NGOA	GEF	72400	Communications Cost	500	1,000	1,000	1,000	3,500	52
	PMO	GEF	74596	Services to Project – GOE for CO	5,266	-	-	-	5,266	53
Project Management Total					**94,406**	**123,690**	**85,640**	**121,584**	**425,320**	
Overall GEF Budget					**663,006**	**4,928,083**	**2,581,016**	**760,315**	**8,932,420**	

Summary of Funds[31]

Fund Source	Amount US$				Total
	Year 1	Year 2	Year 3	Year 4	
Global Environment Facility	663,006	4,928,083	2,581,016	760,315	8,932,420
United Nations Development Programme	22,200	29,100	20,100	28,600	100,000

[31] Summary table should include all financing of all kinds: GEF financing, co-financing, cash, in-kind, etc.

National Government Office Administration *(on behalf of GOC)*, Provincial Government, and Private Companies	6,398,700	38,663,900	18,743,500	6,193,900	70,000,000
TOTAL	**7,083,906**	**43,621,083**	**21,344,616**	**6,982,815**	**79,032,420**

Budget Notes:

No.	Explanation
1	IC Rate = US$670-750/day; 125 person-days; Tasks: Formulation and presentation of new policies and regulations on EMIS for public buildings and evaluation of the energy performance, and verification of reported energy savings (in EMIS), of public buildings; Formulation and presentation of supporting policies and regulations on energy conservation in the Public Sector; Design and conduct of survey of cities on the development of local public building EC&EE policies and regulations; Design of the follow-up plan to promote and implement the replication of the successful applications of the piloted EC&EE and LC policies to other LGOA regions in the country; Technical advice in the enforcement of public sector building energy codes and standards.
2	LC Rate = US$ 85-100/day; 740 person-days (inclusive of 30 person days for study tour for policy research); Tasks: Review of EC&EE and LC policies and regulations for public sector buildings, including current policy and implemented system for energy monitoring and reporting of public buildings. Work with the IC in the implementation of IC tasks; Conduct of advocacy and promotional work to secure approval of the ESVS for public buildings; Evaluation of the results and impacts of public sector energy conservation policies; and Promotion of the successful applications of the piloted EC&EE and LC policies to other major Chinese provinces and cities.
3	5 international round trips @ US$ 2500-3,000/round trip; Local Perdiem Rate = US$ 100-130/day; 245 days; US$ 800-1,000/domestic trip; 23 person-legs
4	PCs and peripheral equipment @ US$ 10000/set, 5 sets
5	US$ 100/print unit; 500 units
6	Estimated sundries, extraordinary expenses.
7	Sub-contract components: (1) Inventory and analysis of public building EC&EE policies; (2) Research, development and implementation of EC&EE roadmaps for the public buildings sector; (3) Design, implementation and evaluation of EC&EE/LC policy pilots; and, (4) Assessment of public buildings EC&EE best practices, and EE building code for public buildings.
8	Total 24 days workshop/training events @ US$2,500-3,000 per W/T
9	Estimated cost for office supplies related to work on policy/regulatory barrier removal activities in Component 1
10	Estimated communications cost for work on policy/regulatory barrier removal activities in Component 1
11	IC Rate = US$ 750/day; 250 person-days; Tasks: Conduct of research on EMIS for buildings in other countries and comprehensive review of the current and planned EC&EE activities in various types of public buildings; Organization and conduct of a workshop on EC&EE in public buildings; Review and development of a public sector Building Energy Audit System (BEAS); Evaluation of BEAS and recommendation of building energy audit program improvements; Finalization of the EMIS framework and mechanisms based on the approved policies and IRRs; Implementation of the EMIS, including provision of technical assistance to public buildings requesting assistance in the reporting process; Evaluation of the results and impacts of the EMIS; Formulation and finalization of the ESMVS mechanisms and supporting regulations; and, Implementation of the ESMVS, including evaluation of its results and impacts.
12	LC Rate = US$ 100/day; 1,770 person-days; Tasks: Work together with the IC in delivering the IC tasks; Preparation of annual reports on the status and trends in the energy supply, demand and consumption in the public buildings sector in China; Publication and dissemination of results of the ESMVS; and, Conduct of

	research/study on the requirements and appropriate procedures for processing, verification, and encoding, as well as data updating.
13	14 international round trips @ US$ 2500/round trip; Local Perdiem Rate = US$ 100/day; 980 days; US$ 800/domestic trip; 92 person-legs
14	5 sets of energy audit instruments/equipment @ US$ 75,000/set; 1 set of required hardware and software package for the EMIS & PBEMIS Database @ US$ 100,000/set
15	US$ 100/print unit; 502 units
16	Estimated sundries, extraordinary expenses.
17	Sub-Contract Components: (2) Development and implementation of Public Building Energy Audit Program; conduct of energy audits, and publication and dissemination of energy audit program results; (2) Design of the EMIS for public buildings; (3) Design of the ESMVS for public buildings; and (4) Design and development of the PBEMIS Database, including capacity building on the use and maintenance of the database.
18	Total 33 days workshop/training events @ US$2,500 per W/T
19	Estimated cost for office supplies related to work on institutional barrier removal activities in Component 2
20	Estimated communications cost for work on institutional barrier removal activities in Component 2.
21	IC Rate = US$ 750/day; 236 person-days; Conduct of analyses of data/information in the PBEMIS Database; Design and establishment of a scheme for the sourcing of EC&EE/LC technology-related information for the PBEMIS Database; Review and updating of the EC&EE and LC technologies directory; Promotion of potential schemes for EC&EE/LC project financing in public buildings; Review and pre-feasibility assessment of the potential demo/pilot projects; Conduct of preliminary discussions for financing of demo projects; Finalization of the line-up of confirmed EC&EE demos/pilots; Establishment and operationalization of the selected financing scheme(s); Conduct of promotional campaigns for the financing/incentive schemes; Capacity development for public buildings; Implementation of the EC&EE/LC demonstrations that are funded by market-based financing schemes; Preparation of the demo project profiles (as case studies); Conduct of an overall performance evaluation of the demo/pilots projects; Technical advice on the implementation of the EC&EE/LC demonstrations; Preparation of the demo project profiles (as case studies) and overall performance evaluation of the demo/pilots projects; Organization and conduct of a seminar-workshop on the results of the demonstrations; Conduct of a workshop on the sharing of EC&EE and LC technologies as applied to public buildings, including documentation of seminar-workshop proceedings; Evaluation of additional capacity development needs on EC&EE/LC technologies; Design of sustainable follow-up plans; and technical assistance in the promotion of the sustainable follow-up program.
22	LC Rate = US$ 100/day; 1,690 person-days; Tasks: Work together with the IC in delivering the IC tasks; Organization and conduct of a workshop on the sharing of EC&EE and LC technologies as applied to public buildings.
23	9 international round trips @ US$ 2500/round trip; Local Perdiem Rate = US$ 100/day; 593 days; US$ 800/domestic trip; 56 person-legs
24	Allocated budget for incremental hardware for the selected EC&EE/LC technology application demonstrations. The preliminarily identified demonstrations are presented in Annex K. The EC&EE/LC technology applications that will be showcased are known including the hardware and software that will be used. However, the specific incremental equipment/instruments that will be required and procured are not yet confirmed. The estimated budget is based on preliminary economic feasibility analyses that were performed for each demo. The incremental budget will be allocated based on the economic viability of the demo with those having lower economic performance to be provided higher incremental budgets. Total estimated incremental costs for the identified demonstrations is roughly US$ 2M.
25	US$ 100/print unit; 711 units
26	Estimated sundries, extraordinary expenses.
27	Sun-contract components: (1) Review of financing schemes for EC&EE/LC projects in China, in general, and buildings sector, in particular; (2) Design of market-based financing and incentive schemes for EC&EE/LC projects in the public buildings; (3) Conduct of pre-feasibility and comprehensive feasibility studies of demo projects; and, (4) Design and promotion of the EC&EE/LC demo projects.

28	Total 33 days workshop/training events @ US$2,500 per W/T
29	Estimated cost for office supplies related to work on financial barrier removal activities in Components 3.1 & 3.2.
30	Estimated communications cost for work on financial barrier removal activities in Components 3.1 & 3.2.
31	IC Rate = US$ 750/day; 180 person-days; Tasks: Conduct of capacity needs assessment on EE&EC/LC technologies; Design and conduct and evaluation of promotional campaigns and capacity development program; Conduct of capacity and information needs assessments of the public buildings sector; Development, establishment and operationalization of a buildings energy technology information exchange (BETIX) service; Assessment of current capabilities and resources for regional capacity development; Establishment and operationalization and performance evaluation of EC&EE/LC management cum education training centers; and, Post-Evaluation of the EC&EE/LC capacity development program.
32	LC Rate = US$ 100/day; 1,035 person-days; Tasks: Work together with the IC in delivering the IC tasks; Provision of technical advice on sustaining and strengthening the BETIX service; and, selection of LGOAs where EC&EE/LC management cum education training centers will be established.
33	6 international round trips @ US$ 2500/round trip; Local Perdiem Rate = US$ 100/day; 368 days; US$ 800/domestic trip; 35 person-legs
34	Cost of the purchase and installation of required package of hardware and software for the Building Energy Technology Information Exchange (BETIX) system @ US$100,000; Cost of the purchase and installation of required office equipment (PCs and peripherals including software) for each EC&EE/LC management cum education training center (5 centers). Includes educational and promotional materials and subscriptions to EC&EE/LC literature and technology databases. US$ 250,000/package @ 1 package per center.
35	US$ 100/print unit; 930 units
36	Estimated sundries, extraordinary expenses particularly for the study tours and set-up of the EC&EE/LC management cum training centers.
37	Sub-contract components: (1) Design and conduct of a comprehensive EC&EE/LC capacity development program for NGOA and LGOA; and, (2) Design and conduct of an EC&EE/LC capacity development program for the public buildings sector.
38	Total 7 days workshop/training events @ US$2,500 per W/T. Special study tours abroad for: (1) NGOA and LGOA technical personnel (12 pax); and, (technical personnel from NGOA-, and LGOA-administered public buildings (15 pax). Cost includes learning institution fee; travel and DSA of trainees; and study tour organizer fees. Training @ 10 days; travel and DSA (US$ 275/day) @ 15 days; US$ 2,500/pax round trip air fare; Learning Institution fee @ US$ 2,000/day. Study tour fee @ 15% total cost.
39	Estimated cost for office supplies related to work on capacity and information barrier removal activities in Component 4.
40	Estimated communications cost for work on capacity and information barrier removal activities in Component 4.
41	IC Rate = US$ 750/day; 70 person-days; Tasks: Mid-term and Terminal Evaluations
42	LC Rate = US$ 100/day; 135 person-days; Tasks: Mid-term and Terminal Evaluations
43	National Project Manager @ US$ 180/day (22 days/mo, 12 mos/yr, for 4 years)
44	Finance/Admin Staff @ US$ 80/day (22 days/mo, 12 mos/yr, for 4 years)
45	2 international round trips @ US$ 2500/round trip; Local Perdiem Rate = US$ 100/day; 40 days; US$ 800/domestic trip; 4 person-legs
46	Project computers @ US$ 2,500/ea (2 units)
47	US$ 100/print unit; 100 units
48	Estimated sundries, extraordinary expenses.
49	Financial audit @ US$4,000/yr; 4 years
50	Total 6 days workshop/training events @ US$2,500 per W/T

51	Estimated cost for office supplies related to project management
52	Estimated communications cost related to project management
53	Agreed UNDP-China and GOC (NGOA) direct project cost. See Annex

XI. LEGAL CONTEXT

This document together with the CPAP signed by the Government and UNDP which is incorporated by reference constitute together a Project Document as referred to in the SBAA [or other appropriate governing agreement] and all CPAP provisions apply to this document.

Consistent with the Article III of the Standard Basic Assistance Agreement, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the implementing partner's custody, rests with the implementing partner.

The Implementing Partner shall:

- Put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried; and,
- Assume all risks and liabilities related to the implementing partner's security, and the full implementation of the security plan.

UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of this agreement.

The implementing partner agrees to undertake all reasonable efforts to ensure that none of the UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/Docs/sc/committees/ 1267/1267ListEng.htm. This provision must be included in all sub-contracts or sub-agreements entered into under this Project Document.

Any designations on maps or other references employed in this project document do not imply the expression of any opinion whatsoever on the part of UNDP concerning the legal status of any country, territory, city or area or its authorities, or concerning the delimitation of its frontiers or boundaries.

XII. MANDATORY ANNEXES

Annex A. Multiyear Work Plan

Annex B. Monitoring Plan

Annex C. Evaluation Plan

Annex D: GEF CCM Tracking Tool (Separate File)

Annex E. Terms of Reference

Annex F. UNDP Social and Environmental and Social Screening Template (SESP)

Annex G. UNDP Project Quality Assurance Report

Annex H. UNDP Risk Log

Annex I. Results of the capacity assessment

Annex J. Additional agreements

Annex K. Description of EC&EE and LC Demonstrations

Annex L. GHG Emission Reduction Estimates

Annex M. Annual Targets

Annex N. Gender Analysis

Annex O. Knowledge Management Strategy

Annex A. Multi Year Work Plan

Outputs	Year 1				Year 2				Year 3				Year 4			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Component 1: Public Sector EC&EE Policy and Regulatory Frameworks																
Output 1.1: Completed comprehensive assessment of applicable foreign and domestic energy conservation and energy efficiency (EC&EE) and low carbon (LC) policies and regulations for public sector buildings.																
Output 1.2: Formulated and promoted EC&EE improvement roadmaps.																
Output 1.3: Formulated and enforced policies including the associated guidance and implementing rules and regulations (IRRs) on energy monitoring and reporting, energy savings verification, and energy conservation in public buildings.																
Output 1.4: Completed demonstrations on the application of EC&EE policies and systems in 3 to 5 local government office administration (LGOA) regions.																
Output 1.5: Developed and approved follow-up plan for the replication of piloted EC&EE improvement policies in public buildings in other provinces.																
Output 1.6: Developed improved and updated public sector building energy standards.																

Project Outputs	Year 1				Year 2				Year 3				Year 4			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Component 2. Energy Performance Monitoring and Evaluation System for Public Sector Buildings																
Output 2.1.1: Reviewed and verified supplemental baseline energy information in the various major types of buildings within the public sector.																
Output 2.1.2: Established public sector building energy audit system.																
Output 2.1.3: Established public sector buildings energy management information system (EMIS).																
Output 2.1.4: Established energy savings measurement & verification system (ESMVS) in public buildings sector.																
Output 2.2.1: Completed minimum 120 energy audits of public sector buildings (40 per year, for 3 years).																
Output 2.2.2: Established Public Buildings EMIS database.																

Outputs	Year 1				Year 2				Year 3				Year 4			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Component 3: EC&EE Improvement Promotion and Demo Programs for Public Sector Buildings																
Component 3.1: Facilitation of EC&EE/LC Technologies Application Demonstrations																
Output 3.1.1: Established scheme for providing information about the features, technical specifications and costs of new EC&EE low carbon technologies (including products) for the public sector.																
Output 3.1.2: Published directory of recommended applicable and cost-effective new EC&EE/LC technologies (systems and products) for public sector building administrators/managers.																
Output 3.1.3: Completed assessment of market-based financing scheme options.																
Output 3.1.4: Designed market-based financing of new EC&EE/LC technology (system and product) applications.																
Output 3.1.5: Selected EC&EE/LC projects (total of 15) in public buildings in selected public sub-sectors.																
Output 3.1.6: Completed feasibility analyses and design of EC&EE and RE technology application demonstrations.																

Outputs	Year 1				Year 2				Year 3				Year 4			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Component 3: EC&EE Improvement Promotion and Demo Programs for Public Sector Buildings																
Component 3.2: Implementation of EC&EE/LC Technologies Application Demonstrations																
Output 3.2.1: Established scheme for market-based financing of new EC&EE/LC technology (system and product) applications.																
Output 3.2.2: Implemented 5 EC&EE/LC projects financed through market-based financing scheme in public buildings in selected public sub-sectors in 2 to 5 cities or regions.																
Output 3.2.3: Implemented EC&EE (and RE) technology application demonstrations in public sector buildings (for 2 sub-sectors).																
Output 3.2.4: Published reports on the impacts of the EC&EE/LC project financing and demonstration program.																
Output 3.2.5: Developed sustainable follow-up plans for the replication of the demonstrated applicable and feasible EC&EE/LC technologies in the public sector buildings in other provinces.																

Project Outputs	Year 1				Year 2				Year 3				Year 4			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Component 4. Public Sector EC&EE Capacity and Awareness Enhancement Program																
Output 4.1.1: Completed 5 project promotional campaigns and 5 workshops in target segments of the public sector.																
Output 4.1.2: Completed 5 training courses and 2 study tours for technical personnel of NGOA and LGOA, and NGOA-, and LGOA-administered public buildings on the implementation of the various EC&EE/LC programs.																
Output 4.1.3: Completed and post-evaluated EC&EE/LC capacity development program for the public buildings sector.																
Output 4.2.1: Established information network for the promotion and dissemination of knowledge on public sector EC&EE and LC technology applications.																
Output 4.2.2: Established 5 public sector EC&EE/LC management cum education training centers in the different climate regions of China.																

Annex B. Monitoring Plan:

The Project Manager will guide the collection of results data according to the following monitoring plan:

Monitoring	Indicators	Description	Data Source or Data Collection Methods	Frequency	Responsible for data collection	Means of Verification	Assumptions
Project Objective: Facilitation of the energy conserving and energy efficient operation of buildings and building services in the public sector in China	Cumulative fossil fuel savings due to project intervention by EOP, ktce	Cumulative energy savings from business as usual via adoption of EE technologies, techniques and practices in the public buildings sector in China	Established and operational Energy Management Information System (EMIS) PBEMIS Database	After 2nd PIR submitted to GEF; After 4th PIR submitted to GEF.	NGOA – as part of operation EMIS and maintenance of the PBEMIS	EMIS & PBEMIS Database To be verified and reported in mid-term review and terminal evaluation reports, and DO tab of the GEF PIR.	Continued and consistent commitment of the government to EC&EE/LC
	No. of new jobs created with the application of EC&EE technologies and techniques in the public buildings sector in China	Number of new businesses and business expansions (and indirectly no. of new job creation or no. of new entrants to labor force) to support increase application of EC&EE/LC projects in the public buildings sector.[32]	Building industry survey Sociological survey	After 2nd PIR submitted to GEF; After 4th PIR submitted to GEF.	Contracted Project Consultants Survey contractors	Building Industry Survey Report Sociological Survey Report	

[32]Examples include more building practitioners (e.g., new architecture and engineering graduates) working on new public building projects (new and retrofits), building materials suppliers, increased demand for EE building materials translates to more work in the local building material manufacturing industry; more ESCOs can get involved in the financing and provision of technical services (project development, engineering design, costing, operations and maintenance) on EC&EE/LC projects to existing public buildings.

Monitoring	Indicators	Description	Data Source or Data Collection Methods	Frequency	Responsible for data collection	Means of Verification	Assumptions
Outcome 1: Strict enforcement of approved enhanced policies and rules and regulations on energy efficiency and low carbon operation and maintenance of public sector buildings.	No. of approved and enforced policies, and associated guidance and implementing rules and regulations (IRRs)	Number of policies that are approved for enforcement in the public buildings sector both at the national and local levels and the associated set of IRRs	Documentation of the approved and enforced policies	Annually, reported in DO tab of the GEF PIR.	PMO, guided by the National Project Manager (NPM)	Government records on the enforced policies and IRRs	NGOA and LGOA regularly oversee the implementation of the policies.
	No. of improved and updated public sector building energy standards.	Number of aspects in which newly adopted policies and energy standards support the widespread application of EC&EE/LC technologies and techniques/practices in public buildings.[33]	Documentation of the approved and enforced energy standards	Annually, reported in DO tab of the GEF PIR.	PMO, guided by the NPM	Government records on the enforced energy standards Periodic media articles reporting on EC&EE/LC policies and standards	Continued and consistent government support to EC&EE/LC applications in public buildings via improved building energy policies and standards
Outcome 2: Better control and enhanced management of the energy performance of public sector buildings	No. of official building energy audits completed each year under the energy audit system starting Year 2	Number of building energy audits that are quality certified in compliance with specific energy audit specifications under the establish public buildings energy audit system.	Building energy audit reports submitted to NGOA/LGOA	To be reported in DO tab of the GEF PIR *twice* throughout the project implementation period[34].	PMO, guided by the NPM	Reviewed building energy audits reports Summary reports of building energy audits	High level of commitment of stakeholders (including the necessary co-financing from government agencies NGOA, LGOA in the implementation of project activities and monitoring systems
	No. of public buildings that regularly submit energy supply and consumption reports annually to the EMIS	Number of public buildings covered by the EMIS, which is operated and maintained by the NGOA	EMIS in NGOA Regular periodic summary reports from the EMIS	To be reported in DO tab of the GEF PIR *twice* throughout the project implementation	PMO, guided by the NPM	Regular periodic summary reports from the EMIS	

[33]Examples of policy aspects include: (1) fiscal incentives for public sector buildings that implement EC&EE/LC projects; (2) fiscal incentives for building technical service providers including those in the building materials industry; (3) Public-private-partnerships in the implementation of EC&EE/LC projects in public buildings; (4) Public buildings availing access to market-based financing mechanisms; (5) other financial incentives for EC&EE retrofits; (6) new energy efficiency standards in building code, e.g., specifications for what an energy efficient building; and, (7) new EE standards for appliances and equipment used in public buildings.

[34] This is at the middle of the project implementation, after 2nd PIR submitted to GEF; and at the end of the project, after 4th PIR submitted to GEF.

Monitoring	Indicators	Description	Data Source or Data Collection Methods	Frequency	Responsible for data collection	Means of Verification	Assumptions
				period			
	No. of public buildings that are classified as energy efficient	Number of public buildings classified by the NGOA as energy efficient based on the set specifications for EE public buildings.	Periodic summary reports from the PBEMIS Database Periodic reports from the ESMVS	To be reported in DO tab of the GEF PIR *twice* throughout the project implementation period.	PMO, guided by the NPM	Periodic summary reports from the PBEMIS Database Periodic reports from the ESMVS	
Outcome 3.1:: Increased availability of resources (technical capacity, information and financing) for EC&EE initiatives in public sector buildings and facilities	No. of public buildings utilizing the established system for providing information about the features, technical specifications and costs of new EC&EE/LC technologies	Number of public buildings that make use of information on applicable EC&EE/LC technologies and techniques made available for them to use through the PBEMIS Database.	Periodic summary reports from the PBEMIS Database Survey of public buildings and building practitioners that make use of the PBEMIS Database for information	To be reported in DO tab of the GEF PIR *twice* throughout the project implementation period.	PMO, guided by the NPM Survey Contractors	Periodic summary reports from the PBEMIS Database PBEMIS Database User Survey Reports	Government and LGOAs provide support in gathering and providing necessary data on measuring and monitoring energy and environmental impacts
	No. of banks that are providing financing for EC&EE/LC projects through the market-based financing scheme for public buildings	Number of banks or financial institutions that agree to support the market-based financing schemes for public buildings. These banks/FIs will provide financing (using their funds) to public buildings that will implement EC&EE/LC projects.	Survey of banks/FIs	To be reported in DO tab of the GEF PIR *twice* throughout the project implementation period	PMO, guided by NPM Survey contractors	Survey reports	Public and commercial banks have public and socio-economic development financing portfolios
Outcome 3.2: Increased application of EC&EE technologies in public sector buildings and facilities	No. of EC&EE/LC projects financed through the market-based financing scheme for public buildings.	Number of EC&EE/LC projects that were implemented by public buildings and funded by the banks/FIs that are implementing the market-based financing schemes developed by the PSBEE Project	Survey of banks/FIs, and project proponents that were granted project financing through the market-based financing schemes.	To be reported in DO tab of the GEF PIR *twice* throughout the project implementation period	PMO, guided by NPM Project Proponents Survey Contractor	Project documents of EC&EE/LC projects Project M&E reports Project activity reports	Steady economic growth in China Local

Monitoring	Indicators	Description	Data Source or Data Collection Methods	Frequency	Responsible for data collection	Means of Verification	Assumptions
	No. of successfully implemented EC&EE/LC technology application demonstrations in public buildings.	Number of successful demonstration EC&EE/LC application projects implemented by host public buildings under the PSBEE Project	Demo project documents Demo project implementation reports	To be reported in DO tab of the GEF PIR *twice* throughout the project implementation period.	PMO, guided by NPM Demo Project Hosts Demo project personnel	Demo Project documents Project M&E reports Project activity reports	Demo projects are implemented and completed as per schedule
	No. of EC&EE/LC projects implemented in public buildings in other provinces that replicate the demonstrations.	Number of replication EC&EE/LC projects that are implemented in public buildings in other LGOA regions. These are projects that are based on the demonstrations and are funded by the project owners themselves or through the market-based financing schemes.	Replication project documents Replication project implementation reports	To be reported in DO tab of the GEF PIR *twice* throughout the project implementation period	PMO, guided by NPM Replication Project Owners Replication project personnel	Replication Project documents Project M&E reports Project activity reports	Government and private industry sector fully support and commit to the replication of successful results of the project
Outcome 4: Enhanced awareness and knowledge of public sector authorities and personnel and the citizenry on the cost-effective application of EC&EE technologies.	No. of trained public building personnel that are qualified and skilled in the design and cost-effective implementation and evaluation of EC&EE projects	Number of building practitioners that were trained and are actively and gainfully employed as skilled technicians or experts in the design and implementation of EC&EE/LC projects in buildings in the public sector and commercial sector)	Certifications of LGOA and public building personnel Reports on energy management programs and the planned and implemented EC&EE/LC projects of public buildings	Continuous data collection, reported in DO tab of the GEF PIR *twice* throughout the project implementation period	PMO, guided by the NPM Project Consultants Survey contractor	Survey Reports Project M&E reports Project activity reports Reports on the operations of the public sector education training centers	NGOA commitment and support for EC&EE/LC
	No. of public buildings with established energy management programs and are implementing EC&EE/LC projects.	Number of public buildings that have established and implement an energy management program, and are implementing EC&EE/LC projects. Public buildings that are not covered by the EMIS but have ongoing programs will also be included.	EMIS in NGOA PBEMIS Database	Continuous data collection, reported in DO tab of the GEF PIR *annually*	PMO, guided by the NPM	Regular periodic summary reports from the EMIS	

Monitoring	Indicators	Description	Data Source or Data Collection Methods	Frequency	Responsible for data collection	Means of Verification	Assumptions
Mid-term GEF Tracking Tool	N/A	N/A	Standard GEF Tracking Tool available at www.thegef.org Baseline GEF Tracking Tool included in Annex.	After 2nd PIR submitted to GEF	Project Consultant	Completed GEF Tracking Tool	Continued and consistent government support to EC&EE/LC applications in public buildings via improved building energy policies and standards
Terminal GEF Tracking Tool	N/A	N/A	Standard GEF Tracking Tool available at www.thegef.org Baseline GEF Tracking Tool included in Annex.	After final PIR submitted to GEF	Project Consultant	Completed GEF Tracking Tool	-Ditto-
Mid-term Review	N/A	N/A	To be outlined in MTR inception report	Submitted to GEF same year as 3rd PIR	Independent evaluator	Completed MTR	-Ditto-
Project Terminal Evaluation	N/A	N/A	To be outlined in Terminal Evaluation inception report	Submitted to GEF same year as 4th PIR	Independent evaluator	Completed Terminal Evaluation Report	-Ditto-
Environmental and Social Risk Management Plans	N/A	N/A	Updated SESP and management plans	At the mid- and at the end of the project	Project Manager UNDP CO	Updated SESP	-Ditto-

Annex C. Evaluation Plan

Evaluation Title	Planned start date Month/year	Planned end date Month/year	Included in the Country Office Evaluation Plan	Budget for consultants[35]	Other budget (i.e., travel, site visits, etc.)	Budget for translation
Mid-Term Review	*Nov 2019*	*Feb 2020*	Yes/No *Mandatory*	*USD 50,000*	*USD 4,000*	*USD 1,000*
Terminal Evaluation	*Sep 2021*	*Dec 2021*	Yes/No *Mandatory*	*USD 50,000*	*USD 4,000*	*USD 1,000*
			Total evaluation budget	USD 87,200		

[35] The budget will vary depending on the number of consultants required (for full size projects should be two consultants); the number of project sites to be visited; and other travel related costs. Average # total working days per consultant not including travel is between 22-25 working days.

Annex D. GEF Tracking Tool *(at baseline)*

Submitted as a separate file.

Annex E. Terms of Reference

This section presents the terms of reference (TORs) for the key personnel positions for the management of the project implementation. These are the TORs for the project management office (PMO) personnel, the National Project Manager (NPM), Chief Technical Adviser (CTA), and the Project Steering Committee (PSC).

During the inception phase of the project, the PMO will prepare the TORs for the key personnel requirements for the implementation of the various activities in each of the components of the PSBEE Project. These are TORs for consultants/experts/specialists, whose services will be engaged in the implementation of the activities in each project component.

PROJECT MANAGEMENT OFFICE (PMO)

National Project Manager (NPM)

Duties and Responsibilities
The National Project Manager (NPM) will be accountable to both the National Project Director and to UNDP-China for the overall management of the PSBEE project for the quality, timeliness and effectiveness of the: (a) Services provided; (b) Activities carried out; and, (c) Use of project funds.

1. Quality, timeliness and effectiveness of the services provided
- Working directly with the NPD, and supervising the PMO for the implementation of the PSBEE project;
- Being the focal point for communications to the participating partners, national and regional organizations, and others concerning the implementation of the project;
- Providing a coordination and management structure for the implementation of the project, functioning in accordance with the rules and procedures of UNDP;
- Agreeing in advance with the project executive board on the exact progress reporting format and periodicity.

2. Quality, timeliness and effectiveness of the activities carried out
- Maintaining constructive relations with key project stakeholders, including GOC institutions, donors and project partners, informing them and consulting with them as appropriate regarding strategic project decisions, and, in discussion with the international and national technical experts/specialists, promoting the balanced participation of different Government entities in project activities and benefits;
- Providing support to UNDP in the preparation of annual reports, incorporating inputs provided by project contractors in relation to each of the project's components;
- Supporting contractors in the preparation of Annual Work Plans and Budgets (AWPBs) and review of the AWPBs prior to their presentation to the PSC for approval, in order to ensure their feasibility, relevance, correspondence with project resource availability and the harmonization of the activities proposed under each component; and,
- Working together with international and national technical advisors review and updating of project risk log and corresponding mitigations strategies.

3. Quality, timeliness and effectiveness of the use of funds
- Reviewing quarterly work plans, expenditure reports and disbursement requests prepared by contractors, and recommendation to UNDP regarding their approval or, where necessary, modification prior to approval; and

- Overseeing the work of the finance and administration officer.

4. Supervision, Teamwork and Administrative Support

The NPM will be supervised by the National Project Director (NPD). Administrative support will be supplied by the PSBEE Team, and where needed by UNDP (e.g. regarding contractual issues). The individual will also work closely with stakeholders from other relevant ministries/agencies, provincial and district representatives, and bilateral and multilateral development partners.

Required Skills and Experience

- Diploma (Master's degree preferable) in Engineering, Environment, Climate Change or other relevant discipline
- At least 5 years' experience with the management of complex projects and programs (UN experience a distinct advantage).
- Experience working with Government counterparts as well as representatives from development agencies.
- Experience in working in the energy sector is an advantage.
- High computer literate with experience of website design and maintenance and working with graphic design packages is an advantage.
- Fluency in spoken and written Mandarin and English.

Chief Technical Advisor

Duties and Responsibilities

1. Support the day-to-day management of the PSBEE project through provision of technical advice, including coordination across agencies and with other programs.
   - Provide close and regular technical backstopping to NPM and PMO, and other implementing entities;
   - Review of relevant policies and development of policy frameworks as well as the institutional arrangements for EC&EE/LC promotion and application in the public buildings sector;
   - Organization of and substantive contributions to a high level policy dialogue on EC&EE/LC promotion and application in the public buildings sector in China;
   - Support the NPM and PMO in coordinating and supervising activities of national and international experts /consultants to secure timely production of planned outputs and the review of these outputs;
   - Together with the NPD, coordinate and liaise with stakeholders – including line ministries, development partners, civil society, indigenous peoples and the private sector, the PSBEE project activities;
   - Ensure coordination with other EC&EE/LC-related initiatives in China;
   - Ensure communication between different government agencies, national and international initiatives and stakeholders.

2. Provide advice to Government counterparts and facilitate knowledge building activities among stakeholders.
   - Provide support and technical advice to the development of policy options and strategy for the design and implementation of the PSBEE project to the GOC (i.e., NGOA and LGOAs) and other development actors including support on the development of the various institutional instruments/systems for EC&EE/LC initiatives promotion and implementation such as:

    - Building Energy Audit System
    - Energy Management Information System (EMIS) and Public Buildings EMIS Database
    - Energy Savings Measurement and Verification System (ESMVS)
    - Other elements of national and local EC&EE/LC initiatives promotion and implementation as required
- Share knowledge on EC&EE/LC technologies by documenting lessons learnt and best practices from the PSBEE project and contributing to the development of knowledge based tools (including policies, strategies, guidelines, etc.);
- Provide timely quality information and technical advice to the NGOA, PSBEE PMO, pertinent GOC national and local government agencies and other partners to ensure effective delivery of the PSBEE project;
- Coordinate institutional capacity assessments of relevant national, provincial and local entities along the EC&EE/LC technologies supply chain; and,
- Provide guidance and assistance in the identification of technical expertise and lead on the preparation of TORs, identification and evaluation of experts, and reviewing project activity and monitoring reports produced.

Required Skills and Experience

- Postgraduate degree in Engineering, Environment, Climate Change or other relevant discipline, or similar;
- At least 7 years of working experience on energy sector specifically on EC&EE/LC technology applications in developing countries is preferable;
- Strong facilitation and coordination skills;
- Experience of working in multinational teams to implement programs involving different funding and implementing agencies;
- Experience of implementing programs across the UN agencies and across government are definite advantages;
- Experience in the development and implementation of EC&EE/LC technologies, especially in buildings (institutional, commercial and residential) at international, national and local levels;
- Knowledge on the UNFCCC negotiations, decisions and guidelines for national systems in the context of energy as well as national and international best practices within RE/EE and other initiatives; and,
- Proficiency in both spoken and written English

Project Administration and Finance Officer

Duties and Responsibilities
The overall objective of the assignment is to ensure efficient and effective implementation of the PSBEE project through the development of effective systems and the building of staff capacity. Specifically, the Officer's role will focus in on financial management and procurement using funds provided by government and development partners for administration by the PMO:

1. Provide support to Project Planning and Coordination
    - Work with the NPM and CTA of the PMO and UNDP program support and operations staff in the preparation of work plans, allocation of budgets and request for budgets;
    - Prepare documentation to ensure the flow of funds for project implementation occurs on a timely basis – this will require development of specific financial documentation and provision of support to exiting government staff in developing similar documentation;

- Participate in work planning and progress reporting meetings with the PSC under the guidance and supervision of the NPM or CTA; and,
- Coordinate financial management between participating agencies as well as coordination with other relevant development partner or government funds.

2. Develop, implement and improve Accounting and Reporting Procedures
   - Review and update Standard Operating Procedures for procurement, inventory and financial management
   - Ensure the establishment of an effective inventory system for all equipment purchased by the PMO
   - Provide oversight of financial management and procurement process processes to ensure they are conducted in accordance with partner guidelines
   - Ensure that petty cash transactions are effectively maintained. This includes writing of receipts, preparation of payment request form, receipt and disbursement of cash and clearance of advances;
   - Prepare project financial reports and submit to relevant officials for clearance;
   - Enter financial transactions into the computerized accounting system;
   - Reconcile all balance sheet accounts and keep a file of all completed reconciliation.

3. Ensure strong financial and operational control
   - Check and ensure all expenditures of project are in accordance with relevant procedures. This includes ensuring that receipts are obtained for all payments and that correct procurement procedures are followed;
   - Check budget lines to ensure that all transactions are correctly booked to the correct budget lines;
   - Ensure documentation relating to payments are duly approved;
   - Ensure Petty Cash is reviewed and updated and records are kept up-to-date;
   - To continuously improve system & procedures to enhance internal controls and satisfy audit requirements.

4. Conduct Bank reconciliation - Bank accounts should be reconciled and reported on a monthly basis

5. Develop Procurement processes and Inventory Register
   - Establish a procurement system and ensure all procurement is conducted according to relevant guidelines;
   - Establish and maintain a proper inventory of project assets register, including numbering, recording, and reporting;
   - Maintain the inventory file to support purchases of all equipment/assets.
   - Ensure program staff are able to maintain inventory of equipment and are updating and managing information on a timely basis

6. Support program administration and coordination
   - Provide oversight to the calculation and preparation of staff time records;
   - Provide assistance to organization of events, including workshops, seminars, and meetings
   - Identify potential opportunities for linkages and synergies between existing programs.

<u>Required Skills and Experience</u>

- Bachelor’s degree in Finance, Economics, Accounting, Administration or a related field.

- At least 7 years of experience in positions relating to providing administrative support and/or with financial and procurement systems, portfolio and administration preferably for foreign funded environmental projects.
- Excellent knowledge of the administrative, finance and procurement procedures of the Government, and development agencies.
- Experience working with Government agencies to build their capacity and support them in the development of nationally implemented programs.
- Experience working in a multinational multi-skilled team and to coordinating activities with other partners to achieve the best long-term results for the program and the Ministries and Agencies with whom they are working.
- Experience in the usage of computers and office software packages for data analysis and technical reports; ability and willingness to work in remote areas.
- Fluency in spoken and written Mandarin and English.

NATIONAL PROJECT DIRECTOR (NPD)

The NPD will be responsible for daily oversight of the project management office (PMO).

Tasks
- Guidance to the PMO team in implementation
- Handling of financial requests and review of financial reports
- Technical coordination in project implementation with NGOA and other stakeholders
- Coordination with the project's provincial partners (e.g., LGOA offices)
- Participation in meetings of the UN/UNDP coordinating mechanisms
- Reporting to the Project Steering Committee on project progress
- Promoting the project to high level officials to gain their buy-in
- Representation of the project at important meetings

Qualifications
- Senior official of NGOA
- Experience in management of development projects
- Strong knowledge on climate change and energy sector
- Knowledge of financial management
- High level of enthusiasm and support for EC&EE/LC technologies applications

PROJECT STEERING COMMITTEE (PSC)

The project board will have responsibility for monitoring of the project at a high level and for providing high-level support and decision-making as needed.

Tasks
- Meeting twice annually, for a total of eight times over the project's four-year lifetime
- High-level monitoring of project progress and, in particular, reviewing of outcome-level and objective-level progress of the project
- Decision-making with regard to major issues facing project that cannot be resolved at the working level
- Provision of high-level support to push progress in certain areas in which such support can make a difference, such as in policy-making
- Holding of end of project review to capture lessons learned; discuss opportunities for scaling up and highlighting of project results; and discuss findings of terminal evaluation

General Qualifications of PSC Members

- Roles as senior level officials and managers within government and other organizations
- Expertise in areas relevant to project, such as energy and energy management, building management, provincial development planning, etc.

Annex F. UNDP Social and Environmental and Social Screening Template (SESP)

Project Information

Project Information	
1. Project Title	**Energy Efficiency Improvement in Public Sector Buildings in China (PSBEE)**
2. Project Number	**PIMS 5395**
3. Location (Global/Region/Country)	**China**

Part A. Integrating Overarching Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Overarching Principles in order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the Project mainstreams the human-rights based approach
While the project has no specific activities on human rights promotion, human-rights principles were integrated into the design of the project. Facilitation of prior informed consent of all beneficiaries, i.e., the public buildings occupants/tenants, private sector, and state institutions was done through the stakeholder consultation processes that promoted gender inclusive participatory approaches involving men, women and youth. During the design phase, the project team met with public building managers, building practitioners in the LGOA regions, as well as those in the buildings industry to discuss proposed EC&EE/LC demonstrations to assess their interest in such initiatives emphasizing the importance of each key players in the successful promotion of the widespread application of EC&EE/LC technologies. A follow-up feasibility study for each of the demonstration sites will also take into account socio-cultural aspects that need to be considered. This will inform implementation of project activities and where appropriate fine-tuning of plans would be necessary. Doing these will in the end contribute to human social and environmental well-being in the public buildings sector of China.
Briefly describe in the space below how the Project is likely to improve gender equality and women's empowerment
The PSBEE Project will impact gender equality and women's empowerment in a positive way since this presents opportunities for the involvement of women working in management and technical units of public sector institutions. To facilitate this, the project design has taken into account principles of gender equality and women's empowerment. Part of the project design strategy is the involvement of women working in both management and technical departments of the NGOA, as well as those in the LGOAs who can play important roles in the design, development and implementation of the project. It is important to ensure that benefits from the enabling conditions that facilitate EC&EE/LC initiatives are enjoyed by both women and men. The activities are designed in such a way to take into account the opportunity for the country to further enhance the role of women in deployment of low carbon technologies and mitigation options, and come up with gender-sensitive policies in the public buildings sector, recognizing the possible contributions of women in the management and implementation of climate change mitigation measures. The implementation of the project's capacity development activities will target high rates of participation from women.
Briefly describe in the space below how the Project mainstreams environmental sustainability
The PSBEE Project will contribute not only to the achievement of China's climate change mitigation targets but is also within the context of the country's sustainable development. In that regard, it will bring about local benefits mainly through contributions to not only the improvement of the specific energy consumption of public buildings, but will also contribute to the protection of the natural environment. The global environmental benefits from the project will mainly come from GHG emission reductions from fossil fuel consumption through the improvement of the energy utilization efficiency, and where feasible displacement of fossil fuel usage with available RE resources, in public buildings. These will be facilitated by the barrier removal approach that this proposed project will employ. Environmental sustainability will also be assured through the synergistic aspect of the integrated way the key stakeholders will be working together, and the higher chances of scaling-up/replication of whatever low carbon development techniques/practices will

be introduced, demonstrated, and promoted under the project. Lastly, since the project is linked and is complementing and supplementing the national development strategy and plans of the country, the sustainability of project outputs can be ensured.

Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Describe briefly potential social and environmental risks identified in Attachment 1 – Risk Screening Checklist (based on any "Yes" responses). If no risks have been identified in Attachment 1 then note "No Risks Identified" and skip to Question 4 and Select "Low Risk". Questions 5 and 6 not required for Low Risk Projects.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5 below before proceeding to Question 6*			**QUESTION 6: What social and environmental assessment and management measures have been conducted and/or are required to address potential risks (for Risks with Moderate and High Significance)?**
Risk Description	***Impact and Probability (1-5)***	***Significance (Low, Moderate, High)***	***Comments***	***Description of assessment and management measures as reflected in the Project design. If ESIA or SESA is required note that the assessment should consider all potential impacts and risks.***
Based on the results of the Social and Environmental Risk screening, there are no clear direct impacts and risks. There may be some but most likely limited in scale and perhaps with a reasonable degree of certainty can be avoided through compliance with the appropriate regulations; or managed through application of standard best practice. There may be possible negative environmental and social impacts, and/or risks associated with the project (or specific project component), but these are predominantly indirect. For example, EE retrofits that will be done in the demos may result in negative impacts if the disposal or the management of the handling of replaced non-EE items are done improperly.	1	Low	The proposed project is intended to remove barriers to the widespread practice of EC&EE/LC technologies in public buildings in China, to enhance the efficiency of energy utilization of these buildings. Obviously, these will be done with the aim of avoiding negative environmental and social impacts to the areas where these public buildings are located. There will be demonstration activities that will be carried out under this project. These demos will be designed in line with the relevant building standards	The environmental and social concerns that may arise from the project are actually manageable and easily controlled during the project implementation. During the final design, engineering and installation of the demos/pilots of this UNDP-GEF project, the existing environmental impact assessments during the project design will be revisited to ensure that these will not only bring about positive impacts, but also will be designed in such a way that the emergence of potential negative impacts are brought to the minimum, if not completely avoided. The potential downstream impacts will also be assessed as to the likelihood of these happening, and also determining the factors that would contribute to them from happening. The project design (particularly during the logical framework analysis) also took into consideration such factors and where possible and applicable come up with the relevant activities that will adequately address them.

			(construction and operation) and environmental requirements, and that there will be no adverse negative impacts and consequences in their implementation and operation.	
Project pose potential risks to community health and safety due to the transport, storage, and use and/or disposal of hazardous or dangerous materials.	1	Low	The demos will involve the replacement of existing building materials, appliances and devices used in the demo buildings. The design of the demos shall include facilitating the proper disposal of such items as required by law.	To prevent potential downstream impacts, possible issues concerning the proper disposal or recycling of existing building materials and devices/appliances will be addressed. For example, the use of LED lamps to replace CFLs and FLs should take into account the proper disposal of these replaced items. Same with air conditioning units that will be replaced – the proper disposal of these items (particularly the refrigerants) have to be considered. Replaced building materials (e.g., insulations) should also be addressed. The capacity building on the application of new EC&EE/LC techniques and practices in public buildings shall also include the proper (i.e., safe and environment-friendly) handling and disposal of waste and recyclable materials. The demonstrations that will be featured in this project shall be designed and implemented taking into consideration the need to reduce environmental impacts in the application of EC&EE/LC technologies in public buildings.
Project potentially result in the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or trans-boundary impacts.	1	Low	The design of the demos shall include facilitating the proper disposal of replaced non-EE in public buildings as required by law.	This rather indirect negative impact shall be avoided through inclusion of proper waste handling and disposal in the EC&EE/LC capacity building activities of the project. The demos that will be featured in this project shall be designed and implemented taking into consideration the need to reduce environmental impacts in the application of EC&EE/LC technologies in public buildings.
	QUESTION 4: What is the overall Project risk categorization?			
	Select one (see SESP for guidance)			**Comments**
	Low Risk	√		Based on the results of the Social and Environmental Risk screening, there are no clear direct impacts and risks. Only indirect impacts that can be avoided by strictly complying with the relevant regulations; or managed through application of standard best practice.
	Moderate Risk	☐		
	High Risk	☐		
	QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are			

	relevant?		
	Check all that apply		**Comments**
	Principle 1: Human Rights	☐	
	Principle 2: Gender Equality and Women's Empowerment	☐	
	1. Biodiversity Conservation and Natural Resource Management	☐	
	2. Climate Change Mitigation and Adaptation	☐	
	3. Community Health, Safety and Working Conditions	☑ ☐	Some indirect negative impacts that can be avoided with proper regulatory compliance, or addressed with proper application of standard best practice.
	4. Cultural Heritage	☐	
	5. Displacement and Resettlement	☐	
	6. Indigenous Peoples	☐	
	7. Pollution Prevention and Resource Efficiency	☑ ☐	Some indirect negative impacts that can be avoided with proper regulatory compliance, or addressed with proper application of standard best practice.

Final Sign Off

Signature	*Date*	*Description*
QA Assessor		UNDP staff member responsible for the Project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver		UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD), Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair		UNDP chair of the PAC. In some cases PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

SESP Attachment 1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks	
Principles 1: Human Rights	**Answer (Yes/No)**
1. Could the Project lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	No
2. Is there a likelihood that the Project would have inequitable or discriminatory adverse impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups? [36]	No
3. Could the Project potentially restrict availability, quality of and access to resources or basic services, in particular to marginalized individuals or groups?	No
4. Is there a likelihood that the Project would exclude any potentially affected stakeholders, in particular marginalized groups, from fully participating in decisions that may affect them?	No
5. Is there a risk that duty-bearers do not have the capacity to meet their obligations in the Project?	No
6. Is there a risk that rights-holders do not have the capacity to claim their rights?	No
7. Have local communities or individuals, given the opportunity, raised human rights concerns regarding the Project during the stakeholder engagement process?	No
8. Is there a risk that the Project would exacerbate conflicts among and/or the risk of violence to project-affected communities and individuals?	No
Principle 2: Gender Equality and Women's Empowerment	
1. Is there a likelihood that the proposed Project would have adverse impacts on gender equality and/or the situation of women and girls?	No
2. Would the Project potentially reproduce discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	No
3. Have women's groups/leaders raised gender equality concerns regarding the Project during the stakeholder engagement process and has this been included in the overall Project proposal and in the risk assessment?	No
4. Would the Project potentially limit women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? *For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being*	No
Principle 3: Environmental Sustainability: Screening questions regarding environmental risks are encompassed by the specific Standard-related questions below	
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management	
1.1 Would the Project potentially cause adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	No
1.2 Are any Project activities proposed within or adjacent to critical habitats and/or environmentally sensitive areas, including legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	No
1.3 Does the Project involve changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	No
1.4 Would Project activities pose risks to endangered species?	No
1.5 Would the Project pose a risk of introducing invasive alien species?	No
1.6 Does the Project involve harvesting of natural forests, plantation development, or reforestation?	No
1.7 Does the Project involve the production and/or harvesting of fish populations or other aquatic species?	No
1.8 Does the Project involve significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	No
1.9 Does the Project involve utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)	No
1.10 Would the Project generate potential adverse trans-boundary or global environmental concerns?	No
1.11 Would the Project result in secondary or consequential development activities which could lead to adverse social and environmental effects, or would it generate cumulative impacts with other known existing or planned activities in the area? *For example, a new road through forested lands will generate direct environmental and social impacts (e.g.*	No

[36] Prohibited grounds of discrimination include race, ethnicity, gender, age, language, disability, sexual orientation, religion, political or other opinion, national or social or geographical origin, property, birth or other status including as an indigenous person or as a member of a minority. References to "women and men" or similar is understood to include women and men, boys and girls, and other groups discriminated against based on their gender identities, such as transgender people and transsexuals.

	felling of trees, earthworks, potential relocation of inhabitants). The new road may also facilitate encroachment on lands by illegal settlers or generate unplanned commercial development along the route, potentially in sensitive areas. These are indirect, secondary, or induced impacts that need to be considered. Also, if similar developments in the same forested area are planned, then cumulative impacts of multiple activities (even if not part of the same Project) need to be considered.	
Standard 2: Climate Change Mitigation and Adaptation		
2.1	Will the proposed Project result in significant[37] greenhouse gas emissions or may exacerbate climate change?	No
2.2	Would the potential outcomes of the Project be sensitive or vulnerable to potential impacts of climate change?	No
2.3	Is the proposed Project likely to directly or indirectly increase social and environmental vulnerability to climate change now or in the future (also known as maladaptive practices)? *For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding*	No
Standard 3: Community Health, Safety and Working Conditions		
3.1	Would elements of Project construction, operation, or decommissioning pose potential safety risks to local communities?	No
3.2	Would the Project pose potential risks to community health and safety due to the transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	Yes - *In cases of improper disposal replaced non-EE items*
3.3	Does the Project involve large-scale infrastructure development (e.g. dams, roads, buildings)?	No
3.4	Would failure of structural elements of the Project pose risks to communities? (e.g. collapse of buildings or infrastructure)	No
3.5	Would the proposed Project be susceptible to or lead to increased vulnerability to earthquakes, subsidence, landslides, and erosion, flooding or extreme climatic conditions?	No
3.6	Would the Project result in potential increased health risks (e.g. from water-borne or other vector-borne diseases or communicable infections such as HIV/AIDS)?	No
3.7	Does the Project pose potential risks and vulnerabilities related to occupational health and safety due to physical, chemical, biological, and radiological hazards during Project construction, operation, or decommissioning?	No
3.8	Does the Project involve support for employment or livelihoods that may fail to comply with national and international labor standards (i.e. principles and standards of ILO fundamental conventions)?	No
3.9	Does the Project engage security personnel that may pose a potential risk to health and safety of communities and/or individuals (e.g. due to a lack of adequate training or accountability)?	No
Standard 4: Cultural Heritage		
4.1	Will the proposed Project result in interventions that would potentially adversely impact sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: Projects intended to protect and conserve Cultural Heritage may also have inadvertent adverse impacts)	No
4.2	Does the Project propose utilizing tangible and/or intangible forms of cultural heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement		
5.1	Would the Project potentially involve temporary or permanent and full or partial physical displacement?	No
5.2	Would the Project possibly result in economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	No
5.3	Is there a risk that the Project would lead to forced evictions?[38]	No
5.4	Would the proposed Project possibly affect land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	No
Standard 6: Indigenous Peoples		
6.1	Are indigenous peoples present in the Project area (including Project area of influence)?	No
6.2	Is it likely that the Project or portions of the Project will be located on lands and territories claimed by indigenous peoples?	No
6.3	Would the proposed Project potentially affect the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the Project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in	No

[37] In regards to CO_2, 'significant emissions' corresponds generally to more than 25,000 tons per year (from both direct and indirect sources). [The Guidance Note on Climate Change Mitigation and Adaptation provides additional information on GHG emissions.]

[38] Forced evictions include acts and/or omissions involving the coerced or involuntary displacement of individuals, groups, or communities from homes and/or lands and common property resources that were occupied or depended upon, thus eliminating the ability of an individual, group, or community to reside or work in a particular dwelling, residence, or location without the provision of, and access to, appropriate forms of legal or other protections.

	question)? *If the answer to the screening question 6.3 is "yes" the potential risk impacts are considered potentially severe and/or critical and the Project would be categorized as either Moderate or High Risk.*	
6.4	Has there been an absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	No
6.5	Does the proposed Project involve the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No
6.6	Is there a potential for forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources?	No
6.7	Would the Project adversely affect the development priorities of indigenous peoples as defined by them?	No
6.8	Would the Project potentially affect the physical and cultural survival of indigenous peoples?	No
6.9	Would the Project potentially affect the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices?	No
Standard 7: Pollution Prevention and Resource Efficiency		
7.1	Would the Project potentially result in the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or trans-boundary impacts?	Yes - *In cases of improper disposal replaced non-EE items*
7.2	Would the proposed Project potentially result in the generation of waste (both hazardous and non-hazardous)?	No
7.3	Will the proposed Project potentially involve the manufacture, trade, release, and/or use of hazardous chemicals and/or materials? Does the Project propose use of chemicals or materials subject to international bans or phase-outs? *For example, DDT, PCBs and other chemicals listed in international conventions such as the Stockholm Conventions on Persistent Organic Pollutants or the Montreal Protocol*	No
7.4	Will the proposed Project involve the application of pesticides that may have a negative effect on the environment or human health?	No
7.5	Does the Project include activities that require significant consumption of raw materials, energy, and/or water?	No

Annex G: UNDP Project Quality Assurance Report

Submitted as a separate file.

Annex H. UNDP Risk Log

OFFLINE RISK LOG

Project Title: Energy Efficiency Improvement in Public Sector Buildings in China	**Project ID:**	**Date:**

#	Description	Date Identified	Type	Impact & Probability	Countermeasures / Management Response	Owner	Submitted /updated by	Last Update	Status (stable/ improving)
1	Municipality, city & county governments in LGOA regions may not support the project implementation promptly and sufficiently	Project Formulation Phase (2016)	Political	Lack of political leadership and commitment will make it hard for the project to get direction and support for policy, legal, and institutional framework and for budgetary allocation for investment in EC&EE/LC applications in public buildings. Probability = 2 Impact = 2 Risk = 2	NGOA and LGOAs will be supported with government-wide stakeholder engagement and outreach program to create awareness and advocate at the highest political and decision-making levels within and among the local government institutions	UNDP-China, NGOA	UNDP-GEF STA	January 2017	Improving
2	Inadequate communication and cooperation between national and local government institutions to effectively plan and implement EC&EE/LC measures in the public buildings sector.	Project Formulation Phase (2016)	Organizational	Lack of proper coordination will create confusion over the responsibilities of each institutions that are key players of the project, thus negatively affecting efforts and investments in EC&EE/LC technologies in the public sector	The project will support NGOA in conducting stakeholder mapping through its leading of the Low Carbon Growth Technical Working Group, which is comprised of development partners and public sector, private sector, and CSO stakeholders. It will further support NGOA in strengthening the existing coordination mechanism within the LGOA	UNDP-China, NGOA	UNDP-GEF STA	January 2017	Improving

#	Description	Date Identified	Type	Impact & Probability	Countermeasures / Management Response	Owner	Submitted /updated by	Last Update	Status (stable/ improving)
				Probability = 3 Impact = 3 Risk = 3	regions.				
3	The committed level of co-financing for specific activities of the project may not become fully available in time.	Project Formulation Phase (2016)	Financial	Lack of ongoing budget allocations would result in ad hoc budgetary allocation, which would limit sustainable financing of renewable energy and energy efficiency initiatives. Probability = 3 Impact = 3 Risk = 3	The project team shall secure central government funding prior to project launching. During project implementation, the project team will closely monitor and ensure the timely availability of co-financing from project partners and co-financers. In case of co-financing shortfall for a particular activity, the NGOA will rely on the support of either the GOC or the local governments.	UNDP-China, NGOA	UNDP-GEF STA	January 2017	Improving
4	Technical capacity on EC&EE/LC in the public buildings sector is inadequate	Project Formulation Phase (2016)	Organizational	Limited pool of national experts will affect technical oversight for and operation and maintenance of EC&EE/LC investments. Probability = 2 Impact = 2 Risk = 2	The project will undertake capacity assessment and target capacity development in NGOA and LGOA regions on EC&EE/LC implementation and O&M. This is expected to generate interest and mobilize support and partnership to promote cost-effective investment in technologies and human resources.	UNDP-China, NGOA	UNDP-GEF STA	January 2017	Improving
5	Low level of participation of the management and personnel of public sector buildings	Project Formulation Phase (2016)	Operational	The main impact will be delays in the scheduled implementation of project activities and could lead also to complications in the schedules of the co-financed activities.	As the national energy management administrator of the public sector, NGOA has enough management and coordination capabilities for ensuring the participation and cooperation of the administrators/managers of	UNDP-China, NGOA	UNDP-GEF STA	January 2017	Improving

#	Description	Date Identified	Type	Impact & Probability	Countermeasures / Management Response	Owner	Submitted /updated by	Last Update	Status (stable/ improving)
				Probability = 3 Impact = 3 Risk = 3	public sector buildings to participate in this project. Fast track recruitment arrangements will be employed to hire the relevant project staff.				
6	Relevant national/local government agencies' failure to approve and enforce formulated policies and regulations	Project Formulation Phase (2016)	Regulatory	The project seeks to remove policy barriers to EC&EE/LC applications in public buildings. But without the enforcement of the proposed policies, not only will the barriers be really removed but the resulting institutional competition is likely to affect leadership and ownership of project results and negatively impact long-term sustainability Probability = 3 Impact = 3 Risk = 3	Policy change through planned intensive communication and advocacy activities will be pursued. The formulated and approved policies and regulations that will be delivered by the project will be enforced by the NGOA. Policy makers will be adequately briefed on the cost advantages of EC&EE/LC and showing them proof of concept with the project demos to ensure they will support EC&EE/LC applications in the public buildings sector.	UNDP-China, NGOA	UNDP-GEF STA	January 2017	Improving
7	Sustaining the outcomes and benefits of GEF investment on the activities implemented will not be fully sustained.	Project Formulation Phase (2016)	Financial	All the enabling frameworks and conditions that the project will establish will all be for naught if these will be not institutionalized and discontinued after the end of project	Sustainable follow-up plan will be developed as part of the project activities (to be supported by the local governments and LGOAs) for the replication of the demonstrated applicable and feasible EC&EE technologies in the public sector buildings in other	UNDP-China, NGOA	UNDP-GEF STA	January 2017	Improving

#	Description	Date Identified	Type	Impact & Probability	Countermeasures / Management Response	Owner	Submitted /updated by	Last Update	Status (stable/ improving)
				Probability = 2 Impact = 2 Risk =2	provinces.				

Annex I: Results of the capacity assessment

Submitted as a separate file.

Summary of the Objective and Risk Rating Profile for China National Government Office Administration (NGOA)

The objective of the micro assessment of the Implementing Partners (IP) is to assess the IP's financial management capacity (i.e. accounting, procurement, reporting, internal controls, etc.) to determine the overall risk rating and assurance activities. The grading system assigned a number to each control area of 1 to 4 where 1 was lower risk and 4 being higher risk. The overall assessment rating for the China National Government Office Administration (NGOA) (hereinafter referred to as NGOA) was 1 with lowest risk. Below is a summary of the issues that were noted from the conduct of the micro assessment. These issues are for information and it is important to note that they do not affect the overall assessment rating.

Summary of risk areas identified and recommendations. The following paragraphs outline the key recommendations resulting from the key risks identified from the micro assessment.

i. Pending Legal Cases - There are currently no litigation against NGOA. The existing UNDP or future UNDP funded projects will not be affected.
ii. Key Finance Position – The Principal Accountant and Principal Administration Officer is overseeing both the Administration and the Finance team.
iii. Senior Finance Officer - The Senior Accountant as per selection criteria in the job description is required to hold a degree in commerce. Currently the Senior Accountant only holds a Certificate in Commerce as per her CV and is familiar with the roles and duties for the position and we recommend that the Senior Accountant continue to pursue further undergraduate qualifications.
iv. There is no retaliation policy for those who report fraud - There is also no retaliation policy to give guidance and outline protection mechanisms to all parties when fraud, waste or misuse of agency resources or property is reported.

HACT Micro Assessment of China NGOA

This will be provided during the Project Inception. This is work in progress.

Annex J. Additional Agreements

国家机关事务管理局公共机构节能管理司

"Energy Efficiency Improvement in Public Sector Buildings in China" Project

Co-financing Letter

***United Nations Development Programme*:**

This is to confirm that the National Government Office Administration (NGOA), the Jilin Provincial Government, the Jiangxi Provincial Government, the Gansu Provincial Government and relevant partner companies are committing co-financing support amounting to US$ 70,000,000 to the implementation of the "Energy Efficiency Improvement in Public Sector Buildings in China" Project.

Co-finance funds amounting to US$ 30,810,000 will be dispatched from the Central Government's budget for the project implementation. From this funds, US$ 25,810,000 cash is earmarked to support research on relevant systems and standards, promotion and application of new energy-efficient technologies and products, providing of publicities and trainings for public institutions, and for demonstration project management expenses. The rest of the funds amounting to US$ 5,000,000 is for in-kind to support to project management, monitoring, evaluation and project coordination.

For the co-financing provided by the partner local governments, Jilin Provincial Government, Jiangxi Provincial Government and Gansu Provincial Government are committing cash co-financing in the amounts US$ 11,700,000, US$ 5,700,000 and US$ 5,790,000, respectively, to support research on local energy-efficiency systems and standards, promotion and hosting of the demonstrations on the application of new energy-efficient technologies and products in public buildings, promotion of market-based mechanisms and establishment of evaluation mechanisms.

In addition, the partner private sector companies will collectively provide cash co-financing of US$ 16,000,000 to support promotion and demonstration on the application of new energy-efficient technologies and products in public buildings.

We anticipate the approval of the project by GEF and we are willing to cooperate with the project partners and utilize these funds over a period of 4 years to implement the "Energy Efficiency Improvement in Public Sector Buildings in China" Project. The support from the NGOA, on behalf of the Chinese Government, to the project, is described in detail in the Project Document and the associated budget. We consider that all of the direct costs that will be paid for using these allocated funds for producing the pertinent project outputs as described in the Project Document qualify as co-financing.

Sincerely Yours,



Department of Energy Conservation for Public Institutions
National Government Offices Administration
December 14, 2016

United Nations Development Programme
联合国开发计划署

Empowered lives.
Resilient nations.

16th, Janurary 2017

Dear Dr. Ishii,

Subject: Proposed Co-financing Commitment Letter from UNDP for GEF project "China – Energy Efficiency Improvement in Public Sector Buildings"

Through the use of PPG fund, UNDP China developed a GEF project on "China – Energy Efficiency Improvement in Public Sector Buildings". The project's aim is to widespread application of energy conservation in public sector buildings of China.

This proposed GEF Project also closely corresponds to the on-going United Nations Development Assistance Framework (UNDAF) 2011-2015 as expressed in the Country's UNDAF document signed by both UNDP and the Government of China in March, 2010. This document provides details of UNDP assistance for "Government and other stakeholders ensure environmental sustainability, address climate change, and promote a green, low carbon economy." UNDP expects to achieve this important outcome through the following relevant outputs:

- Establishment and implementation of public sector energy performance monitoring and evaluation systems.
- EC&EE project financing mechanisms, improving the capacity and awareness of the public sector, and the provision of public information services on building EC&EE and low carbon technologies.

To strengthen our assistance towards achievement of these development objectives, we confirm that the UNDP CO will contribute USD 100,000 co-financing in grant to the project. We are looking forward to the timely approval of this GEF project.

Yours sincerely

Agi Veres
Country Director
UNDP, China

Dr. Naoko Ishii
CEO, Global Environment Facility
Washington DC
U.S.A.

Copy to: Ms. Adriana Dinu, UNDP-GEF Executive Coordinator

2 Liangmahe Nanlu, Beijing, China, 100600, Tel: 86-10-85320800 Fax: 86-10-85320900 www.undp.org.cn
中国北京亮马河南路二号 邮编：100600

STANDARD LETTER OF AGREEMENT BETWEEN UNDP AND THE GOVERNMENT FOR THE PROVISION OF SUPPORT SERVICES

Dear Mr. Zhang Shiliang

Director General of the Department of Energy Conservation for Public Institutions, National Government Offices Administration

1.Reference is made to consultations between officials of the National Government Offices Administration (hereinafter referred to as "the NGOA") and officials of UNDP with respect to the provision of support services by the UNDP country office for nationally managed programmes and projects. UNDP and NGOA hereby agree that the UNDP country office may provide such support services at the request of NGOA through its institution designated in the relevant programme support document or project document, as described below.

2.The UNDP country office may provide support services for assistance with reporting requirements and direct payment. In providing such support services, the UNDP country office shall ensure that the capacity of the NGOA-designated institution is strengthened to enable it to carry out such activities directly. The costs incurred by the UNDP country office in providing such support services shall be recovered from the administrative budget of the office.

3.The UNDP country office may provide, at the request of the designated institution, the following support services for the activities of the programme/project:
(a)Identification and/or recruitment of project and programme personnel;
(b)Identification and facilitation of training activities;
(c)Procurement of goods and services;

4.The procurement of goods and services and the recruitment of project and programme personnel by the UNDP country office shall be in accordance with the UNDP regulations, rules, policies and procedures. Support services described in paragraph 3 above shall be detailed in an annex to the programme support document or project document, in the form provided in the Attachment hereto. If the requirements for support services by the country office change during the life of a programme or project, the annex to the programme support document or project document is revised with the mutual agreement of the UNDP resident representative and the designated institution.

5.The relevant provisions of the Standard Basic Assistance Agreement between the Government of China and the United Nations Development Programme in China signed on January 29 1979 (the "SBAA"), including the provisions on liability and privileges and immunities, shall apply to the provision of such support services. The Government shall retain overall responsibility for the nationally managed programme or project through its designated institution. The responsibility of the UNDP country office for the provision of the support services described herein shall be

limited to the provision of such support services detailed in the annex to the programme support document or project document.

6. Any claim or dispute arising under or in connection with the provision of support services by the UNDP country office in accordance with this letter shall be handled pursuant to the relevant provisions of the SBAA.

7. The manner and method of cost-recovery by the UNDP country office in providing the support services described in paragraph 3 above shall be specified in the annex to the programme support document or project document.

8. The UNDP country office shall submit progress reports on the support services provided and shall report on the costs reimbursed in providing such services, as may be required.

9. Any modification of the present arrangements shall be effected by mutual written agreement of the parties hereto.

10. If you are in agreement with the provisions set forth above, please sign and return to this office two signed copies of this letter. Upon your signature, this letter shall constitute an agreement between your Government and UNDP on the terms and conditions for the provision of support services by the UNDP country office for nationally managed programmes and projects.

Yours sincerely,

Signed on behalf of UNDP
Mr. Patrick Haverman
Deputy Country Director

Date: 2017.2.22

For the Government
Mr. Zhang Shiliang

Director General of the Department of Energy Conservation for Public Institutions, National Government Offices Administration

Date: 2017.1.19

Attachment

DESCRIPTION OF UNDP COUNTRY OFFICE SUPPORT SERVICES

1.Reference is made to consultations between NGOA, the institution designated by the Government of China and officials of UNDP with respect to the provision of support services by the UNDP country office for the nationally managed programme: Energy Efficiency Improvement in Public Sector Buildings in China Between the Government of the People's Republic of China and United Nations Development Programme to enhance the production and widespread application of energy conservation.

2.In accordance with the provisions of the letter of agreement and the programme project document, the UNDP country office shall provide support services for the Programme as described below.

3.Support services to be provided:

Support services (insert description)	Schedule for the provision of the support services	Cost to UNDP of providing such support services (where appropriate)	Amount and method of reimbursement of UNDP (where appropriate)
1.Recruiting 4 specialists	As per AWPs	As Per UPL	Estimated amount: US$ 5,265.60

4. Description of functions and responsibilities of the parties involved is as per the project document. UNDP country office will provide the services as stated above upon the request of the National Government Offices Administration. The reimbursement of the UNDP support cost will be recorded as per transactions based on the established UNDP financial regulations and rules.

Annex K: Description of Demonstrations

A major activity under Component 3 of the PSBEE Project is the design and implementation of EC&EE/LC technology application projects. Based on the stakeholder discussions during the PPG phase, it was agreed that overall, there will be 20 demonstrations. A preliminary line-up of these demonstrations was prepared during the PPG phase.

For the EC&EE/LC technology application demonstrations in public buildings, 10 demos will be implemented - Provincial Government (PG) buildings: 3 office buildings, 3 university buildings, 1 hospital building; and County Government (CG) buildings: 1 university building and 2 hospital buildings. For the EC&EE/LC demonstration projects financed through market-based financing scheme in public buildings in selected public buildings, there will be 10 demos – National Government (NG) building: 1 office building; Provincial Government (PG) buildings: 2 office buildings, 2 university buildings, and 1 hospital building; Municipal Government (MG) buildings: 1 office building and 1 hospital building; and Country Government (CG) buildings: 1 office building, and 1 hospital building. The following tables provide details of the preliminarily identified and evaluated EC&EE/LC technology application demonstrations.

Table 1 presents the initially identified and evaluated demonstrations that will showcase the design, planning, engineering, installation and operation of EC&EE/LC technology application projects in different types of public buildings in China. These are initially planned projects by the demonstration hosts but are enhanced/modified with features that improves the energy performance and resulting GHG emission reduction.

Table 2 shows the initially identified and evaluated demonstrations that will either be financed through the market-based financial scheme(s) that will be developed under the PSBEE Project, or financed by the demo project owners/implementers themselves. These demos are meant not only to showcase the design, planning, engineering, installation and operation of EC&EE/LC technology application projects in different types of public buildings in China, but also how these projects will be financed, apart from the usual government-provided budget.

The pre-feasibility analyses that were done for the abovementioned demonstrations are based on the following assumptions:

Energy Prices: Grid electricity @ US$ 70/MWh; Diesel Fuel Oil @ US$ 0.89/lit
Economics: Discount Rate @ 11.6% (WACC range in China Construction Sector is 8.9% – 14.6%) http://www.waccexpert.com/; Income Tax @ 25% (Standard corporate income tax rate in China); Straight line depreciation.

These demonstrations are the bases for estimating the GHG emission reductions that will be attributable to the PSBEE Project. Please refer to Annex K.

Table 1: List of EC&EE/LC Technology Application Demonstrations

Demo No.	Technology Application	Demo Public Building Type	Location	Description	Investment, US$	Annual Energy Savings, MWh	Annual GHG Emission Reduction, tCO2	Annual Energy Cost Savings, US$	Remarks
EC&EE/LC Technology Application Demonstrations									
1	Solar PV power generation system	PG Office Building	Gansu Province	Installation of solar PV power generation system – integrated in existing electrical system (360 kWe)	400,000	725.3	760.9	50,773	Investment cost @ US$ 1100/kW; grid electricity savings. Annual cost savings = US$ 50,773 @ US$ 70/MWh electricity cost. NPV = US$ 14,123; IRR = 12.2%
2	Solar hot water system	PG University Building	Jiangxi Province	Installation of solar water heating system – integrated in the existing hot water system (500 kWth)	250,000	1,007.4	1,056.8	82,783	Investment cost @ US$ 500/kW; diesel oil savings. Annual energy savings = US$ 82,783 @ US$ 0.89/lit DFO price; NPV = US$ 245,366; IRR = 27.1%
3	Combined heat and power (CHP) system	PG University Building	Xinjiang Province	Installation of CHP system serving the heating and power requirements of specific campus facilities (1,200 kW)	1,506,000	7,942.4	8,331.6	555,968	Investment cost @ US$ 1255/kW; grid electricity savings from increased in system efficiency from 45% to 75%. Annual cost savings = US$ 312,353 @ US$ 70/MWh electricity cost. NPV = US$ 915,266; IRR = 15.1%
4	CHP with liquid absorption refrigeration (LAR) system	PG Hospital Building	Henan Province	Installation of CHP system serving the cooling and power requirements of specific hospital facilities (1,200 kW)	2,640,000	27,438.5	28,783.0	1,920,696	Investment cost @ US$ 2200/kW; grid electricity savings from increase in system efficiency from 55% to 67%. Annual cost savings = US$ 2,495,868 @ US$ 70/MWh electricity cost. NPV = US$ 3,508,266; IRR = 26.6%
5	Building energy management system	PG University Building	Shandong Province	Energy management system for specific campus facilities (40,000 m2 floor area)	35,000	400.0	419.6	28,000	Investment cost includes BEMS unit for 200 measuring points and pertinent software. Includes 3 years annual management cost. Assumes SEC = 200 kWh/m2/yr and 5% savings in annual energy savings. Annual cost savings = US$ 28,000 @ US$ 70/MWh electricity cost. NPV = US$ 115,398; IRR = 70.9%
6	Building energy management system	PG Office Building	Jiangsu Province	Energy management system for specific office facilities (25,000 m2 floor area)	20,000	225.0	236.0	15,750	Investment cost includes BEMS unit for 100 measuring points and pertinent software. Includes 3 years annual management cost. Assumes SEC = 180 kWh/m2/yr and 5% savings in annual energy savings. Annual cost savings = US$ 15,750 @ US$ 70/MWh electricity cost. NPV = US$ 64,784; IRR = 70.0%
7	Boiler system improvements	CG Hospital Building	Jiangxi Province	Improved boiler controls; improved hydronic system balancing; waste heat recovery (2000 kW)	2,000,000	83,395.2	87,481.6	6,853,000	Investment cost @ US$ 1000/kW; diesel oil savings from improved boiler efficiency (80% to 85%). Annual energy savings = US$ 6.853 million @ US$ 0.89/lit DFO price; Heating is 35% of the year.NPV = US$ 11,162,122; IRR = 96.6%
8	Boiler system improvements	CG University Building	Guangxi Province	Installation of condensing boilers; improved boiler controls; improved water treatment system (2,500 kW)	2,500,000	104,244.0	109,352.0	8,566,000	Investment cost @ US$ 1000/kW; diesel oil savings from improved boiler efficiency (80% to 85%). Annual energy savings = US$ 8.566 million @ US$ 0.89/lit DFO price; Heating is 35% of the year. NPV = US$ 13,952,201; IRR = 96.6%
9	Heat pump application	PG Office Building	Sichuan Province	Installation of heat pumps (Geothermal/Air) – integrated into hot water system (30 tons)	80,000	294.9	309.3	24,230	Investment cost @ US$ 2670/kW; diesel oil savings. Annual energy savings = US$ 24,230 @ US$ 0.89/lit DFO price; NPV = US$ 18,145; IRR = 15.4%
10	Heat pump application	CG Hospital Building	Zhejiang Province	Installation of heat pumps (Geothermal/Air) – integrated into hot water system (60 tons)	160,000	589.7	618.6	48,460	Investment cost @ US$ 2670/kW; diesel oil savings. Annual energy savings = US$ 48,460 @ US$ 0.89/lit DFO price; NPV = US$ 36,290; IRR = 15.4%

Table 2: List of EC&EE/LC Projects (Funded by Market-Based Financing Schemes)

Demo No.	Technology Application	Demo Public Building Type	Location	Description	Investment, US$	Annual Energy Savings, MWh	Annual GHG Emission Reduction, tCO2	Annual Energy Cost Savings, US$	Remarks
EC&EE/LC Projects (funded by Market-based Financing Schemes)									
11	Boiler system improvements	CG Office Building	Jilin Province	Boiler control improvements, installation of TSVs, waste heat recovery (150 kW)	150,000	8,935.2	9,373.0	734,000	Investment cost @ US$ 1000/kW; diesel oil savings from improved boiler efficiency (80% to 85%). Annual energy savings = US$ 0.734 million @ US$ 0.89/lit DFO price; Heating is 50% of the year. NPV = US$ 1,233,626; IRR = 135.1%
12	Boiler system improvements	PG University Building	Gansu Province	Replacement of inefficient boilers, improved boiler controls (300 kW)	300,000	12,509.3	13,122.2	1,028,000	Investment cost @ US$ 1000/kW; diesel oil savings from improved boiler efficiency (80% to 85%). Annual energy savings = US$ 1.028 million @ US$ 0.89/lit DFO price; Heating is 35% of the year. NPV = US$ 1,674,408; IRR = 96.6%
13	Air conditioning system improvements	PG Office Building	Guangdong Province	Installation of inverter-type RAC units, improved space cooling controls. (25,000 m2 conditioned area)	125,000	1,023.8	1,073.9	71,633	Investment cost @ US$ 300/TR @ 60 m2/TR; Grid electricity savings. Annual cost savings = US$ 71,633 @ US$ 70/MWh electricity cost. Building SEC = 180 kWh/m2/yr; 65% of this is AC; and AC is used 35% of the year. NPV = US$ 78,412; IRR = 21.8%
14	Air conditioning system improvements	CG Hospital Building	Hubei Province	Improvement of chiller system controls, improvements in air handling units, improved space cooling controls (50,000 m2 conditioned area)	250,000	2,625.0	2,753.6	183,750	Investment cost @ US$ 300/TR @ 60 m2/TR; Grid electricity savings. Annual cost savings = US$ 183,750 @ US$ 70/MWh electricity cost. Building SEC = 300 kWh/m2/yr; 70% of this is AC; and AC is used 25% of the year. NPV = US$ 230,101; IRR = 28.0%
15	Hot water system improvements	PG University Building	Heilongjiang Province	Repair of leaks, improvements in room radiators and other hot water using devices, hydronic balancing (20,000 m2 heated area)	450,000	1,000.0	1,049.0	82,175	Investment cost @ US$ 25/m2; diesel oil savings from 50% energy losses reduction. Annual energy savings = US$ 82,175 @ US$ 0.89/lit DFO price; Building SEC = 200 kWh/m2/yr; Heating is 50% of the year. NPV = US$ 50,356; IRR = 14.0%
16	Biomass/biogas utilization for energy generation	MG Office Building	Beijing	Utilization of organic waste material for biogas generation for use in hot water production (630 kW)	1,200,000	4,415.0	4,631.4	362,806	Investment cost @ US$ 1900/kW; diesel oil savings. Annual energy savings = US$ 362,806 @ US$ 0.89/lit DFO price; Heating is 80% of the year. NPV = US$ 270,521; IRR = 15.4%
17	Electric motor system improvements	PG Hospital Building	Liaoning Province	Improvement of motor loadings and efficiency, use of energy saving devices, replacement with HEMs (2,500 kW)	100,000	766.5	804.1	53,655	Investment cost @ US$ 40/kW; grid electricity savings from increased motor efficiency (3.5% average). Annual cost savings = US$ 53,655 US$ 70/MWh electricity cost. NPV = US$ 62,739; IRR = 23.4%
18	Water system improvements	NG Office Building	Beijing	Repair of leaks, improvement in water treatment system (520 gpm)	78,000	410.0	430.1	28,700	Investment cost @ US$ 150/gpm; grid electricity savings @ US$ 0.0015 kWh/gal. Annual cost savings = US$ 28,700 @ US$ 70/MWh electricity cost. NPV = US$ 30,757; IRR = 18.1%
19	Lighting system improvements	MG Hospital Building	Yunnan Province	Replacement of indoor lighting system with LED lamps, 40,000 m2 floor area	300,000	1,460.0	1,531.5	102,200	Investment cost @ US$ 7/m2; grid electricity savings from LED lamp (60 lm/W) replacements of FL lamps (40 lm/W). Annual cost savings = US$ 102,200 @ US$ 70/MWh electricity cost. Average illumination level = 500 lm/m2. NPV = US$ 119,815; IRR = 19.3%
20	Building wall system improvements	PG Office Building	Chongqing Province	Improvement of wall insulations, replacement of windows (10,000 m2 floor area)	350,000	900.0	944.1	73,957	Investment cost @ US$ 35/m2; diesel oil savings from 50% energy losses reduction. Annual energy savings = US$ 73,957 @ US$ 0.89/lit DFO price; Building SEC = 180 kWh/m2/yr; Heating is 50% of the year. NPV = US$ 96,982; IRR = 17.2%

Annex L: GHG Emission Reduction Estimates

The estimates of GHG emission reductions that are attributable to the project are based on the most recent version of the Calculation Guide for GEF Project Greenhouse Gas Emission Reduction, as well as the 2008 Manual for Calculating GHG Benefits of GEF Projects: Energy Efficiency and Renewable Energy Projects.

1. Basic Assumptions

a. Based on the historical annual energy consumption (million tce) and GHG emissions (million tons CO2) in the public buildings sector in China (NGOA Data 2006-2015), the average annual growth rate of electricity consumption is 3.9%. The annual growth rate in GHG emissions appear to be rather fluctuating. Trend analysis of the data provided indicates that the average annual growth rate is 1.8%. It is assumed that considering a business-as-usual (BAU) scenario these will be the same annual growth rates in electricity consumption and GHG emissions, respectively from 2018 (start-of-project) to 2021 (end –of-project). It is also assumed that these will be the same growth rates, in the next 10 years after the completion of the PSBEE Project, i.e., 2022-2031.

b. Based on the trend analyses of the same set of data, the most likely average annual growth rate in electricity consumption and GHG emissions, considering the alternative scenario that the PSBEE Project is expected to facilitate and establish would be as follows:

Period	Average Annual Growth Rates, %	
	Electricity Consumption	GHG Emissions
2018 - 2021	1.0	0.6
2022 - 2031	0.7	0.4

c. The trend analyses also shows that for the period 2018 – 2021 (PSBEE Project period), the potential % GHG emission reduction (alternative vs. BAU) is about 6.8% p.a., while during the period 2022-2031 (i.e., influence period) is 15.5% p.a.

d. The GHG emission reductions that will be attributable to the PSBEE Project will be derived from the demonstrations and pilots that will be carried out under the project. These will be the 10 EC&EE/LC technology application demos that will be designed and implemented using technical and financial assistance provided under the project. These are projects that are planned to be implemented by the project partners, i.e., demonstration hosts, which are the administrators/managers of selected NGOA-, and LGOA-administered public buildings. The incremental technical and financial assistance that will be provided is for improving the originally planned EC&EE/LC projects of the host public buildings.

e. The other sources of GHG emission reductions that will be attributable to the PSBEE Project are the 10 EC&EE/LC technology application projects that will be financed by the project owner/implementer and/or by banks/financing institutions making use of market-based financing and incentive schemes. These schemes will be developed for, and implemented by, the partner banks/financial institutions. These 10 projects will also form part of the demonstrations under the PSBEE Project. Incremental technical and financial assistance, if required, will also be provided for improving the originally planned EC&EE/LC projects of the host public buildings.

f. The 20 demonstration projects are presented in Annex K. The table below summarizes the estimated annual and incremental direct GHG emission reductions from each project:

Estimated Direct Emission Reductions (DER) from the Demonstration Projects

Demo No.	EC&EE/LC Technology Application	Description	Estimated Annual Emission Reduction (tCO2)	Life Time (Yrs.)	Estimated Lifetime Direct Emission Reduction[39]	Estimated Incremental Lifetime Direct Emission Reduction[40]	
						DER (during PSBEE)[41]	DER (Lifetime)
EC&EE/LC Technology Application Demonstrations							
1	Solar PV power generation system	Installation of solar PV power generation system – integrated in existing electrical system (360 kWe)	760.9	20	15,217.4	155.2	2,160.1
2	Solar hot water system	Installation of solar water heating system – integrated in the existing hot water system (500 kWth)	1,056.8	20	21,135.3	215.6	3,000.1
3	Combined heat and power (CHP) system	Installation of CHP system serving the heating and power requirements of specific campus facilities (1,200 kW)	8,331.6	20	166,631.6	1,699.6	23,653.3
4	CHP with liquid absorption refrigeration (LAR) system	Installation of CHP system serving the cooling and power requirements of specific hospital facilities (1,200 kW)	28,783.0	20	575,659.9	5,871.7	81,714.9
5	Building energy management system	Energy management system for specific campus facilities (40,000 m^2 floor area)	419.6	5	8,392.0	85.6	1,191.2
6	Building energy management system	Energy management system for specific office facilities (25,000 m^2 floor area)	236.0	5	5,900.6	48.1	670.1
7	Boiler system improvement	Improved boiler controls; improved hydronic system balancing; waste heat recovery (2000 kW)	87,481.6	15	2,624,446.9	17,846.2	451,754.8
8	Boiler system improvement	Installation of condensing boilers; improved boiler controls; improved water treatment system (2,500 kW)	109,352.0	15	3,280,558.7	22,307.8	564,693.5

[39] This assumes that the EC&EE/LC project is a new one.

[40] This considers that the assumed percentage improvement from the alternative scenario to that of BAU scenario during the PSBEE Project period (2018-2021) @ 6.8%, and during the 10-year influence period and beyond @ 15.5%.

[41] This assumes that the demo projects are operating during the last 3 years of the PSBEE Project.

Demo No.	EC&EE/LC Technology Application	Description	Estimated Annual Emission Reduction (tCO2)	Life Time (Yrs.)	Estimated Lifetime Direct Emission Reduction	Estimated Incremental Lifetime Direct Emission Reduction	
						DER (during PSBEE)	DER (Lifetime)
9	Heat pump application	Installation of heat pumps (Geothermal/Air) – integrated into hot water system (30 tons)	309.3	20	6,186.2	63.1	878.1
10	Heat pump application	Installation of heat pumps (Geothermal/Air) – integrated into hot water system (60 tons)	618.6	20	12,372.4	126.2	1,756.3
EC&EE/LC Technology Application Demonstrations (TOTALS)					**6,716,501**	**48,419**	**1,131,472**
EC&EE/LC Projects (funded by Market-based Financing Schemes)							
11	Boiler system improvement	Boiler control improvements, installation of TSVs, waste heat recovery (150 kW)	9,373.0	15	281,190.7	1,912.1	48,402.3
12	Boiler system improvement	Replacement of inefficient boilers, improved boiler controls (300 kW)	13,122.2	15	393,667.0	2676.9	67,763.2
13	Air conditioning system improvement	Installation of inverter-type RAC units, improved space cooling controls. (25,000 m2 conditioned area)	1,073.9	20	21,478.3	219.1	3,048.8
14	Air conditioning system improvement	Improvement of chiller system controls, improvements in air handling units, improved space cooling controls (50,000 m^2 conditioned area)	2,753.6	15	82,608.8	561.7	14,219.7
15	Hot water system improvement	Repair of leaks, improvements in room radiators and other hot water using devices, hydronic balancing (20,000 m^2 heated area)	1,049.0	15	31,470.0	214.0	5,417.0
16	Biomass/ biogas utilization for energy generation	Utilization of organic waste material for biogas generation for use in hot water production (630 kW)	4,631.4	20	92,627.5	944.8	13,148.5
17	Electric motor system improvement	Improvement of motor loadings and efficiency, use of energy saving devices, replacement	804.1	10	16,081.2	164.0	2,282.7

Demo No.	EC&EE/LC Technology Application	Description	Estimated Annual Emission Reduction (tCO2)	Life Time (Yrs.)	Estimated Lifetime Direct Emission Reduction	Estimated Incremental Lifetime Direct Emission Reduction	
						DER (during PSBEE)	DER (Lifetime)
		with HEMs (2,500 kW)					
18	Water system improvement	Repair of leaks, improvement in water treatment system (520 gpm)	430.1	20	8,601.8	87.7	1,221.0
19	Lighting system improvement	Replacement of indoor lighting system with LED lamps, 40,000 m^2 floor area	1,531.5	10	30,630.8	312.4	4,348.0
20	Building wall system improvement	Improvement of wall insulations, replacement of windows (10,000 m^2 floor area)	944.1	15	28,323.0	192.6	4,875.3
EC&EE/LC Projects (funded by Market-based Financing Schemes) TOTALS					**986,679**	**7,285**	**164,727**
Overall Total					**7,703,180**	**55,705**	**1,296,199**

2. CO_2 Emissions Reduction Estimates

Direct CO_2 Emission Reductions (DER)

Within the project intervention period, there will be demonstration and replication activities for the application of EC&EE/LC technologies in public buildings. The above table shows that the estimated cumulative direct emission reductions (DER_{EOP}) during the PSBEE Project implementation (Year 2 to Year 4) is about 55,700 tons CO_2. The lifetime direct emission reductions ($DER_{LIFETIME}$) is about 1,296,200 tons CO_2. These are directly attributable to the PSBEE Project.

- **DER_{EOP}** = CO_2 emission reductions due to PSBEE Project intervention (cumulative by end-of-project) = 55,700 tons CO_2
- **$DER_{LIFETIME}$** = CO_2 emission reductions from all demo projects up to their lifetime = 1,296,200 tons CO_2

Direct Post-Project CO_2 Emission Reductions (DPPER)

During the course of the PSBEE project implementation, there will be direct technical assistance provided to selected NGOA-, and LGOA-administered public buildings that will implement other EC&EE/LC technology application projects or will replicate the demonstration projects. These project beneficiaries will implement their EC&EE/LC projects and realize energy savings (and CO_2 emission reductions). However, not all of them are expected to implement their EC&EE/LC projects during the PSBEE project implementation period. Most of them may only implement their EC&EE/LC projects and realize CO_2 emission reductions (from the energy savings) after the PSBEE project completion. The GHG emission reductions from these projects that can still be attributed to the PSBEE Project as part of Direct Post Project Emission Reductions (DPPER). In this regard, as a conservative estimate for part of the DPPER, it is assumed that a replication of the same set of 10 EC&EE/LC Technology Application Demonstrations will be carried out as new projects after the completion of the PSBEE

Project. Since, such projects are planned to be directly provided with technical assistance during their design and development, the direct GHG emission reductions from such projects would be attributed to the PSBEE Project as DPPER. Based on the above table the potential lifetime GHG emission reductions from these is 6,716,500 tons CO_2.

Also, there will be market-based financing schemes that will be developed and implemented by partner banks/financial institutions for financially supporting EC&EE/LC projects in public buildings. During the PSBEE project implementation period, 10 of such projects will be showcased as demo (i.e., design, engineering, commercial financing, installation and operation) EC&EE/LC technology application projects. Such market-based financing schemes are expected to be continued by the partner banks/FIs after the completion of the PSBEE Project. The energy saving and CO_2 emission reductions from the EC&EE/LC projects that will be supported by such market-based financing schemes can still be attributed to the PSBEE Project as part of DPPER. Nonetheless, to be conservative, the estimation of the lifetime direct post project CO2 emission reductions for this aspect (i.e., continuation of established financing scheme) of the DPPER will assume that the financing scheme will not continue, but other similar financing schemes will be developed and implemented by banks and financing institutions, or by the government. Hence, for this DPPER aspect, the DPPER is Zero.

Lifetime DPPER = 6,716,500 + 0 = 6,716,500 tons CO_2

Consequential CO_2 Emission Reductions (CER)

Bottom-up Approach (BUA)

In estimating the Lifetime Consequential Emission Reductions using the bottom-up approach (CER_{BUA}), the sum of the Lifetime DER and Lifetime DPPER is multiplied by a replication factor (RF). As a market transformation and demonstration project, the GEF prescribed RF value of 3 will be used. In this case:

CER_{BUA} = (Lifetime DER + Lifetime DPPER) x 3 = (1,296,200 + 6,716,500) x 3 = 24,038,100 tons CO_2 or 24.0 million tons CO_2

Top-Down Approach (TDA)

The 10-year influence period starts from the year after the PSBEE Project completion. In this case it is the period from 2022 to 2031. Based on the trend analysis of the historical annual energy consumption and annual GHG emissions from the public buildings sector, the following can be deduced:

- Estimated Forecast Total Energy Savings (alternative scenario vs. BAU scenario) during the 10 year influence period (ES_{TDA}) = 942.8 million tce
- Estimated Forecast Total CO_2 ERs (alternative scenario vs. BAU scenario) during the 10 year influence period (ER_{TDA}) = 922.2 million tons CO_2

Considering the: (a) enabling environment that the PSBEE project will create for the public buildings sector; (b) total potential emissions reduction during the 10 year influence period; and, (c) the current work on energy efficiency in commercial buildings in China, a causality factor = 0.4 (i.e., the GEF contribution is modest, and substantial indirect emission reductions can be attributed to the baseline) is deemed conservative.

CER_{TDA} = Lifetime Consequential CO_2Emission Reductions = ER_{TDA} * CF = 922.2 * 0.4 = 368.9 million tons CO_2

Summary of CO_2 Emission Reductions

CO_2 Emission Reduction Type	Quantity, tCO_2
End-of-Project Direct CO_2 Emission Reduction (DER_{EOP})	55,700
Lifetime Direct CO_2 Emission Reduction (DER_{TOTAL})	1,296,200
Lifetime Direct Post Project CO_2 Emission Reduction ($DPPER_{TOTAL}$)	6,716,500
Lifetime Consequential CO_2 Emission Reduction - BU Approach (CER_{BUA})	24,040,000
Lifetime Consequential CO_2 Emission Reduction - TD Approach (CER_{TDA})	368,900,000

Range of Lifetime Consequential CO_2 Emission Reduction: CER_{BUA} & CER_{TDA}: 24.0 – 368.9 million tons.

For the PSBEE Project, the GEF Contribution is US$ 8,932,420.
Unit Abatement Cost = US$ 8,932,420/ (1,296,200 + 6,716,500) = 1.115 US$/ton CO_2
Hubei Carbon Exchange CER Value = 16.93 ¥/ton CO_2 (US$ 2.54/ton CO_2)
Source: Factor CO2 - Climate Daily News, 30 March 2017

Annex M: Annual Targets

Indicator	Baseline	Year 1	Year 2	Year 3	Year 4
Project Goal					
Cumulative CO2 emissions reduction, ktons	0	0	18.6	40.0	55.7
Annual growth rate of GHG emissions by end of project, %	1.5	1.4	1.2	1.0	0.5
Project Objective					
Cumulative fossil fuel savings due to project intervention, ktce	0	1,000	2,174	4500	6,523
No. of new jobs created with the application of EC&EE technologies and techniques in the public buildings sector in China.	0	At least 500	At least 700	At least 1,200	1,200
Outcome 1					
No. of approved and enforced policies, and associated guidance and implementing rules and regulations (IRRs)	0	1	2	3	2
No. of improved and updated public sector building energy standards.	0	0	1	0	1
Outcome 2					
No. of official building energy audits completed each year under the energy audit system starting Year 1	0	4	8	10	12
No. of public buildings that regularly submit energy supply and consumption reports annually to the EMIS.	0	At least 700	At least 1,000	At least 1,750	2,200
No. of public buildings that are classified as energy efficient.	0	At least 500	At least 1,000	At least 1,750	2,200
Outcome 3.1					
No. of public buildings utilizing the established system for providing information about the features, technical specifications and costs of new EC&EE low carbon technologies each year starting Year 2	0	0	At least 35	At least 75	100
No. of banks that are providing financing for EC&EE/LC projects through the market-based financing scheme for public buildings	2	1	2	3	5
Outcome 3.2					
No. of EC&EE/LC projects financed through the market-based financing scheme for public buildings	0	At least 7	At least 10	At least 15	20
No. of successfully implemented EC&EE/LC technology application demonstrations in public buildings.	0	At least 7	At least 10	At least 15	20
No. of EC&EE/LC projects implemented in public buildings in other provinces that replicate the demonstrations.	0	At least 3	At least 5	At least 7	10
Outcome 4					
No. of trained public building personnel that are qualified and skilled in the design and cost-effective implementation and evaluation of EC&EE projects	0	At least 1,000	At least 1,500	At least 2,500	3,600
No. of public buildings with established energy management programs and are implementing EC&EE/LC projects	0	At least 700	At least 1,000	At least 1,500	2,200

Annex N: Gender Equality Analysis

The upholding of the constitutional principle of equality between men and women is a basic state policy for promoting progress in China. Over the years China has progressively improved its laws and regulations, developed public policies, worked out development plans and pressed forward steadily with gender equality and women's development.

The following summarizes the analysis made regarding the gender equity aspects of the PSBEE Project in China touching on the policies on gender equality and women' s development, as well as in the efforts made and courses of actions taken to maintain gender equity in all aspects of the project design and implementation.

Facilitating Gender Equity in Improving Energy Utilization Efficiency

Among the bases for ensuring gender equity in the proposed PSBEE Project is China's national mechanism for promoting the status of women, fully utilizing government resources and effectively mobilizing social resources. Over the past two decades, this mechanism has been constantly improved to allow it to play an increasingly prominent role.

Selected provincial governments that will participate in the project, for the piloting of EC&EE/LC policies and instruments as well as hosting EC&EE/LC technology application demonstrations will also ensure that the activities relevant to such demonstrations shall be in line with the province's program for women's development. These local governments are continuously actively developing and improving on the assessment of laws and policies related to gender equality, to ensure solid institutional guarantee for promoting gender equality and women' s development.

The PSBEE Project is comprised on several activities that can potentially provide opportunities for women to contribute in the energy efficiency market in the country, particularly in the energy conserving design, planning, construction and operation of public buildings (which can also extend to commercial and residential buildings). This is in line with the Chinese Government's concept of equal participation in economic activities and equitable access to economic resources are the basic conditions for the well-being and development of women. While working towards economic development, the government sees to it that the economic interests of women are protected, thereby promoting women's equal participation in economic development and equal access to the fruits of reform and development.

With China's active promotion of equality in education, whereby the structure of education is such that the principles of gender equality and working hard to guarantee equal rights and opportunities for both men and women to access education are adhered to[42], this has produced qualified and capable women who can contribute their technical skills in the implementation of projects such as the PSBEE Project. This is potentially possible because the gender gap in education in the country has been markedly narrowed in the last decade. The implementation of the PSBEE Project activities will require several sets of expertise in the design of systems that will facilitate EC&EE/LC technology applications in public buildings; design and development of financing schemes for the financing of EC&EE/technology application projects; as well as in the design and development of various pertinent instruments to enhance the capacity, awareness and knowledge of the public buildings sector and the buildings sector in China on EC&EE/LC technologies, techniques and best practices.

[42] China implements the Compulsory Education Law and other relevant laws, regulations and policies, and takes practical measures to improve women's education. In 2014, in institutions of higher learning women accounted for about 52% of undergraduate students, about 52% of postgraduate students, and about 37% of students studying for Ph.D. degrees.

Since the principles and concepts of gender equality are gradually extending into commercial and industrial establishments, as well as in teaching, technical and scientific research, there are now more women in China are available to take on technical and management jobs. Compared to previous years, there are now more Chinese professional and technical women who work on financial, technical and applied science-related jobs, as well as occupy positions of decision-making and management in public sector and private sector organizations and establishments[43]. In the context of the PSBEE Project, project implementation requirements that involve technical studies/research and analysis, and in the engineering design and installation of EC&EE/LC technology applications, and in the management of the demonstration projects, present opportunities for qualified and capable women.

Women who are currently working on the various energy conservation programs of NGOA (Energy Conservation by Public Sector; Energy & Resource Conservation Assessment Program; Energy & Resource Conservation Demonstration Program, and, EC&EE Promotion and training Program) stand to benefit from the capacity development activities that will be carried out under the PSBEE Project. They could either be analysts, technologists, engineers, policy specialists or women occupying decision making and management positions in NGOA or in LGOA offices in the provinces. This is somehow due to the fact that compared to previous years, the number of women participating in the management of public offices as well as in state public affairs keeps increasing. In 2013, the female employment ratio in central government agencies like NGOA and their subsidiaries reached about 48%. In recent years, the number of newly employed women civil servants has also increased steadily at local levels.

The PSBEE Project will facilitate the energy conserving and energy efficient operation of buildings and building services in the public sector in China. This leads to energy savings, i.e., reduction of the fossil fuels used in the operation of building services and facilities. The energy savings translate to greenhouse gas emission reduction from the displaced/avoided energy. This is also an area where China attaches great importance to contributing to the achievement of its climate change mitigation targets (e.g., those stated in its Nationally Determined Contributions (NDCs), and also in creating a social and cultural environment conducive to boosting gender equality. Women in China are playing an increasingly prominent unique role in the fostering of social culture, as well as protection of the ecological environment. More and more Chinese women are getting involved in environmental protection, making their contribution to protecting the ecological environment, addressing climate change and maintaining energy and resource security. They have been encouraged to participate in energy conservation and emission reduction efforts, and to adopt low-carbon lifestyle. They take part in environmental protection with enthusiasm, and environmental protection organizations with women playing a dominant role are on the rise. With this general attitude towards maintenance of environmental integrity and sustainability, the PSBEE Project can also benefit from involving women in the planning and implementation of the project activities, but also in getting them (e.g., the female employees of NGOA and LGOA offices, and of the demo host public buildings) involved in the advocacy and promotional campaigns of the PSBEE to the public building occupants and tenants, as well as the general public in embracing the concepts of energy conservation, energy efficiency, and sustainable energy in their lives.

One of the major activities of the PSBEE Project is the establishment and operationalization of a buildings energy technology information exchange (BETIX) service, which is a system of EC&EE and

[43] The Government of China ensures equal employment right for women. The Employment Promotion Law stipulates, among others, fair employment emphasizing gender equality in employment right. The Special Regulations on Labor Protection of Female Employees, also include policies to facilitate the growth of female scientists and promote equal employment opportunities for female college graduates. In 2013, about 45% of the country's labor force is accounted for by women (at about 346.4 million). In 2013, the number of mid-ranked and senior female professional and technical personnel was around 44%, an increase of 9 percentage points from that of 2000.

LC technology information sharing. This system will enable sharing of latest EC&EE and LC technology and market development information among public building managers/administrators, building practitioners in China and building materials and building energy technology developers and suppliers in other countries. This contributes also to the South-South and Triangular Cooperation (SSTrC) initiative of both the Government of China and UNDP-China. The PSBEE Project provides another appropriate opportunity for both partners to pursue and promote this noteworthy endeavor by making use of the experience of China's organization and hosting of international conventions and documents concerning gender equality and women's development, international bilateral and multilateral women's exchanges and cooperation, thus further contributing to China's important role in promoting gender equality and women's development around the globe. Over the past two decades, China has carried out international cooperative projects concerning women in fields such as women and health, women and the economy, women and the environment, etc. Within the framework of South-South cooperation, the PSBEE Project will share to interested developing countries information and technical materials that will be generated from the various project activities, as well as specific information about experiences on women-led or assisted EC&EE/LC technology applications in the public buildings sector in China.

Conclusion

The wealth of experience in improving and progressing the promotion of gender equality and women's development in China over the past two decades, as among the means of achieving economic and social development, can be put to good use, either as a reference/guide in the involvement of qualified and capable women professionals in implementation of the PSBEE Project activities. Such experiences were also considered in the design of the project activities.

The implementation of the PSBEE Project will be guided by the fact that China will continue implementing the basic national policy of equality between men and women, and in that regard, the PSBEE project management will ensure: (a) that where there are opportunities for women's involvement in the project activities these will be seriously considered and, (b) that the enabling conditions that will be facilitated by the project to achieve the project objective will take into account safeguarding women's rights and interests in accordance with the law, and contributes to gender equality and women's development.

The PSBEE Project, while facilitating the energy conserving and energy efficient operation of public buildings in China, will also contribute to the promotion of gender equality and women's development through its activities that will strengthen and expand international information and knowledge exchanges and cooperation.

Annex O: Knowledge Management Plan

The Knowledge Management (KM) Strategy to be applied in the PSBEE Project will be one that address the barrier related to the low level of awareness and knowledge of the public buildings sector in China. This include the concepts, principles and technologies and practices associated with energy conservation, energy efficiency and low carbon (EC&EE/LC) technologies, as well as low carbon development. These are actually essential for the public buildings sector (whose energy consumption translates to about 5.4% of China's total GHG emissions) to contribute to the country's climate change mitigation targets (50% reduction in GHG emissions by 2020, as per the China NDC). The expected outputs that will be delivered by the project will bring about improved awareness and attitude towards EC&EE/LC technology applications in the public buildings sector.

The KM strategy of PSBEE will involve the monitoring/tracking, evaluating and documenting the results of the project activities, and sharing the information generated not only to the public buildings sector in China, but also to building practitioners in the country's building industry, as well as to the buildings sector in other countries. Part and parcel of the project are activities involving the gathering of observed and recorded results of specific project activities (e.g., demonstrations, pilots, and capacity building activities), processing, analyzing and uploading them into a Public Buildings Energy Management Information System (PBEMIS) Database. Gathered and processed information from the demonstrations will also be used in the preparation of each demo project profile that will later on be disseminated to the target beneficiary sectors and entities in China and interested developing countries (through South-South Triangular Cooperation). The activities that are intended to promote the project are also intended to enhance the knowledge of the target beneficiaries about the goal and objective, expected outcomes and outputs of the project thereby enhancing their enthusiasm and ownership of or belongingness to, the project and encourage them to cooperate in the delivery of the target outputs. The KM strategy also serves to bridge the communication gaps between the source and the recipient of the knowledge on EC&EE/LC applications.

Actually, knowledge management is part and parcel of the PSBEE Project design. Component 4 of the project focuses on addressing barriers related to the access and affordability of information and low level of awareness about EC&EE/LC technologies. Under this component of the project, a number of knowledge products will be developed. A key output under this component will be a "building energy technology information exchange (BETIX) system. This system will enable sharing of latest EC&EE and LC technology and market development information among public building managers/administrators, building practitioners in China and building materials and building energy technology developers and suppliers in other countries. Another knowledge product is a set of established public sector EC&EE/LC management cum education training centers in the different climate regions of the country. These are strategically sited centers for the promotion, technical assistance and capacity development in the field of EC&EE/LC technology applications in public buildings. The capacity development program (in-country trainings, and study tours abroad) will also produce several KM products among them are the training materials, information and training kits.

Component 2 also will also produce a very important knowledge product in the form of a public buildings energy management information system (PBEMIS) database. This output is in conjunction with the building energy audit system and energy management information system (EMIS). This output will bring together the many knowledge products of the project (from all components) and put them in a convenient place accessible to persons in China, particularly in the public buildings sector, who wish to develop and implement EC&EE/LC projects in buildings, as well as to others in Asia and world, who may learn from the experience of China. Corollary to this is the potential tie-up of the PSBEE Project to the South-South Triangular Cooperation Program of China and UNDP-China. The knowledge products developed by the PSBEE project can be used to share information about

EC&EE/LC technology applications to other countries. Chinese experience in the development, manufacture and application of energy efficient building materials can be shared to developing countries. In the same token, the information sharing presents opportunities too for China to leverage the experience of other developing nations as it promotes EC&EE/LC in the PSBEE Project. This bilateral exchange is a significant potential avenue by which to leverage South-South cooperation in the area of EC&EE/LC applications in public buildings. Moreover, part of the knowledge management strategy is to consider China's sharing of its experience and knowledge base that will be further developed through the PSBEE Project with other Asian countries. The support frameworks and tools that will be developed and implemented under the PSBEE will be shared with other countries through the earlier mentioned BETIX system. With that, China would be able to share its experiences and lessons learned with other developing countries.

From Component 1 (policy, plans, and institutions), all draft policies, roadmaps, standards, regulations, and plans, as well as their approval status, will be made available on the PBEMIS Database, which will also have, among others, specific modules for these kind of information, including modules for storing and retrieving information about best available EC&EE/LC technologies, techniques and best practices.

From Component 3, information from the demonstrations will be turned into knowledge products such as demo project profiles, including the evaluation reports on the operational, energy, economic and environmental performances of each demonstration. Information on the market-based financing schemes will also be used to generate knowledge products in the form of guidebooks on how the funds and financing schemes work and how to apply to them, as well as case studies of experiences to date with projects that have been funded by them. Also include is a "how-to" guidance on the highest potential funding sources, such as the Green Climate Fund. Knowledge gathering and dissemination is a major key for the success of the project through its components and activities. The facilitation and enabling of the application of EC&EE/LC technologies/techniques to be done by the project will be accomplished through the removal of barriers associated with the low level of enforcement of policies, regulations and institutional mechanisms, limited capacity and knowledge about the application, design, financing and operation of such initiatives in the country's public buildings sector.